



Derivation of guidance values under the EU Drinking Water Directive : the case study of trifluoroacetic acid

Robin Sur¹ · Richard Bartlett² · Bastian Becker¹ · Anja Hueser¹ · Felix M. Kluxen³ · Volker Laabs³ · Jyotigna Mehta⁴ · Stephanie Melching-Kollmuss³ · Ursula G. Sauer⁵ · Emily Scorgie⁶ · Derek F. Wallace⁷

Received: 1 October 2025 / Accepted: 13 August 2026
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Abstract

Trifluoroacetic acid (TFA) is a metabolite of e.g., pesticides, refrigerants and propellants, and it is considered a ‘non-relevant metabolite’ of pesticides under the EU Drinking Water Directive (DWD) as current exposures have not been found to provide a consumer health risk. Since an EU-harmonised concept for the derivation of drinking water guidance values is unavailable, various guidance values have been proposed or implemented on the Member State level for TFA in drinking water. This causes uncertainty with respect to the adequacy of such values. To address this uncertainty, the scientific rationale underlying the different guidance values is reviewed here. It is further shown that TFA should not be included in the DWD definitions for per- and polyfluoroalkyl substances; this is relevant for one of the suggested guidance values. Finally, a science-based approach for deriving guidance values for substances in drinking water is proposed. Application of this approach to TFA as an example yields a health-based guidance value of 294 µg/L TFA in drinking water. To foster harmonisation and acceptance across agencies, this guidance value should be complemented with appropriate risk management measures (such as introduction of an upper limit for hygienic reasons); however, such further considerations are outside the scope of the present article. All stakeholders are encouraged to align on an EU-harmonised approach to derive scientifically justified and robust guidance values for substances in drinking water to adequately protect consumers.

Keywords Plant protection product · Relevant metabolite · Per- and polyfluoroalkyl substances (PFAS) · Exposure assessment · Liver effects · Developmental toxicity

Abbreviations

Σ4PFAS Sum of 4 PFAS (PFHxS, PFNA, PFOA, PFOS) in the context of EFSA CONTAM Panel (2020)

ADI	Acceptable daily intake
AGES	Austrian Agency for Health and Food Safety (<i>Österreichische Agentur für Gesundheit und Ernährungssicherheit</i>)
ALT	Alanine aminotransferase
ANOVA	Analysis of variance
ANSES	French Agency for Food, Environmental and Occupational Health and Safety (<i>Agence nationale de sécurité sanitaire de l'alimentation, de l'environnement et du travail</i>)
AP	Alkaline phosphatase
ARfD	Acute reference dose
AST	Aspartate aminotransferase
BAUA	German Federal Institute for Occupational Safety and Health (<i>Bundesanstalt für Arbeitsschutz und Arbeitsmedizin</i>)
BMDL/U	Lower / upper confidence limit on the

✉ Robin Sur
robin.sur@bayer.com

¹ Bayer AG, Crop Science, Alfred-Nobel-Strasse 50, 40789 Monheim, Germany

² Jealott's Hill International Research Centre, Syngenta, Jealott's Hill, UK

³ BASF SE, Limburgerhof, Germany

⁴ ADAMA Agricultural Solutions Ltd, Reading, UK

⁵ Scientific Consultancy – Animal Welfare, Neubiberg, Germany

⁶ Syngenta, Enkhuizen, The Netherlands

⁷ ISK Biosciences Europe NV, Diegem, Belgium

	benchmark dose		drinking water (acronym used only in equations)
BMR	Benchmark response		
bw	Body weight	P ₀	Parental animals
CLH	Harmonised classification and labelling	PBPK	Physiologically based pharmacokinetic
CLP	Classification, labelling and packaging	PFAS	Per- and polyfluoroalkyl substance(s)
CONTAM Panel	Panel on Contaminants in the Food Chain (EFSA)	PFHxS	Perfluorohexane sulfonic acid
DNEL	Derived no-effect level	PFNA	Perfluorononanoic acid
DTU	Danish Technical University	PFOA	Perfluorooctanoic acid
DWD	Drinking Water Directive	PFOS	Perfluorooctane sulfonic acid
EC ₅₀	50% Effective concentration	PMT	Persistent, mobile, toxic
ECETOC	European Centre for Ecotoxicology and Toxicology of Chemicals	PND	Postnatal day
		PPP	Plant protection product
ECHA	European Chemicals Agency	REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals
EFSA	European Food Safety Authority		
EMA	European Medicines Agency	RIVM	Netherlands National Institute for Public Health and the Environment (<i>Rijksinstituut voor Volksgezondheid en Milieu</i>)
EOGRS	Extended one-generation reproductive toxicity study		
EP	European Parliament	SC	Scientific committee
ESTP	European Society of Toxicologic Pathology	T4	Thyroxine
EU	European Union	TDAR	T-cell dependent and independent antibody responses
F ₁	First-generation offspring	TDI	Tolerable daily intake
GD	Gestational day	TFA	Trifluoroacetate / trifluoroacetic acid
GLP	Good laboratory practice	TMDI	Theoretical maximum intake level
GWD	Ground Water Directive	TWI	Tolerable weekly intake
HCFC	Hydrochlorofluorocarbons	UBA	German Environment Agency (<i>Umweltbundesamt</i>)
HFC	Hydrofluorocarbons		
HFO	Hydrofluoroolefins	UNEP	United Nations Environment Programme
ICH	International Conference for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use	US	United States
		VITO	Flemish Institute for Technological Research (<i>Vlaamse Instelling voor Technologisch Onderzoek</i>)
IPCS	International Programme on Chemical Safety (WHO)		
ITRC	Interstate Technology and Regulatory Council	vPvM	Very persistent, very mobile
IUPAC	International Union of Pure and Applied Chemistry	WHO	World Health Organization
JMPR	Food and Agriculture Organization of the United Nations (FAO)/World Health Organization (WHO) Joint Meeting on Pesticide Residues		
LOAEL	Lowest-observed adverse effect level		
mkd	Mg/kg body weight/day		
MoA	Mode-of-action		
MRL	Maximum residue level		
NO(A)EL	No-observed (adverse) effect level		
OECD	Organisation for Economic Co-operation and Development		
OPPTS	Office of Prevention, Pesticides and Toxic Substances		
P	Fraction of the TDI allocated to		

Introduction

European Union legislation related to drinking water

In the European Union (EU), a risk-based approach has been implemented in the recast Drinking Water Directive (DWD), i.e. *Directive (EU) 2020/2184 on the quality of water intended for human consumption* (European Parliament [EP] and Council 2020), to ensure that drinking water is of high quality and that its consumption does not compromise human health. In accordance with Article 5(1) and Annex I (Part B) of the DWD, parametric values to assess the quality of drinking water are set for specific chemicals

and types of chemicals. For ‘Pesticides’ (individual) and ‘Pesticides Total’, parametric values of 0.1 and 0.5 µg/L, respectively, are set. ‘Pesticides Total’ reflects “*the sum of all individual pesticides... detected and quantified in the monitoring procedure*”. It is noted that ‘Pesticides’ shall include “*metabolites as defined in point (32) of Article 3 of Regulation (EC) No 1107/2009 [i.e. the EU Plant Protection Products (PPP) Regulation; EP and Council 2009a] that are considered relevant for water intended for human consumption. A pesticide metabolite shall be deemed relevant for water intended for human consumption if there is reason to consider that it has intrinsic properties comparable to those of the parent substance in terms of its pesticide target activity or that either itself or its transformation products generate a health risk for consumers.*”

Notably, a metabolite of a pesticide, that is considered relevant in drinking water under the DWD, is not necessarily

Box 1 EU Groundwater Directive (GWD) and GW-relevant vs DW relevant metabolites

In the EU, measures to prevent and control groundwater pollution have been implemented in the Groundwater Directive (GWD), i.e. *Directive 2006/118/EC of the European Parliament and of the Council of 12 December 2006 on the prevention and control of groundwater pollution* (EP and Council 2006a). In April 2026, the GWD has been amended by Directive No 2026/805 (EP and Council 2026) amending the Water Framework Directive (EP and Council 2000), the GWD and the Environmental Quality Standards Directive (EP and Council 2008a).

The parametric values of 0.1 µg/L for individual ‘Pesticides’ and 0.5 µg/L for ‘Pesticides Total’ laid down in the DWD (see above) are aligned with the corresponding groundwater quality standards in Annex I of the GWD. However, these groundwater quality standards shall be reconsidered as per Article 2(7) of Directive 2026/805 (EP and Council 2026).

Concordant with the (still valid) groundwater quality standard of 0.1 µg/L for individual ‘Pesticides’, point 9.2.4.1 in Part A of *Commission Regulation (EU) No 284/2013 setting out the data requirements for plant protection products* stipulates the requirement to assess the relevance of metabolites of PPPs estimated to occur at concentrations above 0.1 µg/L in groundwater (European Commission 2013); see also Guidance Sanco/221/2000 – rev.11 on the assessment of the relevance of metabolites of PPPs in groundwater (European Commission 2021).

Importantly, the definition of a ‘relevant metabolite’ in water intended for human consumption (as per DWD) is based on risk, i.e. the function of hazard and exposure (see above: “*generate a health risk for consumers*”), whereas the definition of a ‘relevant metabolite’ in groundwater is based on human health hazard alone (“*has certain toxicological properties that are considered unacceptable*”) in Article 3(32) of the PPP Regulation (EP and Council 2009a). While a hazard-based definition for relevant metabolites was also applicable in the context of the GWD (EP and Council 2006a; European Commission 2021) prior to its amendment in April 2026, a risk-based definition for GW-relevant metabolites (which additionally considers environmental risks) has now been implemented in the GWD as per Footnote 4 to Annex III of Directive 2026/805 (EP and Council 2026). Since the PPP Regulation definition for relevant metabolite remains hazard-based, the legal implications of the altered GWD-definition are currently unclear.

also considered a relevant metabolite in groundwater (Box 1; see also Laabs et al. 2015; Sur et al. 2024; UBA 2026a). To illustrate this, the terms DW-relevant metabolite and GW-relevant metabolite (GW: groundwater) are used in this article, which focusses on the provisions of the DWD and, therefore, on DW-relevant metabolites (Table 1).

For metabolites of PPPs in drinking water, an EU-wide harmonised concept for the derivation of guidance values has not been provided under the scope of the DWD. Instead, the note in Annex I (Part B) of the DWD indicates that “*Member States shall define a guidance value to manage the presence of non-relevant metabolites of pesticides in water intended for human consumption*”. Also, Article 5(3) of the DWD empowers Member States to set guidance values for substances that are not listed in Annex I: “*A Member State shall set values for additional parameters not included in Annex I, where the protection of human health within its national territory or part of it so requires*” (EP and Council 2020). However, efforts are under way to establish a common understanding of which metabolites are non-relevant in drinking water. To this end, the European Commission has commissioned the World Health Organization (WHO) to develop a list of DW-non relevant metabolites that is expected to be published by 2027 (European Commission 2024a, 2025; UBA 2025).

Finally, the parametric value of 0.1 µg/L implemented in Annex I (Part B) of the DWD only applies to individual ‘Pesticides’ and their DW-relevant metabolites present in drinking water as a consequence of agricultural pesticide or other pest control use, but not if the same substance is present in drinking water (as parent compound or metabolite) due to other uses and/or other sources (e.g. emission of substances from consumer products; further discussed in Sect. “DW-relevant metabolites, GW-relevant metabolites and non-relevant metabolites of PPPs”). Indeed, the DWD so far does not include any guidance value for substances that are identical with DW-relevant metabolites of PPPs but that are present in the drinking water due to such other uses and/or sources. Since a comprehensive EU-wide approach for how to set guidance values for substances in drinking water is currently unavailable, different Member States may define different guidance values for a given substance. This legal situation does not only contradict the EU objective of harmonisation but may also cause uncertainty with respect to the adequacy of consumer protection in different Member States and regions.

Scope of this article

Against this background, this review article presents the case study of trifluoroacetate / trifluoroacetic acid (TFA) to pursue these goals:

Table 1 European Union legislation regarding relevant vs non-relevant metabolites in drinking water vs groundwater

	Drinking water	Groundwater	
Legislation	DWD (EP and Council 2020)	GWD (EP and Council 2006a) as amended by Directive 2026/805 (EP and Council 2026)	PPP Regulation (EP and Council 2009a)
Subsequent guidance / legislation	None	Guidance document SANCO/221/2000-rev.11 on the assessment of the relevance of metabolites in GW of substances regulated under the PPP Regulation (European Commission 2021)	Commission Regulation 284/2013 setting out data requirements for PPPs (European Commission 2013)
<i>Relevant metabolite</i>			
Definition	DWD, Annex I, Part B: <i>‘Pesticides’ ...and their metabolites as defined in... [the EU PPP Regulation] that are considered relevant for water intended for human consumption. A pesticide metabolite shall be deemed relevant for water intended for human consumption if there is reason to consider that... either itself or its transformation products generate a health risk for consumers</i>	Guidance document SANCO/221/2000-rev.11, Sect. "3.2": <i>a metabolite for which there is reason to assume that it has ... certain toxicological properties that are considered severe and unacceptable with regard to the decision-making criteria described in the text</i>	PPP Regulation, Article 3(32): <i>A metabolite is deemed relevant if there is a reason to assume that... it has certain toxicological properties that are considered unacceptable. Such a metabolite is relevant for the overall approval decision or for the definition of risk mitigation measures</i>
Implications of definition	If a metabolite has a CLH for acute toxicity, STOT or CMR, a risk assessment (i.e. considering exposure) is required to determine if it is a DW-rM	If a metabolite has a CLH for acute toxicity, STOT or CMR, it is always a GW-rM (see Guidance document SANCO/221–2000-rev. 11 for details on the classification)	
Thresholds / provisions	DWD, Annex I, Part B: maximum permissible concentration in DW: Individual ‘Pesticides’ & DW-rM: 0.1 µg/L ‘Pesticides Total’ & DW-rM: 0.5 µg/L Note: These guidance values only apply to PPPs and their DW-rM, but not if the same substance is present in DW due to other uses and/or other sources	GWD, Annex I, Point 1: maximum permissible concentration in GW: Individual ‘Pesticides’ & GW-rM: 0.1 µg/L ‘Pesticides Total’ & GW-rM: 0.5 µg/L	Regulation 284/2013, Part A, point 9.2.4: assess relevance of metabolites if > 0.1 µg/L in GW
<i>Non-relevant metabolite</i>			
Definition	No specific definition	Guidance document SANCO/221/2000-rev.11, Sect. 3.4: <i>a metabolite which does not meet the criteria provided for “relevant metabolites” and “metabolites of no concern” [a]. A non-relevant metabolite may be subject, on a case-by-case basis, to an individual GW limit concentration...</i>	Not applicable
Thresholds / provisions	DWD, Annex I, Part B: <i>MSs shall define a guidance value to manage the presence of non-relevant metabolites of pesticides in water intended for human consumption</i> Note: Guidance values set by MSs for non-relevant metabolites only apply to the presence of these substances in DW as metabolites of PPPs (but not their presence due to other sources). By comparison, guidance values set by MSs in accordance with Article 5(3) of the DWD apply to the respective substances regardless of source DWD, Article 5(3): <i>A MS shall set values for additional parameters not included in Annex I, where the protection of human health within its national territory or part of it so requires</i>	Guidance document SANCO/221/2000-rev.11, Sect. 4, Step 5: a refined assessment is necessary if non-relevant metabolites are present in GW at a concentration between 0.75 and 10 µg/L and a full risk assessment if they are present above 10 µg/L	Not applicable

AS: active substance; CLH: harmonised classification and labelling; CMR: carcinogenicity, mutagenicity, reproductive toxicity; DW: drinking water; DWD: Drinking Water Directive; GW: groundwater; GWD: Groundwater Directive; MS: Member State; PPP: plant protection product; rM: relevant metabolites; STOT: specific target organ toxicity

[a] Guidance document SANCO/221/2000-rev.11, Sect. 3.3: Metabolite of no concern: A metabolite, which is “(a) CO₂ or an inorganic compound, not containing a heavy metal; or (b) ...an organic compound of aliphatic structure, with a chain length of 4 or less, which consists only of C, H, N or O atoms and which has no ‘alerting structures’ such as epoxide, nitrosamine, nitrile or other functional groups of known toxicological concern; (c) ...a substance, which is known to be of no toxicological or ecotoxicological concern, and which is naturally occurring at much higher concentrations in the respective compartment.”

1. To present and discuss the implications of the definitions of DW-relevant vs non-relevant metabolites of PPPs in the DWD (EP and Council 2020) as compared to those for GW-relevant vs non-relevant metabolites in the PPP Regulation (EP and Council 2009a; European Commission 2021) and the GWD (EP and Council 2006a; as amended by Directive 2026/805; EP and Council 2026),
2. To review the scientific rationale underlying health-based guidance values for TFA in drinking water that have been set or proposed on the Member State level,
3. To pursue the question whether TFA, on account of its toxicological and environmental fate properties, should be considered as a part of the group of per- and polyfluoroalkyl substances (PFAS), as this question is linked to one health-based guidance value proposed for TFA in drinking water,
4. To propose a harmonised, science-based approach for deriving health-based guidance values for substances in drinking water and to demonstrate the applicability of this approach using TFA as an example.

By comparison, a discussion of precautionary risk management measures is excluded from the scope of this scientific publication as such measures are not based solely upon scientific evidence but also incorporate political and socioeconomic considerations.

Sect. "Trifluoroacetic acid - trifluoroacetate (TFA)" provides an overview of the intended uses, physico-chemical, fate and toxicological properties of TFA; also, the PFAS-related parameters implemented in the DWD are introduced. Sect. "TFA as a non-relevant metabolite of PPPs in drinking water" shows that TFA may be considered a non-relevant metabolite of PPPs in drinking water. Sect. "Derivation of guidance values for TFA in drinking water on the Member State level" presents the details on how health-based guidance values for TFA in drinking water have been set or proposed on the Member State level, referring to the data and information presented in Sect. "Trifluoroacetic acid - trifluoroacetate (TFA)". Sect. "Discussion" and "Conclusion and outlook" provide the discussion and the conclusions, respectively, and a Glossary is included before the bibliography.

Trifluoroacetic acid—trifluoroacetate (TFA)

TFA: identification, uses and environmental fate

Trifluoroacetic acid (Chemical Abstract Service Registry Number [CAS No.] 76-05-1) is a corrosive liquid that has a very low dissociation constant (Table 2). However, due to the high buffer capacities of environmental and

physiological matrices, all relevant environmental and physiological routes of TFA formation directly yield trifluoroacetate (CAS No. 14477-72-6) and not first the free acid. Therefore, TFA does not exert its acidic properties in the environment or in organisms, and the environmentally and physiologically relevant deprotonated form (i.e. its alkali salt, such as sodium-trifluoroacetate [Na-TFA], CAS No. 2923-18-4) has been used for most studies conducted to elucidate the toxicological and ecotoxicological properties of TFA as well as its fate in the environment, plants and animals. In the present article, when referring to TFA, it is implicit that in all aqueous environments, this substance is present in the acetate form.

TFA is a metabolite of several active substances in PPPs that contain a C-CF₃ moiety including e.g., trifluralin (EFSA 2009), saflufenacil (EFSA 2014), flurtamone (EFSA 2017), cyflumetofen (EFSA 2021) and flufenacet (EFSA 2024a). Also, TFA is the terminal degradation product of a range of commercially used substances, such as next-generation refrigerants and propellants in spray cans. These hydrochlorofluorocarbons, hydrofluorocarbons and hydrofluoroolefins have short half-lives in the atmosphere and are, therefore, increasingly replacing the use of persistent fluorine-containing compounds to mitigate global warming potential (albeit at the trade-off of TFA formation); the fluorine-containing refrigerants, in return, had been introduced to replace chlorine-containing, ozone-depleting substances (Solomon et al. 2016; Danish EPA 2024; Garavagno et al. 2024; Hanson et al. 2024; UNEP 2024).

TFA is also widely used in the synthesis of organic chemicals, as a laboratory reagent, solvent and catalyst (López and Salazar 2013; ECHA [European Chemicals Agency] 2025a; Garavagno et al. 2024; PubChem 2026a). As such, it has been registered under *Regulation (EC) No 1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals* (REACH; EP and Council 2006b) with an annual production volume in the EU of ≥ 100 to < 1000 tonnes (ECHA 2025b).

The degradation of TFA under environmental conditions is very slow and difficult to measure (Kim et al. 2000; Tang et al. 2024; see also European Commission 2023a), and TFA has very low sorption potential to soil (ECHA 2025a). From a chemical perspective, TFA is a perfluoroalkyl carboxylic acid that contains a single -CF₃ moiety bound to a carboxyl functional group (Polesello et al. 2023; PubChem 2026a).

As such, TFA may be covered by the definition for 'PFAS Total' provided in Annex I (Part B) of the DWD:

- *PFAS Total*: Parametric value 0.50 µg/L. PFAS Total means "the totality of per- and polyfluoroalkyl substances. This parametric value shall only apply once technical guidelines for monitoring this parameter are

developed in accordance with Article 13(7) [i.e. European Commission 2024b]. Member States may then decide to use either one or both of the parameters ‘PFAS Total’ or ‘Sum of PFAS’” (EP and Council 2020).

By comparison, TFA is not covered by the definition for ‘Sum of PFAS’ provided in Annex I (Part B) of the DWD:

- *Sum of PFAS*: Parametric value 0.10 µg/L. Sum of PFAS includes twenty PFAS, which are covered by the definition “the sum of per- and polyfluoroalkyl substances considered a concern as regards water intended for human consumption listed in point 3 of Part B of Annex III. This is a subset of ‘PFAS Total’ substances that contain a perfluoroalkyl moiety with three or more carbons (i.e. $-\text{CnF}_{2n-}$, $n \geq 3$) or a perfluoroalkylether moiety with two or more carbons (i.e. $-\text{CnF}_{2n}\text{OCmF}_{2m-}$, n and $m \geq 1$)” (Annex I, Part B, in EP and Council 2020). As further indicated in point 3 of Part B of Annex III to the DWD, these twenty PFAS “shall be monitored when the risk assessment and risk management of the catchment areas for abstraction points carried out in accordance with Article 8 conclude that those substances are likely to be present in a given water supply”.

The twenty PFAS that are “listed in point 3 of Part B of Annex III” do not include TFA. However, since TFA may fall under the definition for ‘PFAs Total’ provided in the note to ‘Sum of PFAS’ (see also Glossary), the parametric value of 0.5 µg/L for ‘PFAS Total’ provided in the DWD (EP and Council 2020) may apply to TFA in drinking water, and this regardless of its source of origin.

Toxicological database for trifluoroacetic acid

In most toxicity studies, TFA has been administered as neutral salt, i.e. Na-TFA (Table 2), which matches its speciation in the environment (Sect. “TFA: identification, uses and environmental fate”). Na-TFA is not considered mutagenic or genotoxic, and it has low potential for acute systemic toxicity (Dekant and Dekant 2023; BAuA 2025; ECHA 2025a; EFSA 2025a).

Repeated-dose toxicity studies for TFA

Regarding repeated-dose toxicity, rat oral 28-day and 90-day toxicity studies assessing Na-TFA are available (Sect. 7.5.1 in ECHA 2025a). Consistently, the liver was established as target organ in the rat upon subacute and subchronic exposure to Na-TFA. The key study for oral repeated-dose toxicity indicated in the REACH dossier is a subchronic (90-day) repeated-dose toxicity study using Wistar rats

that was conducted in 2006–2007 in accordance with the Organisation for Economic Co-operation and Development (OECD) Test Guideline (TG) 408 (OECD 1998a), in compliance with the principles on Good Laboratory Practice (GLP; OECD 1998b) and was thus assessed as reliable without restrictions (Study 7.5.1 – 001 in ECHA 2025a). Na-TFA was administered at concentrations of 0, 160, 1,600 and 16,000 ppm in feed, corresponding to intakes of 0, 9.9, 98 and 1,043 mg Na-TFA/kg body weight/day (mkd) in males and 0, 12.2, 123 and 1,216 Na-TFA mkd in females (Table 3). In the mid- and high-dose groups, the major findings included increased absolute and relative liver weight as well as reduced total bilirubin and glucose, recorded in both sexes. Further, the serum activities of the liver enzymes alanine aminotransferase (ALT), alkaline phosphatase (AP) and aspartate aminotransferase (AST) were increased in the males only. All observed ALT increases were minimal, i.e. less than double the upper value recorded in the concurrent controls, and they were not correlated with adverse liver findings.

In the high-dose group and in some males of the mid-dose group, the increased liver weight was associated with minimal to moderate diffuse centrilobular to panlobular hepatocellular hypertrophy. Also, there was a higher incidence of hepatocellular necrotic foci in the males of the high-dose group; these animals further showed an overall reduction in body weight gain. Additionally, ground-glass appearance of the hepatocellular cytoplasm was observed, and this was assessed as indicating peroxisome proliferation (ECHA 2025a; see also Dekant and Dekant 2023). However, there was no difference in severity grading of focal/multifocal hepatocellular necrotic foci between treated and control animals.

Based upon the liver effects observed in the mid- and high-dose groups, the no-observed adverse effect level (NOAEL) from the 90-day study was established at the lowest dose, i.e. 9.9 and 12.2 mkd Na-TFA in the males and females, respectively, corresponding to 8.4 and 10.1 mkd TFA. This study endpoint was considered the most sensitive one in the database to derive health-based reference values in the current EFSA conclusions (EFSA 2014, 2017; further discussed in Sect. “Hazard assessment: establishment of an ADI for TFA by EFSA”).

A rat 14-day exploratory study using Wistar rats supported the conclusion from the 90-day study that TFA may be a weak peroxisome proliferator in rats (BAuA 2025; Dekant and Dekant 2023). In the 14-day study, the males/females were exposed to 0, 43/45, 85/91 and 170/190 mkd Na-TFA in the feed. Dose-dependently increased palmitoyl-coenzyme A oxidation (as biomarker for peroxisome proliferation) was observed in the males of the mid- and high-dose groups. Activation of the peroxisome proliferator-activated

Table 2 Trifluoroacetic acid, sodium-trifluoroacetate and perfluorooctanoic acid: identification, physico-chemical and fate properties, classification

Property	Trifluoroacetic acid		Sodium-trifluoroacetate		Perfluorooctanoic acid (pentadecafluorooctanoic acid)	
	Data, information	Reference	Data, information	Reference	Data, information	Reference
CAS No	76-05-1	PubChem (2026a)	2923-18-4	Surya Prakash and Mathew (2010); PubChem (2026b)	335-67-1	PubChem (2026c)
Molecular formula	C ₂ HF ₃ O ₂		C ₂ F ₃ NaO ₂		C ₈ HF ₁₅ O ₂	
Molecular weight	114.02 g/mol		136 g/mol		414.07 g/mol	
General appearance	Colourless fuming liquid with a pungent odour		White hygroscopic powder		White powder, waxy white solid	
Water solubility, density	Soluble in water, density: 1.53 g/mL at 20 °C		Water soluble		Not water soluble, density: 1.79 g/mL at 20 °C	
Vapour pressure	kPa: 11 at 20 °C		Lower than for the free acid due to ionic structure		3.16 × 10 ⁻² mm Hg at 25 °C	
Henry's law constant	1.11 × 10 ⁻⁷ atm-cu m/mol at 25 °C		Not applicable		3.044 atm-cu m/mol	UNEP (2023)
pH of TFA aqueous solution	0.45 at 10% (w/w), non-oxidising	ECHA (2025a)	Neutral	ECHA (2025a)	2.6	Thermo Fisher Sci. (2024)
Dissociation constant	pK _a : 0.3–0.4	ECHA (2025a); PubChem (2026a)	As reported for TFA		PubChem: pK _a : 1.3–2.8 (–0.5 to 4.2)	Vierke et al. (2012);
Stability	Dissociated in all environmental compartments	ECHA (2025a)			Stable and persistent in environment	Ahmed et al. (2020);
Rate of degradation	< 1%, reaction with OH: too slow to measure					UNEP (2023);
<i>n</i> -octanol/water partition coefficient	log K _{ow} : 0.79 at 25 °C – low potential for bioaccumulation				PubChem: log K _{ow} : 6.3	Wee and Aris (2023);
Organic carbon / water partition coefficient	K _{oc} : 6.22 at 25 °C – very low adsorption to soil				UNEP: K _{oc} : 30,440	ITRC: K _{oc} : 98–1288 (sediment), 50–794 (soil)
Preferred environmental compartment	Water, rather than air, ground or biota				Water, sorption to soil and sediment, bioaccumulation	
Half-life in humans	16 h	UNEP (2023)	Same as for TFA		2.1–10 years	UNEP (2023)

Table 2 (continued)

Property	Trifluoroacetic acid		Sodium-trifluoroacetate		Perfluorooctanoic acid (pentadecafluorooctanoic acid)	
	Data, information	Reference	Data, information	Reference	Data, information	Reference
Current CLH	Acute toxicity 4 (inh.), skin corrosive 1A, aquatic chronic 3	ECHA (2025b, 2026a)	Notified C&L: Acute toxicity 2, aquatic acute and chronic 1	ECHA (2025c, 2026a)	Acute toxicity 4 (inh., oral), eye damage 1, carcin. 2, reproductive toxicity 1B, Lact., STOT RE 1 (liver), PBT, POP, SVHC	ECHA (undated)
Additional /modified CLH	Reproductive toxicity 1B, acute toxicity 4 (oral), acute toxicity 3 (inh.), PMT, vPvM		Reproductive toxicity 1B, PMT, vPvM			

C&L: classification and labelling; Carcin.: carcinogenicity; CAS No.: Chemical Abstract Service Registry Number; CLH: Harmonised Classification and Labelling in accordance with Title V of the EU CLP Regulation (EP and Council, 2008b); CLP: Classification, labelling, packaging; Inh.: inhalation; Lact.: may cause harm to breast-fed children; PBT: persistent, bioaccumulative, toxic; PMT: Persistent, mobile, toxic; POP: persistent organic pollutant; STOT RE: specific target organ toxicity, repeated exposure; SVHC: substance of very high concern; vPvM: very persistent, very mobile; w/w: weight by weight

receptor- α leading to hepatotoxicity in rats is considered a rat-specific mode-of-action (MoA) that is not relevant to humans (Bjork and Wallace 2009; Corton et al. 2018; Felter et al. 2018; Dekant and Dekant 2023; see also ECHA 2024a).

As regards longer-term toxicity, a GLP-compliant one-year repeated-dose toxicity study is available (Table 3; Study 7.5.1 – 003 in ECHA 2025a), which was conducted in accordance with OECD TG 452 (OECD 2018a). Sprague–Dawley rats were exposed to 0, 30, 120 or 600 ppm Na-TFA in the drinking water for 52 weeks followed by a six-week recovery period. According to the information provided on the ECHA website, the selected concentration range was based on the 90-day NOAEL. Since no adverse effects were observed in this study at any dose level tested, the one-year NOAEL was set at the highest concentration, corresponding to 37.8 and 64.0 mkd Na-TFA in the males and females, respectively (ECHA 2025a).

Developmental and reproductive toxicity studies for TFA

The REACH registration dossier includes a GLP-compliant extended one-generation reproductive toxicity study (EOGRTS) that was performed in 2020–2022 in accordance with OECD TG 443 (OECD 2018b), i.e. Study 7.8.1 – 001 in ECHA (2025a) (Table 3). Han Wistar rats were exposed to 120, 600 and 3,000 ppm Na-TFA in the feed (reduced to 60, 300 and 1500 ppm during lactation in the parental females and from postnatal day [PND] 21 to 53 in the first-generation [F₁] offspring). As indicated in the REACH registration dossier, this study did not show any adverse effects that were assessed as treatment-related, and the high-dose level of 3,000/1,500 ppm was established as the NOAEL,

corresponding to test material intakes of 242–265 mkd Na-TFA or 203–223 mkd TFA, across sexes/life stages. The liver findings that were observed at this dose level were assessed as adaptive, i.e. non-adverse. Similarly, reduced serum levels of thyroxine (T₄), which were observed in the high-dose group parental animals and in the mid- and high-dose group offspring, were assessed as non-adverse as the T₄ decrements were generally mild to moderate and within the historical control dose range and further did not coincide with any changes in thyroid histopathology or any other indications of abnormal thyroid function. As an exception, serum T₄ levels were reduced by 52% as compared to the concurrent controls at week 13 in the high dose group F₁ males (i.e. at 242 mkd Na-TFA), and this value was outside the historical control dose range (Study 7.8.1 – 001 in ECHA 2025a; Table 3).

Immunophenotyping of the spleen leukocytes of a subset of the F₁ offspring from the EOGRTS did not yield any treatment-related findings. A decreasing trend was observed in the immunophenotyping parameters in the treated males and females as compared to the control group. While decreased relative CD4⁺ T cell counts and increased relative CD8⁺ T cell counts were observed in the low-dose group females, these findings were assessed as not treatment-related. There were no observable effects on the other immunophenotyping parameters in the spleen leukocytes of either the male or female F₁ offspring exposed to up to 1,500/3,000 ppm Na-TFA (Study 7.8.1 – 001 in ECHA 2025a; Table 3).

A GLP-compliant prenatal developmental toxicity study from 2010 (OECD TG 414; OECD 2001) in which up to 150 mkd TFA (as free acid) was administered to pregnant female Sprague–Dawley rats via oral gavage from gestational day (GD) 6 to GD 19, i.e. from implantation towards

Table 3 Toxicological database for sodium-trifluoroacetate (data from: UBA 2020a; BAuA 2025; ECHA 2025a; EFSA 2025a)

	Rat oral 90-day subchronic repeated-dose toxicity study	Rat oral 1-year chronic toxicity study (with toxicokinetic assessments)	Rat extended one-generation reproductive toxicity study
OECD TG	408, conducted in 2006–2007	452 (417), conducted in 2018–2019	443, conducted in 2020–2022
GLP-compliance	Yes	Yes	Yes
Rat strain	Wistar	Sprague–Dawley	Wistar
No. of animals / sex and dose group	10/sex and group	Main study: 20/sex and group Recovery groups: 10/sex and group Toxicokinetics: 3/sex for control group, 4/sex for dose groups	P ₀ generation: 25/sex and group F ₁ A & F ₁ B cohorts: 20/sex and group
Mode of application	Feed	Drinking water	Feed
Exposure duration	90-day exposure	52-week exposure, 6-week recovery period	P ₀ : (m&f) 10 weeks pre-mating & 2 weeks mating & (f) pregnancy, lactation / (m) 6 weeks F ₁ : in utero & pre-weaning & post-weaning up to w13-w14
Dose range (ppm Na-TFA)	0, 160, 1,600, 16,000	0, 30, 120, 600	0, 120, 600, 3,000 (0, 60, 300, 1,500 during lactation & pups from PND 21–35)
Dose range (mkd Na-TFA)	m: 0, 9.9, 98, 1,043 f: 0, 12.2, 123, 1,216	m: 0, 1.8, 7.6, 37.8 f: 0, 3.3, 12.2, 64.0	P ₀ (m): 0, 9.71, 49.2, 248 P ₀ (f, pre-mating): 0, 10.26, 53.9, 265 P ₀ (f, gestation): 0, 8.65, 44.3, 223 P ₀ (f, lactation): 0, 9.85, 47.5, 233 F ₁ (m): 0, 9.37, 47.3, 242 F ₁ (f): 0, 9.83, 49.4, 248
Major findings	≥ 98/123 mkd (m/f): ↑ abs. & rel. liver weight, ↓ total bilirubin, ↓ glucose ≥ 98 mkd (m): ↑ ALT, AP, AST 1043/1216 mkd (m/f, some m at 98 mkd): minimal to moderate diffuse centrilobular to panlobular hepatocellular hypertrophy 1043 mkd (m): ↑ hepatocellular necrotic foci, overall ↓ in body weight gain	No adverse effects ≥ 7.6/12.2 (m/f): ↑ ALT (reversible during recovery period)	No treatment-related adverse effects 9.83 mkd: F ₁ (f, w13): ↓ rel. CD4+ T-cell, ↑ rel. CD8+ T-cell (low dose group only) 47.3 mkd: ↓ T4: F ₁ (m, PND 22): –17%; F ₁ (m, w13): –32% (within HCD) 223–265 mkd: adaptive liver findings, ↓ T4: P ₀ (m/f): –41%/–32% (within HCD), F ₁ (m&f, PND 22): –37% (within HCD); F ₁ (m, w13): –52%
NO(A)EL [a]	NOAEL: 160 ppm ≙ m: 9.9 mkd Na-TFA / 8.4 mkd TFA f: 12.2 mkd Na-TFA / 10.1 mkd TFA	NOEL: 30 ppm ≙ m: 1.8 mkd Na-TFA / 1.5 mkd TFA NOAEL: 600 ppm ≙ m: 37.8 mkd Na-TFA / 31.8 mkd TFA	NOEL: 120/60 ppm ≙ m&f: 8.65–9.83 mkd Na-TFA / 7.27–8.26 mkd TFA NOAEL: 3000/1,500 ppm ≙ m&f: 242–265 mkd Na-TFA / 203–222 mkd TFA

/ legend: ↓: decreased; ↑: increased or induced; ALT: alanine aminotransferase; AP: alkaline phosphatase; AST: aspartate aminotransferase; f: females; F₁: first generation offspring; GLP: good laboratory practice; HCD: historical control dose range; m: males; mkd: mg/kg body weight/day; Na-TFA: sodium trifluoroacetate; NO(A)EL: no-observed (adverse) effect level; OECD: Organisation for Economic Co-operation and Development; P₀: parental generation; P₁: First generation offspring as parental animals; PND: postnatal day; TFA: trifluoroacetic acid; TG: test guideline; w: week

[a] Conversion factor TFA-sodium salt to TFA: 0.84

the end of the 3-week gestation period in the rat, also did not yield any adverse maternal or developmental findings, and 150 mkd TFA was established as both the maternal and embryo-foetal NOAEL (Study 7.8.2 – 001 in ECHA 2025a).

A further GLP-compliant prenatal developmental toxicity study, from 2020–2021 (OECD TG 414; OECD 2018c), in which pregnant female New Zealand White rabbits were administered 180, 375 and 750 mkd Na-TFA from GD 6

to GD 28, i.e. from implantation up until the end of the 4-week gestation period in the rabbit, via oral gavage did show signs of maternal toxicity and developmental toxicity (Study 7.8.2 – 005 in ECHA 2025a). Specifically, the mid- and high-dose group dams showed moderate bile duct hyperplasia/fibrosis; therefore, the maternal NOAEL was set at 180 mkd Na-TFA (corresponding to 153 mkd TFA). In the foetuses, overall mean foetal weights and consequently

mean total litter weights were dose-dependently reduced at 375 and 750 mkd Na-TFA (318 and 636 mkd TFA). Also, increased incidences of abnormalities (above the concurrent and historical controls), which predominantly affected the eyes (i.e. multiple folded retinas and absent aqueous/vitreous humour), were observed at these dose levels. Since one foetus of the low dose group (180 mkd Na-TFA) also showed these abnormalities, it was concluded that the NOAEL for embryonic/foetal development was below 180 mkd Na-TFA / 153 mkd TFA (ECHA 2025a). A further prenatal developmental toxicity study using pregnant female New Zealand White rabbits (OECD TG 414; OECD 2018c) then yielded a maternal and foetal NOAEL of 60 mkd Na-TFA, i.e. 50.4 mkd TFA (Study 7.8.2 – 006 in ECHA 2025a).

A further *in vivo* study, in which pregnant rabbits were also exposed to up to 750 mkd Na-TFA from GD 6 to GD 28, showed very high Na-TFA levels both in the maternal and foetal serum and in the maternal and foetal aqueous/vitreous humour. Individual maximum Na-TFA concentrations attained up to 20 mM in the serum and up to 1.95 and 2.57 mM in the aqueous/vitreous humour of the dams and foetuses, respectively. Hence, this study showed excessive internal exposure levels of Na-TFA. Vitamin A levels remained unaffected on GD 13, indicating that the test item did not interfere with the rabbits' retinoic acid pathway (Study 7.8.2 – 008 in ECHA 2025a). The retinoic acid pathway plays a crucial role during normal eye development (Thompson et al. 2019). *In vitro*, Na-TFA did not bind to or inhibit human recombinant cytochrome P450 family 26 (CYP26; Study 7.9.4 – 002 in ECHA 2025a). This provided further evidence to exclude retinoic acid pathway interference as CYP26 contributes to retinoic acid metabolism and homeostasis (Isoherranen and Zhong 2019). Further *in vitro* studies supported the hypothesis of increased test substance uptake into the aqueous humour as Na-TFA was transported by human monocarboxylate transporter 1 (MCT1) with a 50% effective concentration (EC_{50}) of 12 mM, an effect that was inhibited by the specific and potent MCT1 inhibitor AR-C155858 (Study 7.9.4 – 001 in ECHA 2025a). MCTs are “essential amongst others for lactate, energy and water homeostasis in eyes” (7.8 Summary in ECHA 2025a).

On account of the ophthalmic abnormalities observed in the rabbit prenatal developmental toxicity studies, the German Federal Institute for Occupational Safety and Health (*Bundesanstalt für Arbeitsschutz und Arbeitsmedizin*; BAUA 2025) has proposed assigning Na-TFA and TFA a harmonised classification and labelling (CLH) for reproductive toxicity category 1B, i.e. presumed human reproductive toxicant, in accordance with Title V of *Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures* (CLP Regulation; EP and Council 2008b) and the Commission Delegated Regulation

amending the CLP hazard classes and categories (European Commission 2023b). Additionally, BAUA proposed assigning Na-TFA and TFA CLHs for persistent, mobile, toxic (PMT) and very persistent, very mobile (vPvM) criteria and, for TFA only, modifying the CLH Acute Toxicity category 3 (toxic if inhaled) and adding the CLH Acute Toxicity category 4 (harmful if swallowed) (BAUA 2025). In June 2026, the ECHA Risk Assessment Committee adopted by consensus all proposed CLH for Na-TFA and TFA, and their uptake into the CLP Annex VI list of CLH is expected for its next adaptation to technical progress (ECHA 2026a, 2026b).

TFA as a non-relevant metabolite of PPPs in drinking water

As is explored below, the risk assessment, i.e. consideration of both hazard and exposure, shows that TFA should be considered a non-relevant metabolite of PPPs in drinking water since its presence in drinking water has not been found to pose “a health risk for consumers” (EP and Council 2020), even at concentrations encountered in raw water for the production of drinking water.

Introduction: establishment of tolerable or acceptable daily intake levels

The starting point for the risk assessment of substances that may be present in drinking water or food (and the setting of the corresponding health-based guidance values; Sect. “Introduction: derivation of guidance values for substances in drinking water”) is the tolerable daily intake (TDI) or acceptable daily intake (ADI) level (EFSA SC [Scientific Committee] 2012, 2021). Both the TDI and the ADI reflect the amount of the substance in food or drinking water that “can be consumed over a lifetime without presenting an appreciable risk to health” (EFSA 2026). Whereas TDI levels are generally established for “chemical contaminants, which usually have no intended function in drinking water”, ADI levels are estimated for “food additives and pesticide residues that occur in food for necessary technological purposes or plant protection reasons” (WHO 2022).

The TDI and ADI are mostly derived from experimental data from animal studies since relevant and reliable human data are generally unavailable (WHO 2022). For substances that are not genotoxic carcinogens (which applies to TFA; Sect. “Toxicological database for trifluoroacetic acid”), the “tolerable daily intake (TDI) should be derived using the most sensitive endpoint in the most relevant study, preferably involving administration in drinking-water” (Sect. 8.2.2 in WHO 2022). The selected relevant study should preferably

include subchronic or chronic substance administration so that the exposure duration spans a substantial part of the animal's lifetime. The TDI is then based upon the NOAEL from the most sensitive endpoint in the key study. If it is not possible to establish a NOAEL (e.g. since adverse effects are observed at all dose levels), a lowest-observed-adverse-effect level (LOAEL) may also be used. Alternatively, the lower confidence limit on the benchmark dose (BMDL) may be selected as this “*could eliminate the need for application of an additional uncertainty factor to the LOAEL*” (WHO 2022). To calculate the TDI, the respective parameter is divided by an appropriate assessment factor (also known as uncertainty factor):

$$\text{TDI} = \frac{\text{NOAEL or LOAEL or BMDL}}{\text{Assessment factor}}$$

The assessment factor is determined on a case-by-case basis using expert judgement. As applicable, the assessment factor may address interspecies and intraspecies variation, the adequacy of studies or databases as well as the nature and severity of effects (WHO 2022). In the EFSA SC *Guidance on selected default values to be used by the EFSA Scientific Committee, Scientific Panels and Units in the absence of actual measured data*, it is recommended to use an overall default assessment factor of 100 (10 for interspecies variability $\times$ 10 for intraspecies variability) if chemical-specific data on toxicokinetics and/or toxicodynamics are unavailable (EFSA SC 2012).

Additional assessment factors may be considered, e.g. to extrapolate from subchronic to chronic study duration in rodents. Different approaches for the determination of assessment factors are available. EFSA recommended to use an assessment factor of 2, “*considering the extent of investigations usually performed in 90-day studies*” (EFSA SC 2012). Additional assessment factors to account for deficiencies in the database may need to be established on a case-by-case basis (ECHA 2012; EFSA SC 2012); see also European Centre for Ecotoxicology and Toxicology of Chemicals (ECETOC) *Technical Report on derivation of assessment factors for human health risk assessment* (ECETOC 2003).

Hazard assessment: establishment of an ADI for TFA by EFSA

At the time of writing this article, considerations from EFSA regarding ADI levels for TFA, as a metabolite of a PPP, have been published in:

- EFSA (2014): Reasoned opinion on the setting of MRLs [maximum residue levels] for saflufenacil in various crops, considering the risk related to the metabolite trifluoroacetic acid (TFA)
- EFSA (2017): Conclusion on the updated peer review of the pesticide risk assessment of the active substance flurtamone
- EFSA (2021): Reasoned opinion on the modification of the existing maximum residue levels for cyflumetofen in various crops
- EFSA (2024a): Peer review of the pesticide risk assessment of the active substance flufenacet

In the reasoned opinion of 2014, EFSA used the TG-conforming, GLP-compliant rat 90-day repeated-dose toxicity study as key study and established the liver as target organ, with effects observed in the mid- and high-dose group (Sect. “[Repeated-dose toxicity studies for TFA](#)”, Table 3). “*For the metabolite TFA, on the basis of toxicological studies that were made available for this assessment, EFSA concluded that it is possible to derive a tentative ADI of 0.05 mg/kg bw per day and a tentative ARfD at the same level (0.05 mg/kg bw) [body weight] [acute reference dose]*” (EFSA 2014). The ADI of 0.05 mkd is based on the lowest NOAEL of 10 mkd Na-TFA in the males (without conversion to the free acid TFA) divided by a total assessment factor of 200, i.e. 100 for intra- and interspecies variation multiplied by 2 to extrapolate from subchronic to chronic study duration (EFSA 2014; Table 4).

Generally, this ADI of 0.05 mkd for Na-TFA was confirmed in the subsequent EFSA conclusions and reasoned opinions (EFSA 2017, 2021, 2024a); however, in the most recent EFSA (2024a) peer review of the risk assessment of flufenacet, EFSA noted that the derivation of guidance values for TFA could be affected by evidence from the one-year toxicity study and the developmental and reproductive toxicity studies, which had not been available for the earlier assessments.

In July 2024, the European Commission issued a mandate to EFSA to review the toxicological reference values for TFA in the light of this new evidence (EFSA 2025b). As an outcome, EFSA published a *Draft statement on consumer health-based guidance values on trifluoroacetic acid* in July 2025 (EFSA 2025a), launching a two-month public consultation. In this draft statement, the serum T4 reduction observed in the EOGRS is identified as the critical effect for the derivation of an ADI; additionally, an ARfD is proposed to account for the developmental findings:

“*Based upon the new evidence, the most suitable reference point for setting the ADI is the parental NOAEL of 8.65 mg/kg bw per day based on the observed changes in thyroid hormones (decreased T4 levels) in the EOGRS... The WG [working group]*

proposed applying the standard uncertainty factor of 100, along with an additional factor of 3 to account for the remaining uncertainties related to the absence of a long-term toxicity/carcinogenicity study and the absence of TDAR [T-cell dependent and independent antibody responses] assay on TFA. The resulting ADI is 0.03 mg/kg bw per day (expressed as sodium trifluoroacetate). TFA showed developmental toxicity in rabbits (different eye malformations) at 250 mg/kg bw per day, with a NOAEL identified at 60 mg/kg bw per day. Based on this NOAEL, an ARfD of 0.6 mg/kg bw (expressed as sodium trifluoroacetate) was established by applying a standard uncertainty factor of 100” (EFSA 2025a; Table 4).

So far, there are no (proposed or established) health-based guidance values for TFA in drinking water that are founded on these newly proposed reference values (Sect. "Derivation of guidance values for TFA in drinking water on the Member State level", Table 4). Also, the newly proposed reference values may be subject to reassessment depending on the final EFSA statement (which is pending at the time of writing this article). As it is the primary goal of this article to present the rationale underlying the (proposed or established) guidance values for TFA in drinking water, and not so much specific numerical values, all further considerations focus on the ADI of 0.05 mkgd Na-TFA (EFSA 2014, 2017, 2021, 2024a). The transparent presentation of all calculation methods and underlying parameters enables the reader to update concrete values at any time.

Exposure and risk assessment: TFA as non-relevant metabolite of PPPs in drinking water

Consistently, all exposure estimations for TFA show that the levels of TFA that consumers are exposed to via drinking water (and food) are lower than the ADI of 0.05 mkgd Na-TFA (EFSA 2014, 2017, 2021, 2024a).

In 2014, EFSA conducted a comprehensive dietary consumer exposure assessment for TFA that considered TFA residues in primary and rotational crops resulting from the use of PPPs and in food resulting from environmental contaminants (EFSA 2014). *“To derive the input values for the exposure assessments, a conservative approach was applied using highest TFA concentrations described in relevant studies... or in the public domain”*. In absence of information, residues in commodities of animal origin were not considered. As regards consumer exposure to TFA via food, *“the total calculated intake accounted for up to 5% of the ADI. The major contributors in this calculation were apples (up to 1.9% of the ADI), wheat (up to 1.4% of the ADI) and oranges and wine grapes (both 0.6% of the ADI,*

respectively)” (EFSA 2014). As regards consumer exposure to TFA via drinking water, EFSA estimated that it was *“expected to be insignificant”*, corresponding to *“0.04% of the ADI derived for TFA”* (EFSA 2014). EFSA based this estimation on the assumption that a 60-kg adult consumes 2 L of water per day and that this water contains 630 ng/L TFA, i.e. the *“highest concentration reported for European surface water”* (EFSA 2014, citing Jordan and Frank 1999) (Table 4, Supplementary Information (SI) Table SI-1).

In the risk assessment of flurtamone, EFSA also concluded that TFA exposure via drinking water was low: *“Since the concentrations of TFA residues in groundwater were estimated to be in the range 3.619–22.13 µg/L in four out of nine scenarios... [estimated via exposure modelling], the potential exposure of consumers to this compound via drinking water was assessed and resulted in a dietary intake accounting for 6.6% ADI for infants, 4.4% ADI for children and 1.5% ADI for adults according to the WHO guidelines”* (EFSA 2017). In 2017, EFSA did not finalise the consumer risk assessment for TFA exposure via food due to prevailing data gaps (EFSA 2017).

In the reasoned opinion on the modification of the existing MRLs for cyflumetofen, EFSA (2021) updated the risk assessment conducted by EFSA (2014) considering residues of TFA measured in rotational crop metabolism studies (i.e. OECD TG 502; OECD 2007a) conducted with cyflumetofen if these data were higher than those previously considered (see page 15 in EFSA 2021 for details). EFSA concluded that it had identified no consumer concern regarding the overall exposure to the metabolite TFA from different sources (via food): *“The total chronic intake accounted for a maximum of 9% of the ADI (NL toddler diet)”* (page 4 in EFSA 2021).

In the conclusion on the risk assessment of flufenacet, EFSA indicated that the highest calculated theoretical maximum daily intake (TMDI) of Na-TFA via food *“from the available rotational crop experiments accounts for 31% of the ADI (UK toddler)”* (page 11 in EFSA 2024a). As mentioned on page 36 of the corresponding list of endpoints (EFSA 2024b), this value was estimated using EFSA PRIMo 3.1 (EFSA 2019); however, further details (e.g. on input values) are not provided in the EFSA conclusion (EFSA 2024a), in the list of endpoints (EFSA 2024b) or in the peer review report on flufenacet (EFSA 2024c). By comparison, in the renewal assessment report for flufenacet, the Rapporteur Member State indicated that the highest TMDI of TFA via food, again estimated using EFSA PRIMo 3.1, was 21% of the ADI for the NL toddler (page 68 in European Commission 2024c), and this finding is supported by input values and details on the outcome of the modelling (pages 63, 65, 482 in European Commission 2024c). In the renewal assessment report, it is also stated that the estimations were based upon data from field rotational crop residue trials (i.e.

Table 4 Trifluoroacetic acid in drinking water: acceptable daily intake and guidance values

Member State / reference	Critical effect	Reference point	Assessment factors	DW-cons	Alloc. factor	Average bw	ADI, guidance value or reference value for TFA
<i>Rat 90-day TFA study as key study to establish ADI and to derive guidance value for TFA in drinking water</i>							
EFSA (2014)	Effects on liver weight and histopathology	NOAEL: 9.9 mkd Na-TFA	Intraspecies: 10 Interspecies: 10 Subchronic → chronic: 2	Not applicable			ADI: 0.05 mkd Na-TFA TFA exposure via DW: 0.04% of ADI [a]
DTU Food, Denmark (Nielsen and Beltoft 2021)	Effects on liver weight and histopathology	NOAEL: 9.9 / 8.4 mkd Na-TFA / TFA	Intraspecies: 10 Interspecies: 10 Subchronic → chronic: 2	0.03 L/kg bw/day	10%	DW-cons. expressed in relation to bw	Guidance value: 100 µg/L TFA Rounded from 140 µg/L (i.e. 0.042 mkd ADI * 0.1 AF / 0.003 L/kg bw/day DW-Cons.)
Denmark (Danish Government 2025)	Set for precautionary reasons to undercut the target concentration of ≤ 10 µg/L TFA implemented by the German UBA (2020a, b) for reasons of drinking water hygiene						Quality requirement for consumer taps: 9 µg/L TFA
Belgium, Flanders (De Clerq and De Brouwere 2024)	Reduced total bilirubin at d 90 [b] relative liver weight [b]	BMDL ₃₀ : 5.2 mkd [c] BMDL ₀₅ : 11 mkd	Intraspecies: 10 Interspecies: 10 Subchronic → chronic: 2 Data limitations: 10 [d]	2 L/day	20%	60 kg	Guidance value: 15.6 µg/L TFA (based on total bilirubin) Guidance value: 33 µg/L TFA (based on rel. liver weight)
<i>Rat 1-year TFA study as key study to derive guidance value for TFA in drinking water</i>							
Germany (UBA 2020a, b)	Increased ALT	NOEL: 1.8 mkd Na-TFA	Intraspecies: 10 Interspecies: 10	2 L/day	10%	70 kg	Guidance value: 60 µg/L TFA (rounded from 63 µg/L Na-TFA) Target: ≤ 10 µg/L TFA Guidance value: 60 µg/L TFA Target: ≤ 10 µg/L TFA
Austria (Austrian Government 2022)	No explicit reference						
France (French Government 2025)	Reference to calculations by UBA (2020a, 2020b)						
Italy (Senate of the Republic of Italy 2025)							
Switzerland (Canton Zurich 2025)							Guidance value: 60 µg/L TFA
Hawaii DoH (2025), based upon advice from Gibb and O'Leary (2024)	Reference to calculations by UBA (2020a, b)						Endorses TDI (0.018 mkd), cites guidance value (60 µg/L) and target value (10 µg/L)
Luxembourg Health Dir. (2025)	Increased ALT	NOEL: 1.8 mkd Na-TFA	Intraspecies: 10 Interspecies: 10	0.75 L/day	10%	5 kg infant	Reference value: 12 µg/L TFA
<i>Use of PFAS-related information to derive guidance value for TFA in drinking water</i>							
Netherlands (RIVM 2023)	TFA is considered as part of PFAS group. Line of arguments [e]: EFSA CONTAM Panel (2020): TDI/TWI for PFOS + PFHxS + PFOA + PFNA = 0.63/4.4 ng/kg bw/week, based upon BMDL ₁₀ from study by Abraham et al. (2020) in which 1-year old infants showed inverse association between serum levels of sum of PFOA + PFNA + PFHxS + PFOS and vaccination response Bil et al. (2021): rat liver effect data for 16 PFAS TFA data (rel. liver weight in 90-day study) used to estimate relative potency of TFA 0.002 as compared to the other PFAS			2 L/day	20%	70 kg	Guidance value (TFA): 2.2 µg/L Guidance value (PFOA): 4.4 ng/L

Table 4 (continued)

Member State / reference	Critical effect	Reference point	Assessment factors	DW-cons	Alloc. factor	Average bw	ADI, guidance value or reference value for TFA
Belgium, Wallonia (SPW Env 2025)	Refers to calculations by RIVM (2023)						Guidance value (TFA): 2.2 µg/L
<i>Rat EOGRTS to propose ADI</i>							
EFSA (2025a)	Serum T4 reduction	Parental NOAEL: 8.65 mkd	Intraspecies: 10 Interspecies: 10 Data limitations: 3 [f]	Not applicable			ADI: 0.03 mkd Na-TFA
<i>Rabbit prenatal developmental toxicity studies to propose ARfD</i>							
EFSA (2025a)	Eye malformations	Foetal NOAEL: 60 mkd	Intraspecies: 10 Interspecies: 10	Not applicable			ARfD: 0.6 mkd Na-TFA

ADI: acceptable daily intake; Alloc. factor: allocation factor; ALT: alanine transferase; ARfD: acute reference dose; BMDL: benchmark dose level; DW(-cons.): drinking water (consumption); EFSA: European Food Safety Authority; Na-TFA: sodium trifluoroacetate; NO(A)EL: no-observed (adverse) effect level; PFAS: per- and polyfluoroalkyl substance; PFOS: perfluorooctane sulfonic acid; PFOA: perfluorooctanoic acid; PFNA: perfluorononanoic acid; PFHxS: perfluorohexane sulfonic acid; TFA: trifluoroacetic acid; TWI: tolerable weekly intake

[a] Assuming that the drinking water contains 630 ng/L TFA, which reflects the “highest concentration reported for European surface water” (EFSA 2014; citing Jordan and Frank 1999)

[b] Data evaluation across 90-day and one-year repeated dose toxicity studies

[d] The BMDL was selected instead of the NOAEL to consider effect size, and the BMDL₃₀ was selected to accommodate physiological homeostasis

[d] Due to unavailability of immunotoxicity data in the rat

[e] See Sect. “Netherlands” for details

[f] Due to unavailability of rat 2-year toxicity data and rat immunotoxicity data

OECD TG 504; OECD 2007b) further considering transfer into animal commodities, however, that monitoring data are available for TFA that “give a realistic picture on TFA residue levels in a broad range of different crops” and yield an ADI usage of 5% for food of plant origin (pages 61–62 and 66 in European Commission 2024c).

As regards exposure to TFA via drinking water, the EFSA conclusion on the risk assessment of flufenacet includes the statement that Na-TFA “contributes with 8, 23 and 35% of ADI for adult, child and infant, respectively” (EFSA 2024a; see also page 36 in EFSA 2024b). Details on the underlying groundwater TFA concentrations are not provided. In the renewal assessment report for flufenacet, it is indicated that TFA intake via drinking water contributes with 3%, 9% and 13.4% of the ADI for adult, child and infant, respectively, when considering the (highest provided) upper-limit groundwater concentration of 44.8 µg/L Na-TFA (page 60 in European Commission 2024c). The Rapporteur Member State concluded “when combining intakes from both sources [food and drinking water], maximum intakes by food of plant and animal origin (15.2% ADI) and intakes by means of drinking water (see above) do not result in an exceedance of the ADI for TFA and thus do not present a consumer health concern” (page 66 of European Commission 2024c).

Taken together, all exposure estimations lead to the conclusion that the possible human exposure to TFA via drinking water (and food) is lower than the ADI of 0.05 mkd Na-TFA and that exposure to TFA via drinking water is generally only a fraction of the exposure via food. Therefore, based on current research, exposure to TFA via drinking water has not been found to pose a health risk to consumers, and TFA should be considered a non-relevant metabolite of PPPs in drinking water also once it is assigned a CLH for reproductive toxicity (Sect. “Developmental and reproductive toxicity studies for TFA”). Indeed, there is support for this conclusion by the German Environment Agency (Umweltbundesamt; UBA 2020a, 2026a, 2026b).

Derivation of guidance values for TFA in drinking water on the Member State level

Introduction: derivation of guidance values for substances in drinking water

Since TFA is a non-relevant metabolite of PPPs in drinking water, Member States may set a guidance value for TFA in drinking water, either in accordance with the corresponding note in Annex I (Part B) of the DWD (EP and Council

2020) or, since a TFA-specific parametric value has not been implemented in Annex I, in accordance with Article 5(3) of the DWD (Sect. "European Union legislation related to drinking water").

As stated in Preamble 14 of the DWD (EP and Council 2020), the approach to set guidance values for substances in drinking water are based on the *WHO Guidelines for drinking water quality* (WHO 2022). The type of guidance value that may be set for a given substance depends upon both technical aspects and the underlying toxicological database:

"A guideline value normally represents the concentration of a constituent that does not result in any significant risk to health over a lifetime of consumption. A number of provisional guideline values have been established at concentrations that are reasonably achievable through practical treatment approaches or in analytical laboratories; in these cases, the guideline value is above the concentration that would normally represent the calculated health-based value. Guideline values are also designated as provisional when there is a high degree of uncertainty in the toxicological and health data" (Section 8.2 in WHO 2022).

Guidance values for substances in drinking water are derived by multiplying the TDI (Sect. "Introduction: establishment of tolerable or acceptable daily intake levels") with the average body weight of the relevant human population and the fraction of the TDI allocated to drinking water (P) divided by the daily drinking water consumption (WHO 2022):

Guidance value, drinking water =

$$\frac{\text{TDI} \times \text{average body weight} \times P}{\text{Daily drinking water consumption}}$$

In the following, this equation is referred to as 'WHO standard calculation method'.

The following assumptions are provided in the WHO guidance: "The default assumption for consumption by an adult is 2 L of water per day, whereas the default assumption for body weight is 60 kg" (page 177 in WHO 2022). By comparison, the EFSA SC recommended using a higher default assumption for body weight: "A body weight of 70 kg should be used as default for the European adult population (aged above 18 years). The Scientific Committee acknowledges that this default value will lead in some cases to less conservative risk assessments compared to using 60 kg but considers that 70 kg is a more realistic estimate of the average bodyweight of the European population" (EFSA SC 2012).

The WHO guidance also provides details on the proportion of the TDI that may be attributed to water consumption:

"In the absence of adequate exposure data or where documented evidence is available regarding widespread presence in one or more of the other media (i.e. air, food, soil or consumer products), the normal allocation of the total daily intake to drinking-water is 20% (floor value), which reflects a reasonable level of exposure based on broad experience, while still being protective (Krishnan & Carrier, 2012). This value reflects a change from the previous allocation of 10%, which was found to be excessively conservative. As chemicals are progressively reassessed, overall exposure will be reconsidered, and a change in the default allocation factor from 10% to 20% will be made, if appropriate" (page 176 in WHO 2022).

In the below overview of guideline values for TFA in drinking water that have been set or proposed at the Member State level, it is distinguished between guidance values that are based on (1) the rat 90-day study as key study; (2) the rat one-year study as key study; and (3) PFAS-related information (Table 4). Focus of this overview is a review of the scientific rationale underlying the respective guidance values. To become legally binding, the respective guidance values need to be implemented in national legislation. To date, this has been undertaken by Denmark (Sect. "Denmark"), France and Italy (Sect. "Countries citing the German guidance values for TFA in drinking water"). Therefore, all other guidance values described below currently only serve advisory purposes.

Rat 90-day study as key study to derive a guidance value for TFA in drinking water

Denmark

A maximum permissible value of 9 µg/L TFA in drinking water has been implemented in Annex 1(b) of the *Danish Executive Order on water quality and supervision of water supply facilities* (Danish Government 2025). As becomes evident from an answer from the Danish Ministry of the Environment to the Danish Parliament (Danish Government 2025), this value is aligned with the target concentration of ≤ 10 µg/L TFA implemented by the German UBA for reasons of drinking water hygiene (Sect. "Germany"). Also, it was set for precautionary reasons but may be subject to change depending on the final evaluation of the ongoing developmental toxicity studies (Danish Government 2025).

By comparison, providing advice to the Danish Environmental Protection Agency (EPA), the Danish Technical University – National Food Institute (DTU Food) suggested a more than tenfold higher health-based guidance value of 100 µg/L TFA in drinking water (Nielsen and Beltoft 2021).

The Danish Ministry of the Environment acknowledged this advice but referred to ongoing studies evaluating the developmental toxicity potential of TFA to justify maintenance of the lower guidance value of 9 µg/L until further data became available (Danish Government 2020).

Nielsen and Beltoft (2021) selected the rat 90-day study as the key study, the observed liver effects as critical endpoint and the NOAEL in males as reference point (point of departure) for the derivation of the TDI. Thus, the resulting TDI is widely concordant with the one derived by EFSA in 2014 and 2017 (Sect. "Exposure and risk assessment: TFA as non-relevant metabolite of PPPs in drinking water"). However, Nielsen and Beltoft considered that the sodium salt of TFA had been administered to the rats in the 90-day study and converted the low-dose level of 9.9 mkd Na-TFA in the male rats (used as NOAEL by EFSA) to a NOAEL of 8.4 mkd of TFA. Dividing this NOAEL by the total assessment factor of 200 (concordant with EFSA) yielded a slightly lower TDI of 0.042 mkd TFA in drinking water (0.05 mkd TFA by EFSA).

To derive the guidance value, Nielsen and Beltoft allocated 10% of the TDI to TFA exposure via drinking water. This decision considered that food is the "major source for exposure to TFA", since the total dietary intake of TFA accounted for up to 5% of the ADI (citing EFSA 2014; see Sect. "Exposure and risk assessment: TFA as non-relevant metabolite of PPPs in drinking water"), whereas adults would at most be exposed to approximately 0.07 µg/kg body weight (bw)/day TFA from drinking water. This estimate was based on the highest concentration for TFA of 2.4 µg/L measured in the 2020 groundwater monitoring (Danish EPA 2021), assuming that the concentration of TFA in drinking water would be the same as in groundwater and setting the adult body weight at 70 kg (Nielsen and Beltoft 2021).

As regards daily drinking water consumption, Nielsen and Beltoft used a body weight-related default value of 0.03 L/kg bw/day as laid down in the *Danish EPA guidance document for the setting of health-based quality criteria for chemical substances in relation to soil, ambient air and drinking water* (Danish EPA 2006). Accordingly, the following variant of the WHO standard calculation method was applied to derive the quality criterion:

$$\text{Quality criterion} = \frac{\text{TDI} \times P}{\text{Daily drinking water consumption in relation to body mass}}$$

With a TDI of 0.042 mkd TFA, 10% of the TDI allocated to drinking water (P) and the default assumption of 0.03 L/kg bw daily drinking water consumption, this yields a quality criterion of 0.14 mg/L, i.e. 140 µg/L rounded to 100 µg/L (Nielsen and Beltoft 2021). (Application of the WHO standard calculation method while assuming an average body weight of 60 or 70 kg and 2 L of daily drinking water

consumption would yield similar guidance values of 126 or 147 µg/L, respectively.)

Belgium – Flanders

De Clercq and De Brouwere (2024) from the Flemish Institute for Technological Research (VITO: (*Vlaamse Instelling voor Technologisch Onderzoek*)) provided advice to the Flemish authorities regarding the setting of a guidance value for TFA in drinking water. Whereas De Clercq and De Brouwere also used the rat 90-day repeated-dose toxicity study as key study, they selected reduced total bilirubin as the critical effect. This selection is founded on the evaluation of the TFA database by the Netherlands National Institute for Public Health and the Environment (RIVM 2023; RIVM: *Rijksinstituut voor Volksgezondheid en Milieu*). When jointly assessing the data from the rat 90-day and one-year repeated-dose toxicity studies, RIVM (2023) only found serum total bilirubin (at day 90) and relative liver weight, but not ALT, AST, AP or absolute liver weight, to be statistically significantly and dose-dependently affected across both studies. In RIVM (2023), it is explained that reduced bilirubin is considered an adverse effect as it appears to play an important role in different diseases.

Based upon this cross-study evaluation by RIVM (2023), De Clercq and De Brouwere (2024) selected altered serum levels of total bilirubin at day 90 as the primary effect for the derivation of the guidance value for TFA. De Clercq and De Brouwere explained that total bilirubin was more sensitive than relative liver weight, which they selected as a secondary effect. Also, De Clercq and De Brouwere selected the BMDL as starting point for the derivation of the TDI, stating that they considered the BMDL preferable to the NOAEL for statistical reasons as it represented an effect size. For the parameter total bilirubin, De Clercq and De Brouwere selected the BMDL₃₀, i.e. 5.2 mkd TFA (and for relative liver weight, the BMDL₀₅, i.e. 11.4 mkd TFA). The relatively high BMDL₃₀ was selected to accommodate the broad range of physiological serum levels of total bilirubin (Sect. 4.1.1 in RIVM 2023).

To calculate the TDI, De Clercq and De Brouwere (2024) added an extra assessment factor of 10 to the general assessment factor of 200 (i.e. $10 \times 10 \times 2 \times 10 = 2000$) due to incompleteness of the data. De Clercq and De Brouwere argued that data on immunotoxicity were unavailable, but that this was the critical endpoint 'for other PFAS', citing the EFSA Panel on Contaminants in the Food Chain (EFSA CONTAM Panel 2020) to support this statement (see Sect. "Netherlands" for details on the CONTAM Panel evaluation). Division of the BMDL₃₀ of 5.2 mkd TFA by the total assessment factor of 2000 yielded a TDI of 0.0026 mkd TFA in drinking water.

For further translation of this TDI into a health-based guidance value for drinking water, De Clercq and De Brouwer were allocated 20% of the total consumption of TFA to drinking water, indicating (1) that this standard allocation factor as per WHO (2022) was retained as TFA appears to be accumulating in the aquatic environment (citing Jordan and Frank 1999); and (2) that RIVM (2023) also applied an allocation factor of 20%.

Further, an average body weight of 60 kg and a daily drinking water consumption of 2 L were assumed. Application of the WHO standard calculation method yields the health-based guidance value of 15.6 µg/L TFA in drinking water for the endpoint total bilirubin (or 33 µg/L TFA in drinking water if relative liver weight is selected as critical endpoint). Subsequently, the Flemish Environment Agency (*Vlaamse milieumaatschappij*) implemented 15.6 µg/L as a precautionary value (VMM 2025).

Rat one-year study as key study to derive a guidance value for TFA in drinking water

Germany

In 2020, the German Environment Agency UBA set a health-based guidance value of 60 µg/L TFA in drinking water, which replaced the health-related indication value from 2016 (3.0 µg/L TFA in drinking water, set in consideration of available in vitro genotoxicity and rat acute and subchronic toxicity data), which in turn had superseded a stricter health-related indication value of 1.0 µg/L set in 2008 (UBA 2020a, b); see SI-2 for overview of substance-agnostic thresholds in drinking water vs groundwater implemented in Germany and on the level of the EU. The UBA re-evaluation of 2020 considers the rat one-year repeated dose toxicity study (Sect. "Repeated-dose toxicity studies for TFA") as the key study. While adverse effects were not observed in this study, UBA reported that the liver enzyme ALT was dose-dependently increased in the mid- and high-dose groups (reversible post-exposure) and that statistical evaluation via an ANOVA-type test showed that this ALT increase was statistically significant between the control group and the mid-/top-dose groups. Similarly, ALT increases (but non-statistically significant) were observed after 14-, 28- and 90-day exposure to TFA (UBA 2020a). UBA selected ALT increase observed in the mid- and top-dose groups of the one-year study as the critical finding and set the lowest dose level (1.8 mkd Na-TFA in the males) as the no-observed effect level (NOEL). (By comparison, in the REACH registration dossier, it is stated that "*there was no test article related serum chemistry change*" noted in the one-year study [ECHA 2025a].)

UBA has provided a general justification to support the selection of non-adverse effect levels, instead of adverse effect levels as indicated in the WHO guidelines (WHO 2022), as critical effects for the derivation of drinking water guidance values:

"In individual cases, LOEL or NOEL may also be used if there is a plausible and well-documented justification that the (non-adverse) effects can lead to subsequent damage as precursor events" (UBA 2026c, translation by authors of present article).

Applying an assessment factor of $10 \times 10 = 100$ to account for inter- and intraspecies differences, a TDI of 0.018 mkd Na-TFA was calculated from the one-year NOEL. Further applying the default assumptions of 70 kg body weight (EFSA SC 2012), 2 L daily water consumption and 10% allocation of total consumption to drinking water, the WHO standard calculation method yielded the health-based guidance value of 63 µg/L Na-TFA, which was rounded to 60 µg/L TFA. (Notably, while the findings from the one-year study relate to Na-TFA [Study 7.5.1 – 003 in ECHA 2025a], UBA did not distinguish between Na-TFA and TFA.)

UBA has also provided a general justification for using a 10% allocation factor as default value:

"Germany has decided to maintain the allocation factor of 10% for drinking water to continue ensuring the high level of protection in accordance with the principle of drinking water purity, while conducting specific allocation assessments in individual cases. A default increase in the standard allocation factor [from 10% to 20%] would ultimately result in increases in established guideline values and limit values, which in turn could result in a deterioration of drinking water quality. This would contradict the principle of minimisation and the precautionary principle" (UBA 2026c, translation by authors of present article).

In addition to the health-based guidance value of 60 µg/L, UBA noted that, for reasons of drinking water hygiene, a target concentration of ≤ 10 µg/L TFA in drinking water should be aimed for (UBA 2020a, b; see also background information in Adlunger et al. 2021). This target concentration is aligned with the generic guidance value of 10 µg/L that has been set in the *German Ordinance on catchment areas of abstraction points for drinking water production* for non-relevant metabolites of PPPs that do not provide any indication for chronic toxicity, genotoxicity, neurotoxicity, immunotoxicity or germ cell damaging effects (German Federal Government 2023; see also SI-2).

Finally, in June 2026 UBA confirmed that the health-based guidance value of 60 µg/L TFA in drinking water is also protective of reproductive toxicity and, therefore, remains valid also after the assignment of a CLH for reproductive toxicity to TFA:

“As regards TFA, the most sensitive endpoint remains a toxic effect on the liver. Based thereupon, the German Environment Agency calculated a health-based guidance value of 60 µg/L in 2020. This existing guidance value is therefore also protective against potential reproductive toxicity, even upon lifelong intake via drinking water” (UBA 2026b; translation by authors of present article).

Countries citing the German guidance values for TFA in drinking water

The French Government has adopted the UBA (2020a, b) guidance values, i.e. the health-based guidance value of 60 µg/L TFA in drinking water and the target concentration of ≤ 10 µg/L, while noting that these values may be replaced on account of ongoing work by ANSES, the French Agency for Food, Environmental and Occupational Health and Safety (French Government 2025). Generally, the current ANSES guidance to derive drinking water guidance values follows a similar approach as implemented by Denmark (Sect. "Denmark"), i.e. to express the selected ADI/TDI (multiplied by an allocation factor) relative to a body weight-related daily drinking water consumption value, which is 0.045 L/kg bw/day as per ANSES (2019); see Sect. "Estimation of daily drinking water consumption" for further discussion.

In Italy, the target concentration of 10 µg/L has been adopted as parametric value in the corresponding national Annex I (Part B) to become applicable no later than 12 January 2027 (Republic of Italy 2025).

The German guidance values are also cited on the website of the Canton of Zurich, Switzerland (Canton of Zurich 2025) and in a report from the Austrian Federal Ministry for Agriculture and Forestry, Regions and Water Management (Austrian Government 2022). As regards the formal adoption of a guidance value for TFA in drinking water, the Austrian Agency for Health and Food Safety (AGES; *Österreichische Agentur für Gesundheit und Ernährungssicherheit*) announces deferring the decision until the findings from the developmental toxicity studies evaluating TFA have become available (AGES 2025). The Department of Health of the US State of Hawaii has adopted the UBA TDI of 0.018 mkg TFA as reference point for the derivation of an 'action level' for TFA in drinking water (18 µg/L), which is recommended for the screening of public water systems

(Hawaii DoH 2025), based upon advice from Gibb and O'Leary (2024).

Luxembourg

In 2024, the Luxembourg Health Directorate set a reference value of 12 µg/L TFA in drinking water (Luxembourg Health Directorate 2024, 2025; see also Luxembourg Ministry of the Environment 2024; Luxembourg Water Management Agency 2024). The starting point for the Luxembourg reference value for TFA in drinking water was also the TDI of 0.018 mkg Na-TFA that UBA (2020a) had derived from the rat one-year study. However, the Luxembourg Health Directorate considered a 5 kg infant consuming 0.75 L of water/day. Thereby, application of the WHO standard calculation method (and an allocation factor of 10%) yields a reference value of 12 µg/L instead of 60 µg/L TFA based on the adult default assumptions as indicated by UBA (2020a). Similarly, in a report on metabolites of PPPs in groundwater and drinking water conducted by the Austrian AGES on behalf of the Austrian Government, it is recommended to base drinking water guidance values on 5-kg infants assuming 0.75 L daily water consumption (page 65 in AGES 2024).

Use of PFAS-related information to derive a guidance value for TFA in drinking water

Netherlands

Consideration of TFA as a part of the PFAS group The Netherlands National Institute for Public Health and the Environment (RIVM) pursued the goal to derive an indicative guidance value for TFA in drinking water, explaining that 'indicative' implies that the derivation is based on existing evaluations whereas underlying studies are, in principle, not assessed. Importantly, RIVM assessed TFA "as part of the group of PFAS" (RIVM 2023; see SI-3.1 for original quote in Dutch). The starting point for the derivation of the RIVM guidance value for TFA in drinking water was an evaluation of four PFAS that had been conducted by the EFSA CONTAM Panel in response to a request from the European Commission for a scientific evaluation on the potential risks to human health related to the presence of PFAS in food (EFSA CONTAM Panel 2020); see Fig. 1 for overview of the RIVM's approach to deriving a guidance value for TFA in drinking water.

Effects of PFAS on immune system in humans – derivation of TWI/TDI for the sum of four PFAS The CONTAM Panel's evaluation is summarised here because it is utilised by the

RIVM in establishing the guidance value for TFA in drinking water. The four PFAS considered by the CONTAM Panel were perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorohexane sulfonic acid (PFHxS) and perfluorooctane sulfonic acid (PFOS). The CONTAM Panel justified this focus because these are the major PFAS found in human blood and because different PFAS mediate similar effects (and toxicokinetics) in animals (EFSA CONTAM Panel 2020).

Based on the overall animal and human evidence, the CONTAM Panel selected associations between maternal and offspring PFAS serum concentrations and vaccination responses in children as critical effects for the risk assessment of PFAS. From the available human observational studies (summarised in Table SI-4), the CONTAM Panel then selected Abraham et al. (2020) as the key study for the derivation of the TDI since it yielded the lowest reference point (see below in this Section), further considering that one-year old infants, as included in the study by Abraham et al., are more sensitive to effects on the immune system than older children, as included in the other studies (Table SI-4).

In brief, upon vaccination against *Haemophilus influenza* type b, diphtheria and tetanus, the one-year-old infants showed an inverse association between serum levels of PFOA alone (Abraham et al. 2020), and of all four PFAS combined (EFSA CONTAM Panel 2020; with support from Abraham and his co-authors), and antibody titres against the three pathogens. For serum levels of PFNA or PFHxS

or PFOS alone, such an association could not be established (Abraham et al. 2020). Applying benchmark dose modelling, the CONTAM Panel (again with support from Abraham and his co-authors because the original data were not available to the CONTAM Panel) established a lowest BMDL₁₀ at 17.5 ng/mL for the sum of the four PFAS (in the following: Σ 4PFAS) in the infants' serum "based on the inverse association between serum levels of the sum of these four PFASs and antibody titres against diphtheria" (EFSA CONTAM Panel 2020).

Based on the mean serum levels of the four PFAS in one-year-old breastfed infants as reported by Abraham et al. (2020), the contribution of each PFAS to the serum level of 17.5 ng/mL was estimated to be 8.47 ng/mL PFOA, 0.30 ng/mL PFNA, 1.06 ng/mL PFHxS and 7.67 ng/mL PFOS, or 8.78 ng/mL PFOA/PFNA and 8.72 ng/mL PFHxS/PFOS (EFSA CONTAM Panel 2020).

Using physiologically based pharmacokinetic (PBPK) modelling, the BMDL₁₀ of 17.5 ng/mL for Σ 4PFAS in the infants' serum was estimated to correspond to a long-term maternal exposure (i.e. TDI for adults) of 0.631 ng/kg bw/day, assuming 12 months of breastfeeding (PFOA/PFNA: 0.187 ng/kg bw/day; PFHxS/PFOS: 0.444 ng/kg bw/day). "The CONTAM Panel decided to apply a breastfeeding duration of 12 months, based on the WHO recommendations to breastfeed exclusively for 6 months with continued breastfeeding along with appropriate complementary foods up to two years of age or beyond" (EFSA CONTAM Panel 2020).

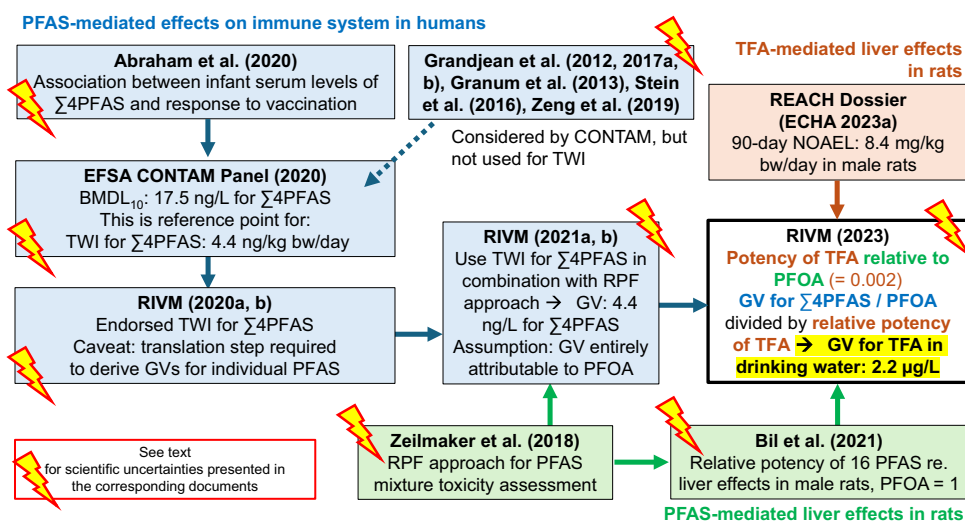


Fig. 1 Overview of approach taken by RIVM (2023) in deriving a health-based guidance value for TFA in drinking water. Abbreviations: Σ 4PFAS: Sum of four PFAS (PFHxS, PFNA, PFOA, PFOS); BMDL: lower confidence limit on the benchmark dose; bw: body weight; CONTAM Panel: Panel on Contaminants in the Food Chain; ECHA: European Chemicals Agency; EFSA: European Food Safety Authority; GV: guidance value; NOAEL: no-observed adverse effect level; PFAS: per- and polyfluoroalkyl substance(s); PFHxS: perfluorohexane

sulfonic acid; PFNA: perfluorononanoic acid; PFOA: perfluorooctanoic acid; PFOS: perfluorooctane sulfonic acid; REACH: Registration, Evaluation, Authorisation and Restriction of Chemicals; RIVM: Netherlands National Institute for Public Health and the Environment (*Rijksinstituut voor Volksgezondheid en Milieu*); RPF: relative potency factor; TWI: tolerable weekly intake. See bibliography for reference details

The applied PBPK model had been developed two years earlier by EFSA CONTAM Panel (2018), based on data from Loccisano et al. (2011), for an evaluation of the human health risk related to the presence of PFOS and PFOA in food. For the evaluation of Σ 4PFAS in 2020, the CONTAM Panel adapted this model to “*estimate the serum levels also in growing children and to include exposure via breast-feeding*” (EFSA CONTAM Panel 2020). To account for accumulation over time, the 24-h-based maternal exposure of 0.63 ng/kg bw/day was then transposed into a TWI of 4.4 ng/kg bw/week for Σ 4PFAS (0.63×7). The CONTAM Panel decided not to apply any assessment factors to derive the TDI/TWI since the BMDL₁₀ had been established for effects in one-year old infants, as a sensitive population, and since the affected parameter, i.e. decreased vaccination response, reflected a risk factor for disease and not an adverse effect (EFSA CONTAM Panel 2020).

Relative potency factor approach and toxic potency of TFA relative to PFOA To convert the EFSA TWI of 4.4 ng/kg bw/week for Σ 4PFAS into a guidance value for TFA in drinking water, RIVM (2023) compared the toxic potency of TFA to that of PFOA. However, the EFSA TWI was based upon human immune response-related data, whereas immunotoxicity data for TFA were unavailable. Similarly, human immune response-related data were only available for a limited number of PFAS (Table SI-4). Therefore, RIVM (2023) decided to base the TFA potency assessment on relative liver weight changes in rats, considering (1) that relative liver weight changes were a critical effect in the rat 90-day TFA study (Sect. “Repeated-dose toxicity studies for TFA”); and (2) that a comprehensive database on PFAS-mediated liver effects in rats was available (Zeilmaker et al. 2018; Bil et al. 2021). This decision was founded on the assumption that a potency ranking based on liver effects in rats could also be applied to other effects and other species.

Zeilmaker et al. (2018) and Bil et al. (2021), from RIVM, had compiled the PFAS hepatotoxicity database to develop a relative potency factor approach to predict the mixture toxicity of PFAS, focussing on effects in male rats as sensitive sex in PFAS risk assessment. The relative potency factor approach involves normalising the dose of each PFAS present in a mixture according to its potency relative to an index substance, which is assigned a relative potency factor of 1. Thereby, each PFAS in the mixture is expressed in terms of its index substance equivalents. The relative potency factor approach is based on the assumptions “*that the mixture components (1) display similar toxicity; (2) differ only in potency (if so, their individual dose–response curves should be parallel on log-dose scale...); (3) do not interact*”, i.e. exhibit dose-additivity (Zeilmaker et al. 2018; citing Bosgra

et al. 2009; van den Berg 2000, 2006). The relative potency factors determined by Zeilmaker et al. (2018) and Bil et al. (2021) can be applied to external PFAS concentrations but not to internal ones as toxicokinetics are not considered (RIVM 2021b; Bil et al. 2022).

The study by Bil et al. (2021) included experimental data on relative liver weight changes for 16 PFAS, covering perfluoroalkyl carboxylic acids (including PFOA and PFNA), perfluoroalkyl sulfonic acids (including PFHxS and PFOS), perfluoroalkyl ether carboxylic acids and fluorotelomer alcohols, and thus also the four PFAS considered by the CONTAM Panel. Bil et al. chose PFOA as the index substance for the potency ranking since it was “*one of the best-studied PFAA [per- and polyfluoroalkyl acids]. Limit values for PFOA (e.g., for drinking water and food) and health-based guidance values are readily available, which enables risk assessment of the summed index compound equivalents*” (Bil et al. 2021).

As a further step in the derivation of the guidance value for TFA in drinking water, RIVM (2023) applied the relative potency factor approach described by Zeilmaker et al. (2018) and Bil et al. (2021) for the relative potency assessment of TFA, focussing on relative liver weight changes. Dose–response modelling using PROAST software Version 70.4 (RIVM 2024) served to ensure that the dose–response curves of TFA and the 16 PFAS were (approximately) parallel on log-dose scale. As indicated above, this is a requirement for applying the relative potency assessment approach, together with the requirements that the substances have a common MoA and do not interact.

This yielded a relative potency of TFA of 0.002 (90% confidence interval: 0.0017–0.0031) as compared to the index substance PFOA (RIVM 2023).

Use of EFSA TWI for Σ 4PFAS to derive drinking water limits for PFAS

Bil et al. (2021) indicated that a drinking water limit of 87.5 ng/L for PFOA was applicable in the Netherlands; this limit value was based upon the steady-state serum concentration of PFOA at the NOAEL observed in a PFOA rat 90-day study, selecting the steady-state serum concentration to account for its bioaccumulation potential (Zeilmaker and Janssen 2016; Box 2A). Four years later, RIVM (2020a, 2020b, 2020c) re-evaluated the evidence for PFAS in the light of the EFSA CONTAM Panel (2020) scientific opinion (Box 2B: overview of RIVM reports and guidance). RIVM (2020b, 2020c) generally approved of the EFSA TWI of 4.4 ng/kg bw/week for Σ 4PFAS as starting point for the derivation of guidance values for individual PFAS, while noting the scientific limitations of the underlying database. Further, RIVM (2020b) highlighted that, since the EFSA TWI related to a total of four PFAS, a translation step was needed to be able to use it as starting point for the

derivation of guidance values for individual PFAS (or other mixtures of PFAS).

The step to translate the EFSA TWI for Σ 4PFAS into a guidance value for individual PFAS was described by RIVM (2021a, 2021b) who proposed applying the EFSA TWI in combination with the relative potency factor approach (Zeilmaker et al. 2018; Bil et al. 2021). To use the EFSA TWI in combination with relative potency factors, RIVM (2021b) defined that *only* PFOA was associated with the EFSA TWI for Σ 4PFAS: “*Specifically, this means that we*

Box 2A Derivation of a health-based guidance value for PFOA in drinking water by RIVM

Zeilmaker and Janssen (2016), from RIVM, calculated the limit value of 87.5 ng/L using the WHO standard calculation method and these parameters: chronic oral daily intake (i.e. TDI) = 12.5 ng/kg bw/day PFOA, 20% allocation factor, 70 kg body weight, 2 L daily drinking water consumption

The TDI of 12 ng/kg bw/day was based upon a RIVM report by Zeilmaker et al. (2016) who used a rat 90-day study by Perkins et al. (2004) as the key study. Perkins et al. established a NOAEL of 0.06 mkd PFOA based on observed liver effects, corresponding to a serum concentration of 7.1 µg/mL PFOA. Zeilmaker et al. (2016) then selected serum concentration of PFOA as the measure of exposure to account for its bioaccumulation potential and applied this formula from an ATSDR report (see footnote # below) to derive the oral daily intake:

Oral daily intake

$$= \frac{C_{ss} \times \text{Daily serum elimination} \times \text{Volume of distribution}}{\text{Gastrointestinal absorption fraction}}$$

C_{ss} : steady-state serum concentration = 7.1 µg/mL in rats (see above); daily serum elimination = $\ln 2/t_{50}$ (serum half-life [t_{50}] of PFOA in humans: 839.5 days); volume of distribution in humans: 0.17 L/kg bw; gastrointestinal absorption fraction = 1

Thereby, the subchronic steady-state serum concentration of 7.1 µg/mL PFOA in rats corresponds to a human subchronic intake of 1 µg/kg bw/day PFOA. Zeilmaker et al. (2016) then divided this value by an assessment factor of 80 (10 to account for intra-human variability $\times$ 8 to extrapolate from subchronic to chronic toxicity, see footnote ## below) yielding a TDI of 12.5 ng/kg bw/day PFOA

Zeilmaker et al. (2016) cited a draft ATSDR report from 2015 that the authors of the present article were unable to retrieve. Therefore, the current version, i.e. ATSDR (2021), is cited in the bibliography

Zeilmaker et al. (2016) recommended an assessment factor of 8 to extrapolate from subchronic to chronic studies: “*The summary of empirically derived assessment factors (...) already shows that for at least 50% of the substances with a subchronic and chronic study available the difference is larger than a factor 2, i.e. when applying a factor of 2 the probability of underestimating the subchronic-chronic differences is at least 50%. Moreover, the empirical studies described in the ECHA guidance show that even 10% of the substances require an assessment factor larger than 10 to account for subchronic-chronic differences. Given the results from the empirical derived assessment factors, and because we want to exclude the possibility of underestimating the true differences between subchronic and chronic exposure, the factor of two is not considered sufficient.*”

deem the EFSA TWI valid for the situation in which people would be exposed to PFOA alone” (unofficial translation from page 7 in RIVM 2021b, see SI-3.2 for original quote). RIVM further described that, by multiplying the concentration of each individual PFAS in a sample by its potency relative to PFOA, the equivalent quantity of PFOA could be calculated for each PFAS and that this value could be compared with the applicable guidance value, which would also be expressed as PFOA equivalents (RIVM 2021b).

RIVM (2021b) derived a guidance value for Σ 4PFAS from the EFSA TWI (or rather the corresponding TDI of 0.63 ng/kg bw/day), using the WHO standard calculation method while assuming an average body weight of 70 kg, 2 L of daily water consumption and a 20% contribution of drinking water to the TDI. This resulted in a drinking water guidance value of 4.4 ng/L Σ 4PFAS, expressed as ‘PFOA equivalents’ or ‘PFOA’ due to the assumption that the immune effects were caused entirely by PFOA (RIVM 2021b). RIVM (2021a) indicated that this guidance value should be used for future risk assessment and the derivation of risk limits for PFAS as it was lower (i.e. twenty-fold lower than the limit value of 87.5 ng/L derived by Zeilmaker and Janssen 2016) and thus offered more protection (page 1 in RIVM 2021a).

Finally, RIVM (2023) established the guidance value for TFA in drinking water by dividing the drinking water guidance value of 4.4 ng/L for Σ 4PFAS / PFOA equivalents / PFOA by the relative potency of TFA (0.002) = 2,200 ng/L or 2.2 µg/L. RIVM (2023) highlighted that this guidance value applies if *only* TFA is present in the water sample in question.

Belgium – Wallonia

The Independent Scientific Committee (CSI) advising the Walloon competent authority critically discussed the shortcomings of the parameters ‘sum of PFAS’ and ‘total PFAS’ as implemented in the EU DWD (EP and Council 2020). Also, the uncertainties related to the selection of assessment factors and the limitations of the toxicological database for TFA were critically discussed (SPW Env 2025). The CSI concluded that it is not (yet) appropriate to derive a TDI for TFA and instead suggested applying the advisory value of 2.2 µg/L TFA in drinking water set by RIVM (2023) since it is based on relative potency factors of PFAS. The CSI further stated that exceedance of the advisory value of 2.2 µg/L TFA would not compromise the potability of the water but would indicate the need for a monitoring plan to determine and eliminate the source of TFA contamination (SPW Env 2025).

Box 2B RIVM guidance and reports relating to PFAS / TFA (see bibliography for reference details)

Zeilmaker et al. (2016): Risk assessment of PFOA emissions for local residents (DuPont/Chemours, Dordrecht, NL)

Zeilmaker and Janssen (2016): Derivation of a PFOA drinking water limit value for lifelong exposure

Zeilmaker et al. (2018): Mixture exposure to PFAS: A Relative Potency Factor approach

RIVM (2020a): Note: Final EFSA opinion on PFAS – scientific considerations for RIVM decision-making based on the EFSA-TWI
RIVM (2020b): Conclusion on the use of the EFSA-TWI for PFAS by RIVM

RIVM (2020c): Note: Status of the EFSA Opinion and the role of health-based limit values in different policy frameworks

RIVM (2021a): Note: Implementation of the EFSA TWI for the sum of four PFAS

RIVM (2021b): Evaluation of the contribution of drinking water and food to exposure to EFSA-4 PFAS in the Netherlands and advice on drinking water guideline values

RIVM (2023): Drinking water guideline value for trifluoroacetic acid

Discussion

Monitoring of TFA in drinking water

The main pathways of entry of TFA into the terrestrial and aquatic environment include precipitation (e.g. from the atmospheric degradation of refrigerants and propellants), direct input and subsequent infiltration into agricultural soils (through pesticides and fertilisers), industrial discharges and municipal wastewater treatment plants (Solomon et al. 2016; UBA 2020a, c; Freeling et al. 2020; Adlunger et al. 2021; Behringer et al. 2021; Freeling and Björnsdotter 2023; Sturm et al. 2023; Danish EPA 2024; Garavagno et al. 2024; Hanson et al. 2024; Joerss et al. 2024; ECHA 2025d). By comparison, evidence to support the hypothesis that TFA in marine waters may also stem from natural sources remains inconclusive (Joudan et al. 2021; Freeling and Mangels 2024; Hanson et al. 2024; UNEP 2024; Lim 2025; Tokranov et al. 2025; see also Footnote to Table SI-1).

The reports by Adlunger et al. (2021), Behringer et al. (2021) and Sturm et al. (2023), which were commissioned by the German UBA, include considerations on possible emission sources for TFA and estimations of TFA quantities emitted into the environment. Overall, all three surveys highlighted the multi-source and often unknown contributions of environmental discharge of TFA (see also Fig. 4 in Adlunger et al. 2021 and Table 6 in Sturm et al. 2023). “*Due to the large number of possible sources and precursors, for which often only insufficient information is available, it is difficult to attribute TFA pollution to specific origins*” (Sturm et al. 2023). The specific estimations depend on the underlying available input data and assumptions, with estimations of the contribution of PPPs to maximum potential

total TFA emissions ranging from approximately 20–80% (Freeling et al. 2020; Adlunger et al. 2021; Behringer et al. 2021; Sturm et al. 2023).

Since TFA is very slowly to basically non-degradable (Kim et al. 2000; Tang et al. 2024; see also European Commission 2023a) and highly mobile, it is being observed in increasing quantities in many aqueous environmental compartments, and especially in terminal water bodies (Zhi et al. 2024). Also, it can only be removed from drinking water with very high technical and financial expenditure (UBA 2020b; Neuwald et al. 2022; Zhou et al. 2022; Brunn et al. 2023; Garavagno et al. 2024; Zhi et al. 2024). In Germany, TFA has been found in 2181 of 2970 raw water extraction points (BDEW/DVGW/IVA 2022); in 14 raw water extraction points, the measured concentrations exceeded the target limit of 10 µg/L set by UBA (2020a) for reasons of drinking water hygiene, and the highest, temporarily measured concentration of 20 µg/L TFA amounted to one third of the health based guidance value of 60 µg/L (UBA 2020b; Freeling and Björnsdotter 2023; Table SI-1). As this example shows, continuous monitoring is required to ensure that critical concentration levels of TFA in drinking water are recorded in a timely manner and that appropriate countermeasures are implemented accordingly. Solomon et al. (2016) concluded that “*current and estimated concentrations of TFA and its salts in the environment... do not present a risk to humans and environment. However, formation of TFA from the degradation of hydrochlorofluorocarbons and hydrofluorocarbons warrants continued attention, in part because of its long environmental lifetime*”. Similarly, the 2024 Assessment Update by the Environmental Effects Assessment Panel of the United Nations Environment Programme concluded that “*the risks to ecosystem and human health from the formation of trifluoroacetic acid (TFA) as a degradation product of ODS [ozone depleting substance] replacements is currently de minimis*” (UNEP 2024).

DW-relevant metabolites, GW-relevant metabolites and non-relevant metabolites of PPPs

Water intended for human consumption must be of high quality and shall not pose a health risk to consumers. Therefore, very strict parametric values and quality standards have been implemented for drinking water in the DWD (EP and Council 2020) and, for groundwater in relation to PPPs, in the PPP Regulation (EP and Council 2009a; b) and the GWD (EP and Council 2006a). The provisions stipulated in the PPP Regulation aim to limit the leaching of PPP-related substances into groundwater; for GW-relevant metabolites, the risk assessment considers acceptable emissions below the precautionary level of 0.1 µg/L (Annex I of the GWD; EP and Council 2006a). Please see Table 1 and Box 1 in

Sect. "European Union legislation related to drinking water" for overview of relevant provisions from the GWD, including its amendments as per Directive 2026/805 (EP and Council 2026) and SI-2 for overview of substance-agnostic precautionary thresholds.

It is important to note that the provisions of the DWD, on the one hand, and those of the PPP Regulation and GWD, on the other hand, serve different purposes (see also UBA 2025).

The DWD has the objective to “*protect human health from the adverse effects of any contamination of water intended for human consumption*” (Article 1 of the DWD, EP and Council 2020). Hence, the parametric values laid down in Annex I of the DWD aim to protect consumers from risks that may be caused by chemicals that are present in drinking water. In addition, proactive measures for the protection of drinking water quality may be implemented as per Article 8 of the DWD, in which a risk assessment and management process for drinking water catchment areas has been established. To ensure protection of drinking water quality, not only the toxicological profile of the substance of interest is addressed but also the actual human exposure. This is reflected in the definition for DW-relevant metabolites of PPPs, i.e. substances that may “*generate a health risk for consumers*” (Annex I, Part B, of the DWD).

The GWD has the scope to prevent and control the input of pollutants into groundwater, aiming to protect it as source for drinking water and as an ecosystem (Article 1 of the GWD; EP and Council 2006a). As such, the corresponding provisions of the GWD serve to assess the chemical status of groundwater and to implement measures to prevent possible future deterioration of the status of groundwater and, hence, to prevent future risk. (For this reason, environmental fate is considered when assessing substances under the GWD, whereas the provisions of the DWD solely address human health-related effects.)

To prevent leaching of pesticides and their metabolites into groundwater, an elaborate and very conservative procedure to approve active substances in PPPs has been implemented in the PPP Regulation (EP and Council 2009a; b) and associated guidance (European Commission 2021). This elaborate procedure also considers the GW-relevant and non-relevant metabolites of pesticides, and the definition for GW-relevant metabolites of PPPs provided in Article 3(32) of the PPP Regulation and focusses on hazard alone (Table 1). While a risk-based definition for GW-relevant metabolites of PPPs (that also includes environmental risks) has now been implemented in the GWD as amended by Directive 2026/805 (EP and Council 2026), the legal implications of the discrepancy between the provisions of the PPP Regulation and those of the amended GWD are currently unclear.

The ECHA Risk Assessment Committee has approved of a CLH for reproductive toxicity category 1B for TFA, and its uptake into the CLP Annex VI list of CLH (EP and Council 2008b) is expected with its next adaptation to technical progress (ECHA 2026a, 2026b; Sect. "Developmental and reproductive toxicity studies for TFA").

Once TFA is assigned a CLH for reproductive toxicity, it by default becomes a GW-relevant metabolite under the PPP Regulation (EP and Council 2009a), but possibly not at the same time under the GWD (EP and Council 2006a). Already in view of the classification intention, the European Commission has concluded that TFA should be considered a GW-relevant metabolite (European Commission 2024d). In line with Article 3(32) of the PPP Regulation, a metabolite is GW-relevant if it “*has certain toxicological properties that are considered unacceptable*” (EP and Council 2009a). The *Guidance document SANCO/211/2000-rev. 11 on the assessment of the relevance of metabolites in groundwater* (European Commission 2021) indicates which toxicological properties need consideration. Specifically, a metabolite shall be considered GW-relevant if it has a classification for acute toxicity, specific target organ toxicity, genotoxicity, reproductive toxicity or carcinogenicity (for details, see European Commission 2021).

For the purposes of the GWD, once TFA is established as a relevant metabolite (i.e. assigned a CLH for reproductive toxicity), exposure levels that comply with the groundwater quality standards specified in Annex I of the GWD (i.e. 0.1 µg/L for individual pesticides and GW-relevant metabolites; see above) will have to be demonstrated for its emissions following use of pesticides containing a C-CF₃ moiety. For substances with several potential sources, such as TFA (Sect. "Monitoring of TFA in drinking water"), it is technically difficult (Sturm et al. 2023), if not impossible, to establish with certainty the specific source(s). Therefore, the authors of the present article consider it more appropriate to lay down a separate, substance-specific threshold value for TFA in Annex I of the GWD, as such a threshold value would be applicable regardless of source.

By comparison, for the purposes of the DWD, once TFA is assigned a CLH for reproductive toxicity, a risk assessment will need to be conducted, considering both hazard and exposure, to determine whether TFA is a DW-relevant metabolite of PPPs to ensure its presence in drinking water does not pose “*a health risk for consumers*” (EP and Council 2020). If TFA would be established as a DW-relevant metabolite, the parametric values of 0.1 µg/L for individual ‘Pesticides’ and 0.5 µg/L for ‘Pesticides Total’, and their relevant metabolites (as implemented in Annex I, Part B, of the DWD), would apply. However, these parametric values would also only apply to the fraction of TFA in drinking water that is attributable to the degradation or transformation

of PPPs but not to the fraction of TFA due to other sources. Clearly, this would not make sense from the perspective of consumer protection. However, the risk assessment for TFA shows that it should be considered a non-relevant metabolite of PPPs in drinking water as it is not considered to pose a health risk to consumers at the encountered concentration range (Sect. "[Exposure and risk assessment: TFA as non-relevant metabolite of PPPs in drinking water](#)"). Indeed, UBA has confirmed that the overall risk assessment shows that the health-based guidance value of 60 µg/L TFA in drinking water, established in 2020 (Sect. "[Germany](#)"), is protective of reproductive toxicity effects and, therefore, remains valid also after the ECHA Risk Assessment Committee's approval of the CLH for reproductive toxicity for TFA (UBA [2026b](#)).

The DWD does not include specific provisions for how to determine that the metabolite of a PPP shall be considered DW-relevant (see also UBA [2025](#)). The European Commission Directorate-General Environment is launching a programme to assess the health effects of PFAS and pesticide metabolites in drinking water. It is one objective of this action "*to assess the relevance for human health of commonly identified pesticide metabolites. Methodologies with the potential for determining the relevance of pesticide metabolites in a consistent way will be assessed*" (page 130–131 in European Commission [2024a](#)). This action has been commissioned to the WHO, and its outcome is expected to be published by 2027 (European Commission [2024a, 2025](#); UBA [2025](#)).

Just as the parametric values for individual 'Pesticides' and 'Pesticides Total', and their relevant metabolites, only apply to the fraction of a DW-relevant metabolite that is attributable to the degradation or transformation of PPPs, the guidance values that Member States may set for non-relevant metabolites in accordance with Annex I, Part B, of the DWD (EP and Council [2020](#)) also only apply to their presence in drinking water as metabolites of PPPs but not to their presence due to other sources.

Guidance values for TFA in drinking water based on toxicological studies

A comprehensive EU-wide approach for how to derive health-based guidance values for substances in drinking water is unavailable, and the Member State competent authorities and advisory bodies have pursued a broad spectrum of different approaches to derive such thresholds for TFA in drinking water (Table [4](#)). This does not only contradict the EU objective of harmonisation (see e.g. European Commission [2023c](#)) but also causes uncertainty with respect to the adequacy of consumer protection efforts.

This section includes a discussion of those parameters that are relevant to derive health-based guidance values. The

exact TDI (Sect. "[Introduction: establishment of tolerable or acceptable daily intake levels](#)") and the exact drinking water guidance value (Sect. "[Introduction: derivation of guidance values for substances in drinking water](#)") that is set for a given substance is dependent on several distinct decisions:

1. The selection of the key study
2. The selection of the relevant, critical effect from this study
3. The consideration of NOAEL, LOAEL or benchmark response (BMR) for the BMDL (and the robustness of the underlying dataset in supporting the selected statistical approach to derive the reference point)
4. The setting of assessment factors
5. The estimation of the average body weight of the relevant human population
6. The estimation of the daily drinking water consumption
7. The estimation of the fraction of the TDI that shall be allocated to drinking water consumption

In the following, these aspects are further explored using the corresponding guidance from the WHO *Guidelines for drinking water quality* (WHO [2022](#)) as a starting point since the elements of the risk-based approach to water quality assessment implemented in the DWD (EP and Council [2020](#)) are based upon these. This section only considers those guidance values that are based on studies evaluating TFA. The RIVM ([2023](#)) approach to consider TFA as a part of the PFAS group for the derivation of the guidance value is discussed in Sect. "[Guidance value for TFA in drinking water based on relative potency to PFAS](#)".

Selection of the key study and relevant toxicological endpoint for the derivation of the TDI

Generally, the TDI (or ADI) is the amount of a substance that "*can be ingested over a lifetime without appreciable health risk, and with a margin of safety*" (WHO [2022](#)). Therefore, longer-term studies are preferable to shorter-term studies as key studies since they better reflect lifetime exposures. Further, as the TDI shall be derived from a NOAEL, LOAEL or BMDL (Sect. "[Introduction: establishment of tolerable or acceptable daily intake levels](#)"), i.e. a measure reflecting adversity, the selected key study should preferably also yield an adverse finding that can be selected as the critical effect. Hence, the aspects of (longer) exposure duration and relevant adverse finding need to be balanced against each other in selecting the key study for the derivation of the TDI.

The database for (Na-)TFA does not include information on lifetime exposures in rodents. The rodent two-year bioassay uses very large numbers of animals, its scientific limitations are well-established, and integrated approaches

are available to assess chronic toxicity and carcinogenicity without the two-year bioassay (Cohen 2010; Cohen et al. 2019; Hilton et al. 2022, 2023; Strupp et al. 2023; Louekari and Jacobs 2024). In deriving the health-based guidance value for TFA in drinking water, the absence of long-term toxicity data is addressed by an appropriate assessment factor (Sect. "Assessment factors"). Therefore, the authors of the present article maintain the view that the questionable added scientific value of data from the two-year bioassay does not balance against the legal mandate to undertake vertebrate animal testing only as a last resort (EP and Council 2006a, 2010).

Liver findings

The liver was the main target organ in all rat studies evaluating TFA. In regulatory toxicity studies, substance-mediated effects on the liver are generally identified via histopathological organ changes and/or altered liver enzyme levels in the serum. If such effects are observed, it needs to be established whether they are indicative of an adverse effect (hepatotoxicity) or of an adaptive reaction (liver enzyme induction).

The participants of a European Society of Toxicologic Pathology (ESTP) workshop provided details on how to generally establish whether liver findings should be considered as adaptive or adverse: "*Hepatocellular hypertrophy without histologic or clinical pathology alterations indicative of liver toxicity was considered an adaptive and a non-adverse reaction*" (Hall et al. 2012). However, if hepatocellular hypertrophy in the absence of histological changes coincided with alterations in relevant clinical pathology parameters, a weight-of-evidence approach was recommended to determine whether the liver effects should be considered adverse; for this weight-of-evidence evaluation, at least two of the following liver parameters should be dose-dependently and biologically significantly affected: (1) at least two- to threefold increase in ALT; (2) a biologically significant change in other liver enzymes indicating hepatobiliary damage, e.g. AP, AST; and (3) a biologically significant change in another clinical pathology marker indicating liver dysfunction, e.g. albumin, bilirubin, triglycerides (Hall et al. 2012).

In line with these recommendations from the ESTP workshop, liver adversity was observed in the rat 90-day repeated-dose toxicity study (i.e. histopathological alterations and changes in clinical pathology indicative of liver damage), but not in the rat one-year toxicity study as the latter only showed increased ALT (at ≥ 1.8 mkd Na-TFA). As a further evaluation of the Na-TFA database shows, the serum ALT increase observed at this low dose level reflects an adaptive reaction of the liver as first-pass organ upon

oral substance exposure (for details, see SI-5). There was also no evidence to suggest that the adaptive liver enzyme induction might eventually lead to adversity as the findings did not increase in incidence or severity over the one-year exposure period and were further reversible within the six-week recovery period included in the one-year study. Also, individual animals with relatively high ALT (as compared to the mean of the test group) also did not exhibit more than two-fold increased ALT or more severe or adverse liver histopathological findings (SI-5).

Therefore, the 90-day study, and not the one-year study, should be selected as key study to derive health-based guidance values for TFA, based on liver adversity. UBA acknowledged that a NOEL (i.e. not a NOAEL) could be established from the one-year study when selecting increased ALT as the critical effect and then used this NOEL (1.8 mkd Na-TFA) as the reference point to derive the guidance value for TFA (UBA 2020a). By comparison, the NOAEL from the one-year study is 37.8 mkd Na-TFA, i.e. its top dose level since there was no adversity, which is higher than the NOAEL of 9.9 mkd Na-TFA derived from the 90-day study (but lower than the 90-day LOAEL, i.e. 98 mkd Na-TFA in the males). With respect to the UBA approach, Nielsen and Beltoft (2021) concluded that the increased ALT activity observed in the one-year study was "*not toxicologically relevant because of the low magnitude of the changes (lower than or borderline to +50% of the control value)*" (similarly in RIVM 2023).

RIVM (2023) evaluated liver effects across the 90-day and one-year Na-TFA studies. Across both studies, the only statistically significant dose-response effects were reduced total bilirubin (at 90 days) and increased relative liver weight. Based upon this evaluation by RIVM, De Clercq and De Brouwere (2024) from the Flemish VITO selected total bilirubin and relative liver weight as critical effects for the derivation of a health-based guidance value for TFA in drinking water. Notably, this selection deviated from the adversity criteria set by Hall et al. (2012) and potentially also from the 'appreciable health risk' criterion for reference values as defined by WHO (2022), as statistical significance alone does not necessarily indicate biological relevance (Wasserstein and Lazar 2016; Amrhein et al. 2019). Then, De Clercq and De Brouwere (2024) suggested the lower value (i.e. 15.6 $\mu\text{g/L}$, based on reduced total bilirubin, as compared to 33 $\mu\text{g/L}$, based on relative liver weight) as limit value for TFA in drinking water, i.e. further deviating from common adversity definitions. Also, the selected two studies (i.e. the 90-day and one-year studies) show inherent differences, which impair their joint evaluation. First, Na-TFA was administered via drinking water in the one-year study but via feed in the 90-day study. These different modes of application lead to differences in toxicokinetics

and, thus, different internal or target site exposures to the test item (Goehl 1997). Also, the two studies were conducted in different rat strains (Wistar vs Sprague–Dawley; Table 3), which have different physiological properties, and in different laboratories, which may also affect study outcomes. Taken together, the approach taken by De Clercq and De Brouwere does not appear scientifically defensible.

Derivation of the reference point: NOAEL vs BMD

Since both the NOAEL and the BMD (or its lower confidence interval, i.e. the BMDL) were applied for the characterisation of numerical endpoints to derive the TDI underlying different guidance values for TFA in drinking water (Table 4), their differences with respect to statistical assumptions and approaches are briefly discussed here.

The BMD is generally considered to be toxicologically more informative as it is based on a dose–response relationship (instead of a comparison of individual test groups) and yields a dose with associated uncertainty (indicated by its confidence interval) that relates to a specific, toxicologically relevant response size, i.e. the BMR (Kluxen 2020; EFSA SC 2022). However, there is no international consensus for exactly how to determine the BMD or BMDL, and, for many studies, it may be difficult to derive a robust BMD or BMDL (Haber et al. 2018; Jensen et al. 2019) due to current internationally accepted toxicity test designs, for example if the number of test doses is limited, dose-spacing is inconvenient or a clear dose–response is lacking. This can result in very wide BMD confidence intervals, which makes the derivation of a suitable reference point challenging or implausible, depending on the underlying data.

The NOAEL is typically based on pairwise comparison of individual test groups to the control group, using statistical significance to identify test groups showing adversity. This approach is typically more robust than the BMD approach if the number of test doses is limited, if dose-spacing is inconvenient and/or if there is no clear dose–response. However, statistical significance is highly dependent on the selection of the hypothesis, the statistical test and the potential adjustment for confounders. Harmonised guidelines for how to conduct statistical analyses are unavailable. In addition, if only pairwise tests are considered, potential false-positive groups might be considered as positive.

Both the NOAEL and the BMD approach have in common that they need to include an assessment of the biological/toxicological relevance of the findings. Hence, for the NOAEL approach, the effect responses should be assessed for their biological relevance in conjunction with statistical significance. Likewise, the BMR utilised in the BMD approach should correspond to a toxicologically relevant response size. However, there is no consensus on how to

establish a toxicologically relevant BMR. While generic values for BMR selection make BMDs comparable, they do not necessarily make them toxicologically meaningful. If either the NOAEL is based on statistical significance alone or the BMR is selected based on a generic (historical default) value, the resulting reference values may not be biologically justified.

Findings related to immune system function

The major incentive for RIVM to base the health-based guidance value for TFA on grouping and read-across to PFAS (Sect. "Guidance value for TFA in drinking water based on relative potency to PFAS") was the unavailability of immunotoxicity data for TFA in light of immunotoxicity concerns expressed for PFAS (RIVM 2023). Most importantly, the developmental immunotoxicity cohort 3 was not included in the EOGRTS (OECD TG 443) as concerns for immunotoxicity were not indicated based on the available rodent data at the time of study design. In accordance with the OECD TG 443 test protocol, the optional cohort 3 includes assessments for rat TDAR in the F₁ offspring (currently the only specific immunotoxicity assessment included in any OECD TG; ECHA 2024b). After completion of the assessments in live animals in the EOGRTS evaluating TFA, immunophenotyping was conducted using spleen tissue from the F₁ offspring. Generally, these assessments did not yield any findings assessed as treatment-related (Sect. "Developmental and reproductive toxicity studies for TFA").

Formal guidance for how to evaluate immunophenotyping data for the regulatory risk assessment of chemicals is unavailable. As regards preclinical safety testing of human medicines, the *International Conference for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) S8 Guidance on immunotoxicity studies for human pharmaceuticals* (EMA 2006) indicates that immunophenotyping may provide supporting evidence. By comparison, the TDAR assay, as an immunological challenge test, provides more conclusive evidence on the immunotoxicity potential of pharmaceuticals (EMA 2006; Ehrlich et al. 2023; Nandre and Terse 2025), and it has been adopted as US EPA TG OPPTS 870.7800 (US EPA 1998). However, in a retrospective analysis of the toxicological database for 78 active substances in PPPs from 46 separate chemical classes, the NOAEL from the TDAR assay was always higher than the NOAEL selected as reference point for the human health risk assessment (Gehen et al. 2014). Gehen et al. concluded that the “data strongly support employment of alternative approaches including initial weight-of-evidence analysis for immunotoxic potential prior to conducting functional immunotoxicity testing for pesticide active ingredients”. Also, ECHA cautioned that the TDAR assay

requires further standardisation to enhance its regulatory applicability in the context of the EOGRTS developmental immunotoxicity cohort 3 (ECHA 2024b).

Multiple studies indicate that C₄ PFAS and longer chained PFAS elicit immunotoxicity in humans and laboratory animals (Ehrlich et al. 2023; Antoniou and Dekant 2024; Bline et al. 2024). By comparison, there is currently no indication that TFA (C₂) may also exhibit immunotoxic potential in either laboratory animals or humans. Based on the results from the TG-conforming rat studies available, there is no indication for immunotoxic responses in rats upon exposure to TFA. While changes in lymphocyte counts were observed for TFA in high-dose group females of the 14-day study, i.e. at 190 mkd Na-TFA (BAuA 2025), similar changes were not observed in the longer-term studies. Specifically, there were no treatment-related findings observed in any immune responsive organs/parameters (bone marrow, thymus, spleen, lymph nodes, white blood cell count, differential blood count) in any of the 28-day, 90-day and one-year repeated dose toxicity studies conducted with Na-TFA (ECHA 2025a; BAuA 2025). Therefore, the altered lymphocyte count observed in the female rats from the 14-day study is an isolated finding that is interpreted as incidental.

This conclusion is supported by toxicokinetic considerations. Very high Na-TFA plasma levels were observed in the rat one-year repeated-dose toxicity study. Already at the low dose, mean maximum serum concentrations of 15.3 ± 2.99 and 15.8 ± 1.54 mg/L Na-TFA were measured in the males and females, respectively (ECHA 2025a). These serum levels are similar to the BMDL₁₀ of 17.5 ng/mL (= 17.5 mg/L) at which PFOA-mediated effects on vaccination response were observed (Sect. "Netherlands"). Since changes in immune-responsive organs and parameters were neither observed at this dose level nor at higher dose levels in the one-year study, an immunotoxic potential for TFA seems very unlikely.

Taken together, from the limited immune system-related data available in the comprehensive toxicological database for TFA, there is not sufficient evidence to suggest immunotoxicity potential of TFA. Therefore, the adverse liver effects observed in the 90-day study should be used as critical effects to derive the health-based guidance value for TFA in drinking water.

Findings related to thyroid function

In the draft statement on consumer health-based guidance values on TFA, EFSA concluded that the serum T4 reduction observed in the EOGRTS was adverse at the mid and top doses both in the offspring at weaning (i.e. on PND 22) and in the adult rats since "*the effects were above the threshold of concern (20%...), were statistically significant*

and showed a dose–response relationship and consistency in direction" (EFSA 2025a). Therefore, EFSA selected the EOGRTS as key study and serum T4 reduction as the critical effect for the derivation of the newly proposed ADI of 0.03 mkd Na-TFA (EFSA 2025a; Sect. "Developmental and reproductive toxicity studies for TFA").

Generally, adversity resulting from thyroid disruption may be (1) thyroid carcinogenicity upon long-term exposure to the disruptor and (2) developmental neurotoxicity if the thyroid disruption occurs in the offspring during the critical window of susceptibility; in the rat, this is during in utero development and during the three weeks of lactation (Melching–Kollmuss et al. (2025); Marty et al. 2022).

Ad. 1: TFA is unlikely to induce thyroid carcinogenicity, as there were no histopathological changes in the thyroid gland either in the EOGRTS or in any repeated-dose toxicity study up to one-year exposure duration, which are usually observed in short-term studies for substances inducing thyroid tumours in rats. This conclusion is consistent with the state-of-the-science (Huisinga et al. 2020) and with the *ECHA and EFSA Guidance for the identification of endocrine disruptors* (ECHA/EFSA 2018). In Appendix A of this guidance, it is stated that isolated thyroid hormone changes without concomitant histopathological changes of the thyroid gland "*would still present a potential concern for neurodevelopment*" (but no concern for thyroid carcinogenicity) and, thus, should "*act as a trigger for further studies*" (ECHA/EFSA 2018).

Ad. 2: The overall weight-of-evidence leads to the conclusion that TFA also does not exhibit potential for developmental neurotoxicity. In the EOGRTS evaluating Na-TFA, the T4 reductions observed in the offspring on PND 22, i.e. at weaning and thus at the end of the critical window of susceptibility (see above), were within the historical control dose range, specifically: -17% and -37% in the mid- and top-dose groups, respectively (Table 3). This outcome speaks against adversity. "*A difference is less likely to be an effect of treatment if the difference is within normal biological variation (i.e. within the range of historical control values...)*" (ECETOC 2002). Similarly, all serum T4 reductions observed in the parental animals were within the historical control range.

In addition, the specific measurements of -17% and -37% T4 reduction on PND 22 in the mid- and top-dose groups, respectively, are assessed as moderate and, since they are further not accompanied by thyroid histopathological findings, as non-adverse. In a comprehensive review of 51 rat studies investigating the thyroid- and brain-related effects of in utero and/or lactational exposure to altogether 14 substances (and dietary iodine deficiency), Marty et al. (2022) observed that pronounced offspring serum T4 reductions of $\geq 50\%$ and $\geq 60\%$ in the lower-dose and top-dose

groups, respectively, were associated with an increased likelihood for statistically significant brain-related findings in the rat (excluding electrophysiological findings, see Box 3 at the end of this section). The serum T4 reductions of -17% and -37% in the mid- and high-dose groups on PND 22 clearly undercut these threshold values.

The only T4 reduction outside the historical control dose range was measured in the top-dose F₁ males in week 13 (-52%), i.e. *after* the critical window of susceptibility to developmental neurotoxicity, and it also undercut the $\geq 60\%$ threshold for the top-dose group observed by Marty et al. (2022).

In the draft statement on consumer health-based guidance values on TFA, EFSA indicated that the serum T4 reduction observed in the EOGRTS was considered to be adverse in both offspring and “*adult ages*” since the effects were above 20%, statistically significant and showed a clear dose–response relationship (EFSA 2025a). EFSA did not relate the T4 decrements to the historical control value range and did not indicate which concern for thyroid-related adversity could be relevant for the adult rats (as there were no thyroid histopathological changes and the only T4 decrement outside the historical control dose range was measured 10 weeks after the critical window of susceptibility for developmental neurotoxicity). To support the suggested threshold of 20% serum T4 reduction as indicating a concern for thyroid-related adversity, EFSA referred to (1) the US EPA document *Integrated approaches to testing and assessment strategy: use of new computational and molecular tools* (US EPA 2011) and (2) the EFSA SC opinion on *Risks to human and animal health from the presence of bromide in food and feed* (EFSA SC 2025). In these two documents, the thyroid-related adversity considered is developmental neurotoxicity. The further evaluation of these two documents shows that the underlying evidence is not sufficiently robust to support the view that 20% serum T4 reduction indicates a concern for developmental neurotoxicity in rats, further considering that this level is below the level of hormone changes that can be reasonably detected in regulatory toxicity studies (Box 3).

Taken together, based on the overall weight-of-evidence reviewed here, the authors of the present article conclude that the T4 reduction observed in the EOGRTS evaluating Na-TFA does not meet commonly accepted adversity criteria for either thyroid carcinogenicity or developmental neurotoxicity and that the T4 decrement should, therefore, not be used as critical effect for the derivation of a guidance value for TFA in drinking water based on currently available evidence. Notably, this view is contested by EFSA (2025a), with the final EFSA statement on consumer health-based guidance values on TFA pending at the time of writing this article.

Developmental findings

The rabbit prenatal developmental toxicity studies showed dose-dependent increased incidences of ophthalmic abnormalities in foetuses upon in utero exposure to Na-TFA. No signs of developmental toxicity, including ophthalmic abnormalities, were observed in the rat. Together, the current evidence from the rabbit studies suggests a threshold-based mechanism in rabbits for triggering a disruption of vitreous body structures and for relevant inhibition of MCT-mediated substance transport into the ocular tissues (maternal / foetal NOAEL of 60 mkd Na-TFA). By contrast, Na-TFA showed no evidence for direct interference with the retinoic acid pathway, which is crucial for normal eye development (i.e. Na-TFA had no effect on Vitamin A levels in rabbits in vivo, and it neither bound to nor inhibited CYP26 in vitro, a central enzyme in the retinoic acid pathway; Sect. “[Developmental and reproductive toxicity studies for TFA](#)”). In the maternal rabbits and foetuses, the recorded internal Na-TFA concentrations ranged at or even exceeded the concentration required to attain the EC₅₀ of 12 mM at MCT transporters in vitro (Sect. “[Developmental and reproductive toxicity studies for TFA](#)”). TFA is known for its chaotropic properties (see e.g. Vemić et al. 2013), i.e. it breaks the structure of water, thereby changing the solubility of hydrophobic compounds (Assaf and Nau 2018). Thus, high concentrations of TFA in the aqueous/vitreous body may affect protein integrity leading to impairment of the development of the vitreous body in rabbits (manuscript in preparation).

The window of susceptibility to substance-mediated developmental effects is generally restricted to the foetal/embryonal and early postnatal period. Therefore, less than lifetime exposures need to be considered when reference values are based on developmental toxicity. This is addressed by the derivation of ARfD values (Solecki et al. 2005; WHO 2022; Majid et al. 2025); see also Moxon et al. (2020) for critical assessment of ARfD values based upon rabbit developmental toxicity studies. In the draft statement on consumer health-based guidance values on TFA, EFSA has newly proposed an ARfD, which is based on the developmental findings from the available rabbit prenatal developmental toxicity studies (Sect. “[Hazard assessment: establishment of an ADI for TFA by EFSA](#)”). Specifically, using the foetal NOAEL of 60 mkd Na-TFA as reference point and applying the standard assessment factor of 100 yielded an ARfD of 0.6 mg/kg bw (EFSA 2025a).

A comparison of the reference point for the ARfD and the reference point from the rat 90-day toxicity study (NOAEL: 8.4 mkd TFA) shows that the maternal / foetal NOAEL of 60 mkd Na-TFA (50.4 mkd TFA) is sixfold higher than the 90-day NOAEL. Therefore, as regards lifetime exposures to TFA, a health-based guidance value that is based on the

Box 3 Secondary literature from EFSA (2025a) to support the view that 20% serum T4 reduction indicates potential concern for developmental neurotoxicity

In the US EPA document on integrated approaches to testing and assessment, it is stated that “there are some data that associate the degree of disruption of thyroid function to adverse neurodevelopmental outcomes where the perturbations are mild to moderate” (page 73 in US EPA (2011)). Two human observational studies, i.e. Haddow et al. (1999) and Henrichs et al. (2010), are cited to support this view. However, these two studies do not provide robust evidence to support a 20% serum T4 reduction threshold as indicating a concern for neurodevelopmental impairment

The study by Haddow et al. (1999) included 124 mothers with normal thyroid function (control group) but only 62 mothers with hypothyroidism (i.e. reduced serum T4 with concomitant increased serum thyroid stimulating hormone). Hence, the overall cohort size was very small. Of the 62 mothers with hypothyroidism, 14 mothers were treated for this condition during the pregnancy under the study, whereas 48 mothers were not, and these were assessed separately in evaluating child neurodevelopment (neuropsychological testing by professionals at the age of 7–9 years), further reducing the cohort size (and increasing the incongruence between control and effect group). In the mothers with hypothyroidism, total and free T4 were reduced by 30% and 27%, respectively, as compared to the control group. Hence, regardless of the very small cohort size, which strongly limits the biological relevance of the statistical findings, these data do not support a 20% serum T4 reduction threshold. Also, in the study by Haddow et al., the women with hypothyroidism exhibited high serum concentrations of antibodies against thyroid peroxidase, which has been identified as potentially affecting foetal neurodevelopment (further discussed in Sauer et al. 2020). Finally, Haddow et al. measured maternal hormone levels during the second trimester of pregnancy, i.e. during a period when the human embryo is no longer fully dependent upon maternal thyroid supply (Sauer et al. 2020). Thus, it is unclear whether other factors, apart from maternal hypothyroxinaemia, may have contributed to the neurodevelopmental findings. As reported by Haddow et al., in the children of the mothers with untreated hypothyroidism, 9 of 15 neuropsychological parameters were statistically significantly affected as compared to the controls

The study by Henrichs et al. (2010) included 3,659 mother–child pairs and distinguished between mild and severe hypothyroxinaemia, which Henrichs et al. defined as free T4 below the 10th and 5th percentile of the normal range, respectively, with concomitant normal thyroid stimulating hormone. The overall mean of free T4 was reduced by at least 24% in the mothers with mild hypothyroxinaemia as compared to the mean of all mothers included in the cohort. Mild hypothyroxinaemia was statistically significantly associated with expressive language delay when evaluated across 18 and 30 months, but not when evaluated for each age separately. The other neurodevelopmental parameters were not statistically significantly affected. In this study, the mothers themselves, but not professionals, conducted the neurodevelopmental surveys (by checklists). In a comprehensive review of human observational studies addressing the impact of maternal hypothyroxinaemia on offspring neurodevelopment, Sauer et al. (2020) concluded that “the evidence for neurodevelopmental impairment recorded in the human studies was mostly weak. Often, statistically significant effects were only recorded in a few of many parameters that were further assessed without consideration of the clinical relevance of findings, or of an a priori hypothesis”. Henrichs et al. did not consider thyroid peroxidase antibody positivity of the mothers, and neither Haddow et al. nor Henrichs et al. considered maternal alcohol consumption during pregnancy as further confounder potentially affecting child neurodevelopment

US EPA (2011) also noted “that a 10% change is within normal experiment-to-experiment variation in control values, whereas the BMR of 20% provides a more reliable BMR. The range of 10–20% is interpreted as a balance between the biological and statistical significance of the reported changes in T4 and associated adverse outcomes” (page 73 in US EPA (2011)). However, given the protocols of OECD TGs, only T4 decrements of 25% as compared to the concurrent controls can be “considered ‘reasonably detectable’ in a common regulatory setting” (Marty et al. 2021; citing Li et al. 2019; see these publications for underlying calculations)

The US EPA continued its argumentation that mild thyroid perturbations are associated with adverse neurodevelopmental outcomes by indicating that the 10–20% range “is consistent with previous work in both animals and humans for changes in thyroid hormones and precursor events such as iodine uptake inhibition (Strawson et al. 2004; Zorrilla et al. 2009; Zhou et al. 2001; Paul et al. 2010)” (page 73 in US EPA (2011)). However, the cited references do not support the hypothesis that 10–20% serum T4 reduction is related to developmental neurotoxicity. The studies by Zorrilla et al. (2009) and Paul et al. (2010) evaluated thyroid- and liver-related effects elicited in rats upon in utero / lactational exposure and pubertal exposure, respectively, to the antibacterial agent triclosan. Neither study considered developmental neurotoxicity as adverse outcome. The study by Zorrilla et al. included androgen-dependent endpoints, which, however, were not affected. The references Strawson et al. (2004) and Zhou et al. (2001) are not included in the bibliography of US EPA (2011) and thus cannot be traced back with certainty. If they relate to the corresponding references provided in the bibliography of the present article, these also did not consider developmental neurotoxicity as adverse outcome. In addition, the 10–20% range determined by the US EPA relates to maternal T4 and not offspring T4 (further, it relates to free T4, whereas it is total T4 that is measured in rats). In the comprehensive review of 51 rat studies, Marty et al. (2022) did not find a causal relationship between maternal serum T4 and statistically significant brain-related effects in the offspring, whereas the extent of offspring serum T4 reduction was indicative of such adversity (see above)

In the EFSA SC opinion on risks related to the presence of bromide in food and feed, altered serum T4 levels were identified as critical effect, and a BMR of 20% was set to provide a potential reference point (EFSA SC 2025): “In rats, a 10%–17% decrease in T4 during pregnancy and lactation is associated with neurodevelopmental toxicity in rat offspring (Gilbert 2011; Gilbert et al. 2016). Heterotropia, an abnormal cluster of neurons within the normal neurons, sparse white matter of the corpus callosum have been observed in fetuses of dams with serum T4 decrements of approximately 20% (Hassan et al. 2017). In humans, a 25% decrease in T4 (either tT4 or fT4) during pregnancy (during second trimester) results in neurodevelopmental and cognitive deficits in children (Haddow et al. 1999)” (EFSA SC 2025). (See above in this box for details on Haddow et al. 1999.)

Box 3 (continued)

The studies by Gilbert (2011), Gilbert et al. (2016) and Hassan et al. (2017) were conducted at the National Health and Environmental Effects Research Laboratory of the US EPA. In all three studies, rats were exposed to the thyroid peroxidase inhibitor propylthiouracil during pregnancy (and lactation). The slightest statistically significant serum T4 reduction reported by Gilbert (2011) was 25% in the PND 22 weanlings exposed to 1 ppm propylthiouracil in the drinking water. In the study by Gilbert et al. (2016), all serum T4 reductions slighter than 20% were not statistically significant. In both studies, the mild serum T4 decreases were associated with specific electrophysiological findings in the rat brain. It has not yet been established which electrophysiological findings in the rat should be considered as adverse (Marty et al. 2022). Further, in the comprehensive review of 51 rat studies investigating the thyroid- and brain-related effects of in utero and/or lactational exposure to altogether 14 substances (and dietary iodine deficiency), Marty et al. found that all studies including electrophysiological investigations were from the US EPA National Health and Environmental Effects Research Laboratory: *“While electrophysiological investigations are apparently sensitive methodologies to evaluate substance-mediated brain-related effects..., knowledge gaps and limited proficiency across laboratories currently prevent their wider application for toxicity testing”* (Marty et al. 2022)

The study by Gilbert (2011) also included assessments of fear conditioning, and the study by Gilbert et al. (2016) included behavioural assessments. These neurodevelopmental assessments only yielded findings at high concentrations of propylthiouracil, which also elicited pronounced offspring serum T4 decrements of at least 70% as compared to the concurrent controls

As regards the association between offspring serum T4 reduction by 61% on GD 20 and periventricular heterotopia observed by Hassan et al. (2017) at 1 ppm propylthiouracil, Marty et al. noted that this finding was not reproduced in other propylthiouracil studies, which followed the identical study design: *“In some studies, periventricular heterotopia was observed even though offspring serum T4 reduction was not pronounced, and in other studies, periventricular heterotopia was not observed even though offspring serum T4 reduction was pronounced”* (Marty et al. 2022)

90-day NOAEL for liver effects is sufficiently protective also with respect to maternal and developmental effects (see also Sect. [“Average human body weight”](#)). This has also been confirmed by UBA:

“As regards TFA, the most sensitive endpoint remains a toxic effect on the liver. Based thereupon, the German Environment Agency calculated a health-based guidance value of 60 µg/L in 2020. This existing guidance value is therefore also protective against potential reproductive toxicity, even upon lifelong intake via drinking water” (UBA 2026b; translation by authors of present article).

Assessment factors

It is common practice to apply assessment factors of 10 each to account for intraspecies and interspecies variability. Similarly, an assessment factor of 2 to extrapolate from subchronic to chronic (i.e. lifetime) exposure duration is commonly accepted (EFSA SC 2012; WHO 2022). These assessment factors should be used to derive a health-based guidance value for TFA in drinking water based upon the 90-day NOAEL. Considering the exposure duration, it is important to keep in mind that no adverse effects were seen in the one-year rat study, further supporting the sufficiency of the 2× assessment factor.

In addition, there is not sufficient evidence to suggest immunotoxicity potential of TFA. Therefore, it is not considered appropriate to assign an additional assessment factor for incompleteness of database related to immunotoxicity as was undertaken by De Clercq and De Brouwere

(2024) (Sect. [“Belgium – Flanders”](#)) or by EFSA in the draft statement on consumer health-based guidance values on TFA (EFSA 2025a). EFSA applied an extrapolation factor of 3 to account for *“remaining uncertainties related to the absence of a long-term toxicity/carcinogenicity study and the absence of TDAR assay on TFA”* (EFSA 2025a).

Average human body weight

The average body weight of the relevant human population is considered in the WHO standard calculation method to derive health-based guidance values in drinking water (Sect. [“Introduction: derivation of guidance values for substances in drinking water”](#)). For this purpose, 60-kg adults are indicated as the default assumption in the WHO guidance (WHO 2022). EFSA applied this value for the derivation of the ADI of 0.05 mkg Na-TFA (EFSA 2014), just as the Flemish VITO derived its guidance value for TFA in drinking water for 60-kg adults (De Clercq and De Brouwere 2024; Table 4). While the WHO guidance does not include further details on how the average body weight of 60 kg was established, it is supported by a review from Walpole et al. (2012) who estimated the adult body mass across different continents and regions and indicated an average body mass of 62.0 kg for ‘world’. As this review also shows, the average body mass differs considerably between continents, and Walpole et al. (2012) indicated 70.8 kg as the average body mass for ‘Europe’. This is reflected in the EFSA SC (2012) statement on the derivation of health-based guidance values who recommended applying 70 kg as default assumption for the EU population. The EFSA SC also presented and discussed the underlying data, retrieved from the EFSA

Comprehensive Database. All means from different surveys were higher than 60 kg with the lowest mean of 66 kg being recorded for females above 75 years of age (EFSA SC 2012). Consistent with the advice from the EFSA SC, the German UBA and the Dutch RIVM applied 70 kg as default value to derive the respective guidance values for TFA in drinking water (Table 4).

While the guidance values for TFA in drinking water were generally derived for adults, the Luxembourg Health Directorate based its guidance value on 5-kg infants (Luxembourg Health Directorate 2025), thereby yielding a reference value of 12 µg/L as compared to 60 µg/L as published by UBA (2020a) despite use of the same reference point from the 90-day study (Table 4).

Guidance values should be based on children if they are more sensitive than adults (Goeden 2018). This premise is also reflected in the WHO guidance: *“In some cases, the guideline value is based on children, where they are considered to be particularly vulnerable to a particular substance. In this event, a default intake of 1 L is assumed for a body weight of 10 kg; where the most vulnerable group is considered to be bottle-fed infants, an intake of 0.75 L is assumed for a body weight of 5 kg.”* (page 177 in WHO 2022). The numeric values provided by the WHO take account of the circumstance that infants have a higher drinking water intake relative to their body weight than older children or adults.

The reference points (NOAELs, BMDLs) from different studies should be compared to determine whether infants are particularly vulnerable to a substance (Goeden 2018). For example, since the toxicity of manganese is driven by neurodevelopmental effects observed in rat pups, bottle-fed infants are the most sensitive subpopulation with respect to drinking water exposure, and the corresponding drinking water guidance values are based on infants (pages 420–421 in WHO 2022; see also Valcke et al. 2018).

For TFA, the NOAELs recorded in the EOGRTS were 203–223 mkd TFA (242–265 mkd Na-TFA) across life stages. In the rabbit prenatal developmental toxicity studies, a foetal NOAEL of 50.4 mkd TFA (60 mkd Na-TFA), after 20-day in utero exposure, was recorded. By comparison, the NOAEL from the 90-day toxicity study was 8.4 mkd TFA (9.9 mkd Na-TFA), i.e. 25-fold lower than the NOAEL range across life stages in the EOGRTS and six-fold lower than the foetal NOAEL in the rabbit developmental toxicity studies. This comparison shows that infants are not more sensitive to TFA exposure than adults, and that a guidance value for TFA in drinking water that is based on adults is also protective for early life stages. Consistent with this conclusion, all currently proposed or implemented guidance values for TFA in drinking water (apart from the Luxembourg value) are based on adults.

Estimation of daily drinking water consumption

Generally, the default value of 2 L drinking water consumption per day was applied for the derivation of the guidance values for TFA in drinking water (Table 4), and this parameter and value are included in the WHO standard calculation method (Sect. [“Introduction: derivation of guidance values for substances in drinking water”](#)). As an exception, Nielsen and Beltoft (2021) from DTU Food used a body weight-related value of 0.03 L/kg bw/day as had been implemented by the Danish EPA (Danish EPA 2006), thereby using a variant of the WHO standard calculation method. The resulting guidance value for TFA in drinking water is widely identical with a guidance value that is based on a 70 kg adult (Sect. [“Denmark”](#)).

In the *ANSES Opinion on the determination of maximum health values for various pesticides and pesticide metabolites in water intended for human consumption*, the French ANSES has related the daily drinking water consumption to body weight, implementing a default value of 0.045 L/kg bw/day (ANSES 2019). This value considers higher water consumption by infants and children.

Since the WHO approach is the best standardised and internationally accepted, it appears preferable to maintain the default value of 2 L drinking water consumption per day for the derivation of guidance values for substances in drinking water for the general human population.

Allocation of TFA uptake via water

The allocation factor reflects the proportion of the daily exposure to a substance that is allocated to drinking water as compared to other sources, such as food, consumer products and airborne particles (as relevant). Allocation factors are considered for the derivation of drinking water guidance values to ensure that the total exposure to the substance from all possible sources does not exceed the ADI/TDI (WHO 2022; Greene et al. 2025). For example, allocation factors of 10% or 20% imply that 90% or 80% of the TDI, respectively, are attributable to non-water sources. As indicated in the WHO (2022) guidance, 10% was the former default allocation factor. Since it was *“found to be excessively conservative”*, it was suggested to change it to 20% *“as chemicals are progressively reassessed”* (WHO 2022; Sect. [“Introduction: derivation of guidance values for substances in drinking water”](#)). A two-fold higher allocation factor implies that the resulting guidance value is two-fold higher since the TDI is multiplied with the allocation factor following the WHO standard calculation method (Sect. [“Introduction: derivation of guidance values for substances in drinking water”](#)). If substance-specific exposure data are available (as is the case for TFA), these data may

be considered in setting the allocation factor. It has been criticised that an internationally agreed approach for setting allocation factors is currently unavailable: *“In regulatory practice these values appear to be determined through a mixture of science- and policy-based decisions that are often poorly documented and, in some instances, may conflict between regulatory agencies”* (Greene et al. 2025).

As regards TFA, the Dutch RIVM (RIVM 2023) and the Flemish VITO (De Clerq and De Brouwere 2024) selected an allocation factor of 20% for the derivation of the respective guidance values for TFA in drinking water, whereas the Danish DTU Food (Nielsen and Beltoft 2021), the German UBA (UBA 2020a) and the Luxembourg Health Directorate (2025) used an allocation factor of 10%. The decision by the Danish DTU food considered (1) that food is the *“major source for exposure to TFA”*; (2) that the total dietary intake of TFA corresponded to up to 5% of the ADI (citing EFSA 2014; Sect. *“Exposure and risk assessment: TFA as non-relevant metabolite of PPPs in drinking water”*); and (3) that adults would be exposed to up to 0.07 µg/kg bw/day TFA from drinking water, based on the highest concentration measured during ground water monitoring in 2020, i.e. 2.4 µg/L TFA (Sect. *“Denmark”*). By comparison, UBA has published a generic, substance-unrelated justification on their website: *“Germany has decided to maintain the allocation factor of 10% for drinking water to continue ensuring the high level of protection in accordance with the principle of drinking water purity, while conducting specific allocation assessments in individual cases”* (UBA 2026c, translation by authors of present article; Sect. *“Germany”*). The authors of the present article are unaware of a justification by the Luxembourg Health Directorate for selecting the allocation factor of 10% (Sect. *“Luxembourg”*).

Generally, comprehensive data on dietary burden are available for active substances in PPPs and their metabolites if relevant under the PPP Regulation (EP and Council 2009a). This stands in contrast to e.g. industrial chemicals and metals regulated under the REACH Regulation (EP and Council 2006a), for which comprehensive exposure data may not be available. Indeed, substance-specific exposure data are available for TFA as metabolite of active substances in PPPs containing a C-CF₃ moiety. These comprehensive data on dietary burden indicate that the major proportion of the human exposure to TFA does not come from water but from other sources of dietary uptake and that the possible human exposure to TFA via drinking water (and food) is lower than the ADI of 0.05 mkg Na-TFA (EFSA 2014, 2017, 2024a; Sect. *“Exposure and risk assessment: TFA as non-relevant metabolite of PPPs in drinking water”*).

As regards consumer exposure to TFA via drinking water, EFSA (2014) and EFSA (2017) estimated that this corresponded to 0.04% and 1.5–6.6%, respectively, of the

ADI derived for TFA. By comparison, it is unclear how the data provided by EFSA (2024a) came about, i.e. that Na-TFA intake via drinking water *“contributes with 8, 23 and 35% of ADI for adult, child and infant, respectively”* (EFSA 2024a; see also page 36 in EFSA 2024b). These values do not match the corresponding information from the Rapporteur Member State, indicating that TFA intake via drinking water contributes with 3%, 9% and 13.4% of the ADI for adult, child and infant, respectively (page 60 in European Commission, 2024c); see Sect. *“Exposure and risk assessment: TFA as non-relevant metabolite of PPPs in drinking water”* for detailed discussion.

Taken together, on account of the available substance-specific data for exposure to TFA via drinking water and food, the authors of the present article conclude that it is protective to set an allocation factor of 20% for the derivation of a health-based guidance value for TFA in drinking water.

Guidance value for TFA in drinking water based on relative potency to PFAS

TFA should not be included in the DWD definitions for PFAS

EU definitions for PFAS versus OECD definition for PFAS TFA may be covered by the DWD parameter ‘PFAS Total’, but it is currently not included in the DWD parameter ‘Sum of PFAS’ (EP and Council 2020; see Sect. *“TFA: identification, uses and environmental fate”* and Glossary for definitions). However, in its *Scientific Opinion on draft environmental quality standards for PFAS Total under the Water Framework Directive* (i.e. EP and Council 2000), the European Commission’s Scientific Committee on Health, Environmental and Emerging Risks (SCHEER) advised to also add TFA to the DWD parameter ‘Sum of PFAS’ (SCHEER 2025); see Sect. *“EU rationale to include TFA in the definitions for PFAS”* for details on the SCHEER rationale.

The definition for ‘PFAS Total’ in the DWD is consistent with the definition for PFAS provided on the ECHA website regarding the restriction proposal for PFAS (ECHA 2026c). There, it is stated that a PFAS is *“any substance that contains at least one fully fluorinated methyl (CF₃-) or methylene (-CF₂-) carbon atom (without any H/Cl/Br/I attached to it)”* (ECHA 2026c). Both the definition for ‘PFAS Total’ in the DWD and the definition for PFAS in the restriction proposal are based upon the OECD definition for PFAS provided in the *Recommendations and practical guidance – reconciling terminology of the universe of PFAS* (OECD 2021), with the definition in the restriction proposal excluding few fully degradable PFAS subgroups (ECHA 2025e).

It is important to note that the OECD definition was not established to support regulatory actions but to provide a coherent and consistent definition that is based on chemical structure alone. Also, the OECD definition “*does not inform whether a compound is harmful or not*” (page 8 in OECD 2021). The OECD document includes practical guidance to governments and other stakeholders on how to use the PFAS terminology to define user-specific PFAS subgroups that are appropriate for the specific purpose (e.g. a specific risk assessment goal) and to provide transparency on the underlying context and rationale (OECD 2021). The need to define user-specific PFAS subgroups is enhanced by the circumstance that only a small proportion of the PFAS that are covered by the broad OECD definition are commercially relevant (Buck et al. 2021; Anderson et al. 2022).

EU rationale to include TFA in the definitions for PFAS At the EU level, the definition for PFAS in the restriction proposal was selected with the goal to cover “*a broad chemical scope, thereby avoiding regrettable substitution of one PFAS by another PFAS (some of which may not have even been engineered yet)*” (page 2 in ECHA 2026c). TFA is explicitly included in this broad definition with the justification that “*trifluoromethyl fragments are also linked via degradation to trifluoroacetic acid (TFA) which has been demonstrated to be a persistent substance with harmful properties in a comparable way as other PFASs with longer fluorinated alkyl chains. Support for the outlined justification for grouping may be found in the scientific literature*” (ECHA 2025d). As regards “*scientific literature*”, Cousins et al. (2020), Kwiatkowski et al. (2020) and Wang et al. (2017) are cited.

Similarly, SCHEER justified its advice to include TFA in the DWD parameter ‘Sum of PFAS’ (Sect. “*EU definitions for PFAS versus OECD definition for PFAS*”) by the circumstance that both TFA and PFAS elicit liver effects: “*The SCHEER notes the large overlap in known health effects of TFA (liver metabolism and liver weight) that results in a cumulative effect with the other 24 PFAS in this chemical group*”, while acknowledging that the potency of TFA is much lower than the potency of the other PFAS (SCHEER 2025). In the *Draft EQS* [Environmental Quality Standards] *dossier on PFAS-Total*, the European Commission Joint Research Centre referred to the option to include TFA in the list of ‘Sum of PFAS’ since “*the relative potency factor (0.002, calculated by RIVM (2023)) is available*” (Footnote 48 in JRC 2024).

Thus, the decisions to consider TFA as a part of the PFAS group provided by ECHA (2025d, 2025e, 2026c)

in the context of the restriction proposal for PFAS and by SCHEER (2025) and JRC (2024) with respect to the DWD parameter ‘Sum of PFAS’ are founded on two major arguments, i.e. (1) that TFA is a degradation product for many PFAS; and (2) that TFA and PFAS have comparable harmful properties, i.e. comparable effects on liver metabolism and liver weight, resulting in a cumulative effect.

As regards scope of the definition (OECD 2021), PFAS were defined broadly in the restriction proposal to prevent regrettable substitutions (ECHA 2025e). However, this same goal is irrelevant for the monitoring of drinking water, which addresses actual human exposures (Sect. “*DW-relevant metabolites, GW-relevant metabolites and non-relevant metabolites of PPPs*”). Any definition for PFAS under the DWD should serve to facilitate meaningful monitoring to ensure that drinking water does not pose a health risk to consumers. The authors of the present article maintain the view that it is neither scientifically defensible nor meaningful from a practical perspective of enforcement to consider TFA as a part of ‘PFAS Total’ or ‘Sum of PFAS’ under the DWD. This is further explored below (Table 5).

The discussions below, just as the ongoing DWD-related discussions (JRC 2024; SCHEER 2025), address the question whether TFA should be included as a further substance in the DWD parameter ‘Sum of PFAS’, which currently encompasses 24 PFAS. By comparison, since TFA may already be included in the current definition of the DWD parameter ‘PFAS Total’ (Sect. “*TFA: identification, uses and environmental fate*”), it may require a legal interpretation to clarify the exact chemical space covered by this parameter.

TFA should not be grouped together with PFAS The approach to consider a substance as a part of a group because it is a degradation product of many substances from the group is not meaningful from a biochemical perspective (e.g. amino acids, which are building blocks and breakdown products of proteins, are not themselves proteins). Also, this approach does not stand in line with the guidance for grouping provided in the ECHA Read-Across Assessment Framework (ECHA 2017). Therein, it is stated that ‘similarity’ to support grouping may imply the “*likelihood of common breakdown products through physical and/or biological processes which result in structurally-similar degradation products*” (ECHA 2017). Accordingly, substances that have common breakdown products may be grouped together. However, this does not imply that the degradation products themselves are also part of the group. First, the metabolic processes addressed to investigate similarity within a group occur before the degradation products are formed. Also, similarity cannot be defined by one breakdown product that

is common to all group members since common breakdown products may be formed via different metabolic processes. Most importantly, grouping and read-across are means to fill toxicological data gaps by predicting the properties for a data-poor ‘target substance’ using the available data from a group of structurally similar substances (OECD 2014; ECHA 2017). A comprehensive toxicological database is available for TFA (Sect. "Toxicological database for trifluoroacetic acid") – and these data show differences to PFAS – see below. Therefore, TFA should not be grouped together with PFAS for read-across, but it should be evaluated based on its individual properties.

PFAS bioaccumulate and TFA does not As regards toxicological profile, the available scientific evidence demonstrates that the view that TFA might have harmful properties “in a comparable way” as PFAS (ECHA 2025d) is incorrect. PFAS have the potential to bioaccumulate (the key concern in discussions around PFAS; see e.g. Khan et al. 2023) whereas TFA does not (Table 2). Singh and Papanastasiou (2021), referring to Kwiatkowski et al. (2020), i.e. one of the papers cited by ECHA (2025d), see above, rebut

the claim that all substances with at least one aliphatic perfluorocarbon moiety may have harmful properties:

“TFA is vastly different from the larger perfluorocarboxylic acids... TFA is a well-studied compound that is completely miscible with water. As one would expect with most very water-soluble chemicals, its octanol–water partition coefficient shows there is a negligible risk of bioaccumulation; therefore, any future regulatory action on the premise of bioaccumulation should exclude TFA and its anthropogenic precursors” (Singh and Papanastasiou 2021; see also Buck et al. 2021; Wallington et al. 2021).

Zeilmaker et al. (2016) and Zeilmaker and Janssen (2016), from RIVM, derived a health-based guidance value for PFOA in drinking water using its steady-state serum concentration at the NOAEL for liver effects observed in a rat 90-day study as the reference point to account for its bioaccumulation potential (Box 2A in Sect. "Netherlands"). Following this approach, TFA does not form part of the PFAS group since the available data do not support the conclusion that it bioaccumulates. The differences in bioaccumulation

Table 5 Key consideration for why TFA should not be included in the PFAS definitions under the EU Drinking Water Directive

Section [a]	Key consideration		TFA	PFAS ($\geq C_4$)
EU definitions for PFAS versus OECD definition for PFAS	Scope of risk assessment and international alignment	OECD (2021) PFAS definition	Included in this chemical structure-based definition; guidance to refine PFAS definition per scope of risk assessment	
Definitions for PFAS that do not include TFA		UNEP and US definitions for PFAS [b]	Not included	Included
TFA should not be grouped together with PFAS	Consideration of TFA as part of the PFAS group	Group TFA as common degradation product of PFAS?	Not meaningful from biochemical perspective Not aligned with ECHA RAAF [c]	Not applicable
PFAS bioaccumulate and TFA does not	Bioaccumulation and environmental fate	Bioaccumulation potential Accumulation potential in soils/sediments	Low Low	High High
TFA should not be grouped together with PFAS	Toxicity profile and mode-of-action	Toxicological knowledge	Data rich	Partly very poor
TFA has much lower toxic potency than PFAS		Toxicological potency	Low	High
Mechanisms of toxicity		(Limited) knowledge on mechanisms of toxicologically relative unspecific liver effects	Weak peroxisome proliferator	Peroxisome proliferation, steatosis
Practical perspective of enforcement	Practical enforcement	Recovery in PFAS multi-residue methods Exposure in drinking water	Low High	Sufficient Low

ECHA: European Chemicals Agency; OECD: Organisation for Economic Co-operation and Development; PFAS: per- and polyfluoroalkyl substance(s); RAAF: Read-Across Assessment Framework; TFA: trifluoroacetic acid; UNEP: United Nations Environment Programme; US: United States

[a] See Sect. "TFA should not be included in the DWD definitions for PFAS", for details

[b] UNEP (2023); US EPA (2021, 2022); ITRC (2023)

[c] ECHA (2017)

potential are also reflected in the serum half-lives for TFA vs PFAS. For TFA, the elimination half-life in rabbits following enteral administration was estimated to be 16.7 h (ECHA 2025a). For PFOA, Zeilmaker et al. (2016) provided a serum half-life of 839.5 days in humans. In a comprehensive review of toxicokinetic data available for different PFAS, Pizzurro et al. (2019) showed that PFOA and PFOS (both: C_8) exhibited higher liver-to-serum ratios in rats (1.5 and 10.9, respectively) than PFHxS (C_7 : 0.44) and perfluorobutanoic acid (C_4 : 0.24).

In addition to the differences in bioaccumulation potential (as indicated by the low octanol–water partition coefficient of TFA), TFA is highly mobile (as demonstrated by the high aqueous solubility and the low organic carbon–water partition coefficient). By comparison, PFAS with $\geq C_4$ are much less mobile and water soluble (Zhi et al. 2024; see also example of PFOA in Table 2). Therefore, TFA distributes in all aqueous compartments, but it does not accumulate in sediment and soil. Substance-specific data on TFA biomagnification potential are currently unavailable, also due to limitations of analytical methods with respect to the detection of ultra-short PFAS (Neuwald et al. 2022; Brunn et al. 2023). However, on account of its physico-chemical and toxicokinetic properties, there is no indication that TFA might biomagnify across the food chain (Brunn et al. 2023; Zhi et al. 2024). By comparison, PFAS do accumulate in sediment and soil, and they have the potential to biomagnify across the food chain (see e.g. Brunn et al. 2023; Hopkins et al. 2023; Ecke et al. 2024; Wang et al. 2025).

Taken together, the differences in bioaccumulation, mobility and biomagnification potential provide support to not group TFA together with PFAS under the DWD.

TFA has much lower toxic potency than PFAS TFA has a much lower overall toxicity profile than PFAS (see Table 2 for CLH categories assigned to TFA vs PFOA) just as it has much lower toxic potency, i.e. 140-fold when comparing rat 90-day NOAELs for liver effects: The liver effects elicited by TFA occur at much higher dose levels than those elicited by PFAS. The 90-day TFA study yielded a NOAEL of 8.4 mkg/kg bw/d for liver effects (Table 3). By comparison, a NOAEL of 0.06 mkg/kg bw/d was derived from a rat 90-day study evaluating PFOA (Perkins et al. 2004; Box 2A in Sect. "Netherlands").

Similarly, TFA has much lower potency for developmental effects than PFOA. For Na-TFA, a foetal NOAEL of 60 mkg/kg bw/d was established in rabbits based upon the current evidence indicating eye abnormalities in this species (Sect. "Developmental and reproductive toxicity studies for TFA"). By comparison, across developmental toxicity studies evaluating PFOA, a foetal NOAEL of 5 mkg/kg bw/d was established in mice based upon the predominant effect of

impaired development of the mammary gland (Hoadley et al. 2023).

Mechanisms of toxicity As indicated above, the SCHEER referred to effects on liver metabolism and liver weight to support the view that there may be a large overlap of harmful effects between PFAS and TFA (SCHEER 2025). The liver is the first-pass organ upon oral uptake. Many xenobiotics, which are taken up orally, only or predominantly cause adaptive liver enzyme induction or (adverse) hepatotoxicity (Joseph 2017). Hepatotoxicity is an indication that one or more hepatic pathways for substance metabolism and detoxification have been overwhelmed, and a broad spectrum of mechanisms of hepatotoxicity are known (Hall et al. 2012; Joseph 2017). Concerns for cumulative effects due to co-exposures need to be addressed for all pollutants that may be present in drinking water and this regardless of their possible chemical similarity. Considering the differences in bioaccumulation potential and hepatotoxic potency between TFA and PFAS, the overlap for relatively unspecific toxicological findings (i.e. effects on liver metabolism and liver weight) and the concern for cumulative effects due to co-exposures are not sufficient to justify grouping of TFA together with the PFAS.

A meaningful grouping of PFAS should preferably consider the MoA(s) that underlie the observed effects. The EFSA SC asserted that mechanisms of toxicity are an important element to support grouping for the risk assessment of combined exposures to chemicals:

“MoA and AOP [adverse outcome pathway] data ideally provide a strong scientific basis to group chemicals, but all chemical substances acting via a common MoA are not necessarily included in an assessment group. For example, including all chemicals that act via a specific target molecule would make the assessment unnecessarily resource intensive if there is evidence that some will not contribute to the combined effects based on exposure and potency considerations” (EFSA SC 2019).

Hence, even if substances share a common MoA (or if their mechanisms of liver toxicity cannot be established due to prevailing knowledge gaps; see also EFSA 2025c), their grouping may not be meaningful if potencies differ considerably. Investigations into the mechanisms of PFAS-mediated and TFA-mediated liver toxicity appear as a pertinent research need. Similarly, an independent expert panel providing insight and guidance on the grouping of PFAS for human health risk assessment concluded that information on mechanisms of toxicity for PFAS was needed “such that

sub-groups can be constructed with common toxicological endpoints and mechanisms/modes of action" (Anderson et al. 2022).

As regards MoA of the liver effects caused by TFA, evidence from a 14-day study using rats suggests that TFA may be a weak peroxisome proliferator in rats (Dekant and Dekant 2023; BAuA 2025). By comparison, using high-throughput transcriptomics analysis, Rowan-Corall et al. (2025) showed that four PFAS (PFOA, PFOS, perfluorobutane sulfonic acid and perfluorodecane sulfonic acid) exhibited complex hepatic MoAs, with induction of peroxisome proliferator-activated receptor- α being only one relevant MoA. PFOA exhibited a unique transcriptomic profile, which indicated that, due to its high bioaccumulation, it may induce liver cholesterol enzymes thereby leading to a higher risk of metabolic dysfunction-associated steatohepatitis (Rowan-Corall et al. 2025; see also Peters and Gonzalez 2011). Based upon a systemic review and meta-analysis of published rodent and human studies addressing liver effects upon exposure to PFAS (mainly PFOA, PFOS, PFNA and PFHxS), Costello et al. (2022) concluded: *"In rodents, PFAS exposures consistently resulted in higher ALT levels and steatosis."* Since TFA does not bioaccumulate, it appears very unlikely that it would also elicit steatosis.

In vitro investigations provide further evidence to support the view that the MoAs of TFA and PFAS are not comparable. Specifically, Sodani et al. (2025) revealed differences in the in vitro toxicological profiles of TFA (C_2) and perfluoropropionic acid (C_3) as compared to those of six PFAS (C_4 , C_5 , C_6 , C_8 , C_9 , C_{10}). First, Sodani et al. assessed all test items in ToxProfiler, a human cell-based assay *"to visualise the major cellular stress response pathways"* (Toxys 2025a). The C_8 - C_{10} PFAS exhibited cytotoxicity and predominantly induced endoplasmic reticulum and oxidative stress responses (at 130 μ M), and the C_4 - C_7 PFAS primarily induced autophagy (at 300 μ M). By comparison, TFA (C_2) and perfluoropropionic acid (C_3) were not cytotoxic and did not activate any of the ToxProfiler stress response reporters at their maximum tested concentrations (10 mM). Second, Sodani et al. submitted TFA and PFOA (C_8) to ReproTracker, a human stem cell-based assay that *"captures the key cellular events of stem cell differentiation [into cardiomyocytes, hepatocytes and neuronal cells] and early embryonic development"* (Toxys 2025b). PFOA elicited a concentration-dependent decrease in the hepatocyte-specific and neuroectodermal biomarker genes and disrupted the morphology of the hepatocytes and neuronal cells at 30 and 60 μ M, respectively. By comparison, TFA had no significant effect on biomarker expression, and it did not affect the morphology/functionality of the three differentiated cells (Sodani et al. 2025).

Practical perspective of enforcement From a practical perspective of enforcement, an operational PFAS definition that does not include TFA is critical due to the many challenges related to the analytical detectability of TFA in multi-residue analytical methods (see e.g., Neuwald et al. 2022; Brunn et al. 2023). While the European Commission has proposed three such methods to enforce the parameter 'PFAS Total' in drinking water, they *"have not been fully validated for ultra-short-chain PFAS (e.g. TFA) ... Specifically, the recovery for TFA can vary significantly between sample pretreatments"* (European Commission 2024b). This means that, due to the low recovery of the three proposed multi-residue methods for TFA, the procedure to report 'PFAS Total' will typically yield *"inconclusive"* results (Sect. 4.2.2) in European Commission 2024b), because the difference of 'PFAS Total' (based on the multi-residue methods) minus TFA (as determined with a targeted method) will usually be negative. A report commissioned by the German UBA has confirmed that the first method proposed by the European Commission (2024b), i.e. the Total Oxidisable Precursor Assay, is not capable of determining TFA (Dahme and K  chler 2024). Neither is the alternative Extractable Organofluorine Combustion Ion-Chromatography method, as it has an extraction efficiency of only 1% for TFA (K  rrman et al. 2021). Finally, the third method, i.e. Liquid Chromatography – High Resolution Mass Spectrometry, also suffers from poor retention and low ionisation efficiency of TFA (Camdzic et al. 2023).

As regards drinking water monitoring, it is also noteworthy that concentrations of the (low-toxicity, non-bioaccumulative) TFA in drinking water are orders of magnitude higher than concentrations of the (higher-toxicity, bioaccumulative) PFAS. Across studies and countries, TFA concentrations of approximately 0.1–3 μ g/L (and up to 20 μ g/L in a single instance) in drinking water have been recorded during monitoring (Table SI-1). By comparison, for the sum of 10 PFAS, which did not include TFA, Gao et al. (2024) estimated that concentration levels in tap water were approximately 2.7 and 9.2 ng/L in the United Kingdom and China, respectively.

Together with the scientific needfulness to conduct separate risk assessments for TFA and PFAS, these aspects of practical enforcement of parametric values further speak against the grouping of TFA together with PFAS under the DWD.

Definitions for PFAS that do not include TFA The United Nations Environment Programme (UNEP) Environmental Effects Assessment Panel explicitly excludes TFA from the class of PFAS when addressing the environmental effects of these substances: *"We are of the opinion that the properties of TFA indicate that it should not be included in this class*

[i.e. the class of PFAS] for the purposes of generic regulatory risk assessment” (page 279 in UNEP 2023).

In the USA, TFA is not included in the Drinking Water Contaminant List (US EPA 2022) as PFAS are defined as:

“Chemicals that contain at least one of these three structures: (1) $R-(CF_2)-CF(R')R''$, where both the CF_2 and CF moieties are saturated carbons, and none of the R groups can be hydrogen; (2) $R-CF_2OCF_2-R'$, where both the CF_2 moieties are saturated carbons, and none of the R groups can be hydrogen; (3) $CF_3C(CF_3)RR'$, where all the carbons are saturated, and none of the R groups can be hydrogen” (US EPA 2022).

TFA does not fulfil any of these structural criteria. The US EPA has also excluded TFA from the National PFAS Testing Strategy as it *“is a well-studied non-PFAS”* (US EPA 2021).

The US Interstate Technology and Regulatory Council (ITRC) defined PFAS as having a perfluoroalkyl chain tail of C_nF_{2n+1} , where n is ≥ 2 (thereby excluding TFA where $n = 1$), and an attached functional group head (ITRC 2023). In the OECD guidance on PFAS terminology, the ITRC definition is provided as an example for how *“individual users may define their own PFAS working scope for a specific activity according to their specific needs”* (page 31 in OECD 2021).

The varying perceptions between different jurisdictions and organisations on what constitutes a PFAS have prompted a current International Union of Pure and Applied Chemistry project (IUPAC) on *Terminology and Classification of PFAS*, which aims to provide *“a rigorous definition for PFAS, and standardizing terminology, classification, and nomenclature... to help national and global regulation and policy decisions”* (IUPAC 2024; see also Secundo et al. 2025). The IUPAC PFAS definition shall be based on *“physical–chemical properties of the different classes of fluorinated compounds”* (IUPAC 2024). This activity appears very meaningful as was demonstrated here regarding the fundamental differences between TFA and the typical PFAS.

Conclusion not to consider TFA in the DWD definition for PFAS

The question whether TFA should be included as a 25th substance in the DWD definition for ‘Sum of PFAS’, or not, can be addressed from a scientific perspective and from a political and regulatory perspective. TFA and PFAS exhibit profound differences in bioaccumulation and biomagnification, mobility/water solubility, acidity, adsorption potential, toxicological profile and toxic potency. Due to the vast differences in physico-chemical properties, TFA distributes and in turn occurs in different environmental compartments

at much different concentrations compared to typical PFAS compounds with $\geq C_4$. In addition, the available scientific evidence indicates that PFAS and TFA may exhibit liver effects by different MoAs. Similarly, the authors of the present article maintain the view that the assignment of a CLH for reproductive toxicity to (Na-)TFA does not strengthen the rationale for grouping TFA together with PFAS due to differences in observed types of developmental effects and potencies (Sect. *“TFA has much lower toxic potency than PFAS”*).

All of this evidence speaks against the grouping of TFA together with PFAS. Similarly, the European Association of Drinking Water Suppliers (EurEau) considers TFA as an ‘atypical PFAS’ and advocates for setting a specific health-based limit value for TFA in drinking water rather than grouping it together with ‘PFAS Total’ (EurEau 2025). Colnot and Dekant (2022) propose splitting PFAS into two groups, i.e. perfluoroalkyl carboxylates and perfluoroalkyl sulfonates, and to exclude the short-chain PFAS from these groups: *“Short chain PFAS and other PFAS structures should not be included in the groups due to their low potency and rapid elimination. This approach is in better agreement with scientific and regulatory guidance for mixture risk assessment”* (Colnot and Dekant 2022). The aforementioned IUPAC project to establish a standardised terminology and classification for PFAS (IUPAC 2024) appears very meaningful as is demonstrated here regarding the fundamental differences between TFA and the typical PFAS.

In conclusion, the grouping of PFAS for analytical determination and risk assessment does have its merits given the sheer number of compounds covered by this term. It may be a good approach to enhance analytical efficiency and toxicological assessment in case substance-specific data are unavailable. Therefore, it is a good approach for the majority of longer-chain PFAS with $\geq C_4$. However, TFA stands out as it is quite unique in terms of chemical properties compared to PFAS with $\geq C_4$. Therefore, the separate monitoring of TFA, found to have lower toxic potency, but higher concentration levels in drinking water, and PFAS, found to have higher toxic potency but lower concentration levels in drinking water, would facilitate a focussed identification of critical effect levels of both TFA and PFAS. Such an approach better suits the intention of the DWD to protect consumers from health risks caused by the presence of chemicals in drinking water than consideration of TFA within the parameter ‘Sum of PFAS’ as implemented in the DWD. (Since TFA may already be included in the DWD parameter ‘PFAS Total’, it may require a legal interpretation to clarify the exact chemical space that is currently covered by this parameter; Sect. *“EU rationale to include TFA in the definitions for PFAS”*).

Beyond analytical convenience, grouping may also serve as a precautionary regulatory strategy in contexts where scientific uncertainty remains unresolved, a rationale that, while not applicable to TFA given the available substance-specific data, reflects a legitimate regulatory logic.

Derivation of a guidance value for TFA in drinking water based upon its relative potency to PFAS

The approach by the Dutch RIVM to derive the guidance value for TFA in drinking water on account of its toxic potency relative to PFOA is based upon the premise that TFA may be considered as a part of the PFAS group (RIVM 2023). If the scientific evidence that speaks against this premise (Sect. "TFA should not be included in the DWD definitions for PFAS") is addressed in the regulatory definition for 'PFAS Total' (or 'Sum of PFAS') in the DWD, the RIVM approach is per se invalidated.

In addition, this section includes a discussion of the approach taken by RIVM to derive the threshold value of 2.2 µg/L for TFA in drinking water (which has been endorsed by SCHEER 2025). Focus of the discussion is on:

1. The reference point selected by RIVM (2023) for the derivation of the guidance value for TFA in drinking water, i.e. the TDI/TWI established by EFSA CONTAM Panel (2020) for Σ 4PFAS (i.e. PFHxS, PFNA, PFOA, PFOS), including an appraisal of:
 - a. The robustness of the underlying human data
 - b. The robustness of the CONTAM Panel approach to derive the BMDL₁₀ and TDI/TWI
2. The establishment of the relative potency of TFA as compared to 16 PFAS, including an appraisal of:
 - a. The robustness of the relative potency approach for the 16 PFAS
 - b. The robustness of the comparison of the potencies of TFA vs PFAS

In brief, this discussion considers that the absence of robust or rather concerning immunotoxicity data adds cumulative uncertainty to the overall robustness and thus biological relevance of the grouping, read-across and modelling approach applied by RIVM (2023). Necessarily, the derived guidance value for TFA in drinking water, therefore, depends on substantial assumptions, which have not been scientifically validated.

Re. 1a: Robustness of the human data underlying the TDI/TWI for the sum of four PFAS

The human observational

study by Abraham et al. (2020), i.e. the key study considered by EFSA CONTAM Panel (2020) to derive the TDI/TWI, has several scientific limitations that impair the reliability of its findings. First, this study only included a total of 101 infants, which were further separated into two sub-groups, i.e. 21 children that were breast-fed for a maximum of two weeks and 80 children that were exclusively breast-fed for at least four months. The cohort included 27 children from a "dioxin hotspot" (Abraham et al. 2020). The investigated samples relate to an earlier study conducted to investigate dioxin exposure, which is available in German language (Abraham 2002). Due to the study's aim, socio-economic factors were not considered. However, such factors could affect both PFAS exposure and the investigated endpoint to derive a TDI (i.e. vaccination response). For example, the recruitment approach may indicate uncontrolled heterogeneity in socio-economic status as "recruitment was performed via announcement in newspaper and via paediatricians" (Abraham et al. 2020). For selection, it was required that the child had normal development until sampling, absence of infections at sampling, no exposure towards smoking, and parents from the German-speaking region. Potential further sources of PFAS, other than smoking, were not considered.

RIVM (2020a) and, similarly, De Clercq and De Brouwere (2024) from the Flemish VITO cautioned that the group of children considered by Abraham et al. was heterogeneous and questioned whether the statistical analysis of the data was sufficiently robust to overcome possible confounding effects: "Our concern is that the entire study population consists of two different subgroups, and that the statistical analysis is not sufficient to overcome possible distortive effects due to those differences (such as differences in other contaminants in the blood, socio-economic status, breast-feeding/bottle-feeding)" (RIVM 2020a; unofficial translation, see SI-3.3 for original quote).

Second, the evidence available from different human studies is not consistent. The findings reported by Abraham et al. (2020) indicate that PFOA may be more potent: "While associations between levels of PFOA and levels of vaccine antibodies as well as interferon γ production after specific stimulation were consistently related to each other, no relevant associations were observed with levels of PFOS suggesting a higher immunotoxic potency of PFOA in humans" (Abraham et al. 2020). As regards liver effects-related potencies, it is exactly the other way around, i.e. PFOS has twice the potency of PFOA (RIVM 2025; see below, discussion of Bullet 2a). The findings from the Faroe Island study by Grandjean et al. (2012, 2017a, 2017b), which was also considered by the CONTAM Panel (Table SI-4), indicate that PFOS may be more potent than PFOA in affecting

vaccination responses: “Among perfluorinated compounds in maternal pregnancy serum, PFOS showed the strongest negative correlations with [diphtheria] antibody concentrations at age 5 years” (Grandjean et al. 2012). Abraham et al. noted that, since serum PFOS levels were much higher than the PFOA levels in the Faroe Island children (at 5 years of age: mean PFOS / PFOA: 16.7 / 4.06 µg/L), the higher immunotoxic potency of PFOA may have been masked by PFOS (Abraham et al. 2020). In the study by Abraham et al. (2020), mean serum PFOS levels were also higher than PFOA levels in the 21 one-year old, formula-fed infants (PFOS / PFOA: 6.8 / 3.8 µg/L) but almost equal in the 80 one-year old infants that had been exclusively breast-fed for at least four months (PFOA / PFOS: 15.2 / 16.8 µg/L). Hence, the currently available evidence on PFAS potency in eliciting impaired vaccination responses in children is inconsistent and thus inconclusive.

The CONTAM Panel also highlighted uncertainties and inconsistencies with respect to the potencies of different PFAS in impairing vaccination responses: “Since PFOA and PFOS serum concentrations are higher as compared to serum concentrations of PFNA and PFHxS, and PFOA highly correlates with the serum levels of the other PFASs, it is uncertain whether PFOA has a higher potency for this critical endpoint as compared to the other PFASs and thus drives the association... a potential higher potency of PFOA seems to contradict the outcome of various studies with animals, which thus far showed effects at much lower serum levels for PFOS than for PFOA, keeping in mind that a thorough comparison was never made” (page 140 in EFSA CONTAM Panel 2020).

Further, Abraham et al. admitted that, while adjusted antibody levels were reduced by 53–86% directly after vaccination, “a 50% lower antibody level is already reached after one half-life which is several weeks on average in the 1-year-old children of our study after two to three vaccinations against Hib, tetanus or diphtheria” (Abraham et al. 2020). However, Abraham et al. provided no explanation on how the antibody serum half-lives may have impacted the study outcome.

Finally, Abraham et al. themselves also pointed to the scientific limitations of the findings from their study: “...since most studies in this field are cross-sectional, data need to be interpreted with caution. More insight is needed into possible mechanisms of action, dose–response relationships and the clinical relevance. Therefore, further intensive laboratory and epidemiological research (including prospective studies) is necessary to improve the understanding” (Abraham et al. 2020).

Taken together, the available human evidence hints at a correlation between maternal / offspring serum levels of PFAS and impaired vaccination responses (Table SI-4),

however, without providing robust evidence to support a plausible causal relationship. Further, considering that impaired vaccination responses are not adverse effects, the authors of the present article maintain the view that the scientific evidence underlying the EFSA TWI/TDI is not sufficiently robust to serve as a critical effect for the derivation of a legally binding guidance value for TFA (or PFAS).

Currently, there is no consensus whether vaccination responses should be used as critical effects for the risk assessment of PFAS. Bline et al. (2024), i.e. six authors from three US universities and the US Natural Resources Defense Council, a non-governmental organisation advocating for the reduction of PFAS, expressed support to such use. By comparison, an international collaboration of 24 scientists representing academia, authorities and consultants from eight countries, convened by the Alliance for Risk Assessment™ (ARA 2025) to establish a safe dose range for PFOA, concluded that vaccination responses in humans were not sufficiently robust to serve as reference points for the risk assessment of PFAS (Burgoon et al. 2023):

“Certain effects appear to be irrelevant for the determination of a safe dose in the absence of MoA information relevant to humans, specifically differences in cholesterol and vaccine response... Existing human observational studies cannot be used reliably for developing the critical effect in the absence of mechanistic data relevant to humans at serum concentrations seen in the general public. Existing human observational vaccine findings are not primary immune responses and not of clinical relevance” (Burgoon et al. 2023; see also Reinikainen et al. 2024).

Re. 1b – The CONTAM Panel approach to derive the BMDL₁₀ and TDI/TWI The EFSA CONTAM Panel (2020) received BMD modelling data on all possible combinations of vaccination responses from Abraham and his co-authors, validated the data based on data derived from figures presented in the original publication and then selected the BMDL₁₀ for Σ 4PFAS in relation to diphtheria antibody titres because this was the lowest estimated BMDL₁₀ (17.5 ng/mL). In this regard, Haber et al. (2018) highlighted that “it is important to take confidence in the model into account in choosing the BMDL, rather than simply choosing the lowest value.” As explained above (Sect. “Guidance values for TFA in drinking water based on toxicological studies”), the selected BMR and corresponding BMD should be biologically relevant: “The selection of any critical adverse effect should not solely be based on statistical procedures and considerations of its biological relevance for human risk assessment are key” (EFSA SC 2022). The CONTAM Panel provided no such immunotoxicological reasoning for the derivation of

the BMDL₁₀ for Σ 4PFAS (or for the selection of the lowest estimated BMDL₁₀) but instead solely based it on statistical considerations: “It was decided to use a critical effect size of 10% instead of the default 5% considering a large variation in the response” (page 143 in EFSA CONTAM Panel 2020). (The BMDL₀₅ should also not be considered as a ‘default’ in the absence of an associated biologically relevant response.)

Moreover, the data underlying the BMDL₁₀ for Σ 4PFAS do not show an obvious dose–response but are distributed in a broad scatter (Figs. 3a–c in Abraham et al. 2020; Figures K.1–K.3 on page 343 and page 351–353 in EFSA CONTAM 2020). This shortcoming was also highlighted by RIVM (2020a):

“RIVM has indicated that the available human studies, including the Abraham et al. study, show dose–response relationships that are too uncertain to allow for a robust BMD analysis... EFSA considers that the BMD modeling performed with PROAST does show a clear dose–response relationship for the Abraham et al. study...” (RIVM 2020a; unofficial translation, see SI-3.4 for original quote); see also UK COT (2022).

The EFSA view that the BMD modelling performed with PROAST showed a dose–response does not invalidate the RIVM criticism. EFSA does not state how a clear dose–response has been defined or supported, and a dose response is not apparent to the authors of the present article from looking at the graphs (see also Anscombe 1973). In this regard, the EFSA SC asserts that the dose–response analysis is “traditionally [...] done by visual inspection together with conventional statistical tools” (EFSA SC 2022).

The view that the Abraham et al. study shows “dose–response relationships that are too uncertain to allow a robust BMD analysis” (RIVM 2020a) and, hence, that the BMDL₁₀ of 17.5 ng/mL for Σ 4PFAS is not robust, is supported by the outcome of the model averaging. In providing further details on the large variation of the data, the CONTAM Panel stated:

“The model averaging provided BMDL–BMDU intervals of, respectively, 2.1–45.4 and 3.9–49.5 ng/mL for diphtheria and tetanus. The Panel noted that these BMDLs of 2.1 and 3.9 ng/mL are eight- and five-fold lower than the lowest BMDLs from the individual models. This seems to be caused by the fact that several of the 500 bootstrap runs performed for model averaging provided curve fits that deviated from the individual

models. Furthermore, these BMDLs obtained with model averaging were below the observed range of the serum levels of the 4 PFASs in this study” (page 143 in EFSA CONTAM Panel 2020).

If model averaging does not yield a reliable BMDL, it is not appropriate to use the lowest modelled BMDL (also since the corresponding model might have led to the averaging failure). Instead, the biological relevance of the outcome of the statistical analysis should be reconsidered based on the available data, i.e. a plausibility check should be undertaken (Anscombe 1973; Fox and Landis 2016; Kluxen and Hothorn 2020). In addition to the overriding prerequisite of pre-defining a biologically relevant BMR (see above), the database must be sufficient to support a BMD estimate and an uncertainty analysis for the selected BMD, i.e. to establish the BMDL and BMDU. The data underlying the BMDL₁₀ for Σ 4PFAS (or PFOA, see below, discussion of relative potencies) are too scattered to confidently derive a plausible BMDL.

Overall, the derived relationship between PFOA concentrations and impaired vaccination response does not seem robust but instead seems driven by a few high-concentration responses in the Abraham et al. (2020) database. As a further evaluation of the Fig. 3a–c scatter plots in Abraham et al. (2020) shows, only about 25% of the data were in the PFOA concentration range with a discernible average response reduction (i.e. > 20 µg/L PFOA). Of those values, only some were reduced compared to the overall response, whereas many were increased, i.e. vaccination responses were above average despite high PFOA concentrations. Since the original data have not been published either by Abraham et al. (2020) or by EFSA CONTAM Panel (2020), it is not possible to identify possible reasons for outliers, etc.

The CONTAM Panel then applied PBPK modelling to derive the TDI/TWI for Σ 4PFAS from the BMDL₁₀ (i.e. TDI: 0.63 ng/kg bw/day; TWI: 4.4 ng/kg bw/week). Reiniainen et al. (2024) cautioned that the derivation of the EFSA TWI “involved conservative assumptions, such as exclusive breastfeeding for the first 12 months of the infant.” RIVM (2020a) criticised that it is unclear whether further, potentially immunotoxic PFAS contributed to this TWI. Taken together, RIVM (2020a) outspokenly questioned the scientific relevance of the EFSA TWI:

“The expert group is of the opinion that, scientifically speaking, the foundation for the EFSA TWI is not very robust due to uncertainties in the PoD, the assumption of equipotency, and the limitation to 4 PFAS. This

raises the possibility that the TWI may not be conservative enough, or even unnecessarily conservative. We are not in a position to suggest an alternative, more robust TWI in the short term, nor make a well-founded proposal for the size of an additional uncertainty factor” (RIVM 2020a; unofficial translation, emphasis in bold by authors of the present article, see SI-3.5 for original quote).

Despite substantial scientific concerns, RIVM (2020a) endorsed the EFSA TWI due to the absence of an alternative guidance value for PFAS while noting that one expert from the group did not support this “*pragmatic option*” as he/she considered “*the uncertainties surrounding the TWI too great and, therefore, the TWI not informative for interpreting possible risks*” (RIVM 2020a; unofficial translation, see SI-3.6 for original quote).

Reinikainen et al. (2024) critically discussed the regulatory implications of the EFSA TWI:

“The EFSA’s current TWI may be justified in its policy context and as per the precautionary principle. Nevertheless, it seems legitimate to question whether such a rigorous value – below the average European background exposure – is an appropriate threshold for defining maximum concentrations of substances that are ubiquitous in the environment. Furthermore, the scientific basis of the TWI does not indisputably indicate an increased probability of adverse health effects at this exposure level, nor even a correlation with the vaccine antibodies in the case of three of the four PFAS covered by the threshold” (Reinikainen et al. 2024).

As compared to the EFSA TDI of 0.63 ng/kg bw/day for Σ 4PFAS, Burgoon et al. (2023) provisionally estimated 10–70 ng/kg bw/day as the PFOA safe dose range “*based on information from the various consensus calls regarding PFOA’s underlying MoA for various effects, its likely critical effect(s), and the extrapolation of experimental or human data to the presumed sensitive subgroup*” (Burgoon et al. 2023). This provisional safe dose is comparable to the TDI of 12.5 ng/kg bw/day for PFOA applied by Zeilmaker and Janssen (2016), from RIVM, to derive a health-based guidance value of 87.5 ng/L PFOA in drinking water (Box 2A in Sect. “Netherlands”).

In conclusion, the scientific meaningfulness of the decision by RIVM (2020a) to endorse the EFSA TWI, despite its substantial shortcomings, for the risk assessment of data-poor PFAS appears questionable. However, it is clearly not

defensible for the risk assessment of the data-rich substance TFA, for which a comprehensive substance-specific database is available.

Re. 2a – Relative potency approach for 16 PFAS The estimation of the relative potencies of PFAS as compared to PFOA and use of the relative potency factor approach to derive health-based guidance values for PFAS (and TFA) are based upon a number of strong assumptions, which impair the scientific robustness of this approach.

Since a ranking of PFAS by effects on vaccination response in humans was not possible due to overall limitations of the database and inconsistencies in the findings (see above and Table SI-4), RIVM used a ranking by effects on relative liver weight caused in male rats upon exposure to 16 PFAS (Bil et al. 2021) to derive health-based guidance values for PFAS (RIVM 2021b) and TFA (RIVM 2023). In pursuing this approach, RIVM assumed that the differences in potencies of the 16 PFAS were both comparable between toxicological effects (immunotoxicity vs hepatotoxicity) and between species (human vs rat), thereby implicitly also assuming a functional relationship.

By comparison, the CONTAM Panel decided not to base its guidance value for PFAS in food on relative liver potencies on account of the prevailing mechanistic uncertainties: “*The CONTAM Panel did not apply relative potency factors based on effects in liver because there is too much uncertainty on a common MoA behind both immune effects and those on the liver*” (EFSA CONTAM Panel 2020).

Bil et al. (2021) cautioned that further research with respect to the comparability of PFAS-mediated effects was necessary to facilitate the regulatory applicability of the potency ranking of the 16 PFAS:

“...it may be relevant to validate whether relative potency factors based on other endpoints, such as developmental toxicity or immunotoxicity, will result in comparable estimates of relative potencies and if the potency differences for liver toxicity may be a proxy for differences in the overall toxicity between PFAS. Further substantiation of such a hypothesis requires elucidating the mechanism(s) underlying PFAS-mediated multiorgan toxicity or validation of the in vivo liver relative potency factors with more data for other effects, sex, life stages and species considered” (Bil et al. 2021).

Apparently, RIVM conducted a preliminary analysis of animal studies from the US National Toxicology Program,

which showed that potency differences between PFAS are present in, for example, organ weights, hormone levels, clinical chemistry, white blood cell parameters and pathology endpoints and that the potency ranking was generally comparable with the ranking by liver effects (RIVM 2021a). To the best of the knowledge of the authors of the present article, this analysis is not publicly available and thus cannot be retraced.

Bil et al. (2023a) evaluated the data from eight 28-day repeated-dose toxicity guideline studies assessing eight PFAS ($\geq C_4$) including PFHxS, PFNA, PFOA and PFOS. Toxicokinetic modelling was applied to convert external exposures to internal exposures and dose–response modelling to ensure fitting of the dose–response curves. Internal relative potency factors were determined for absolute and relative thymus weight, absolute spleen weight, globulin concentration and relative liver weight in the adult male rats. As concluded by Bil et al., the outcome of this study showed that the internal relative potency factors for PFAS “based on decreased rat lymphoid organ weights are similar to those previously established for increased rat liver weight” (Bil et al. 2023a). The authors of the present article maintain the view that relative potencies based on internal exposures and altered lymphoid organ weight in adult male rats provide very limited support to the ranking of PFAS by maternal external exposures and effects on vaccination response in human offspring.

As regards immunotoxic potencies of PFAS in humans, the current evidence is not consistent (see above). Also, there is no consensus on how the available evidence shall be interpreted. The CONTAM Panel founded its evaluation on the assumptions (1) of equal potencies by default of the four PFAS on immune outcomes in humans (page 7 in EFSA CONTAM Panel 2020); and (2) that, while a relationship between serum concentration and vaccination response was not established for PFHxS, PFOS or PFNA, on their own, the overall effect may also have been caused by these three PFAS (see also Sect. 4.1.2, page 10, in RIVM 2023). By comparison, Abraham et al. (2020) assumed that PFOA may exhibit the highest immunotoxicity potential of the four PFAS (see also Table 3 in Abraham et al. 2020), and RIVM (2021b, 2023) presupposed that *only* PFOA was associated with the TWI of 4.4 ng/kg bw/week that the CONTAM Panel had derived for $\Sigma 4$ PFAS. Bil et al. (2023b) highlighted that since “the serum concentration of one PFAS may be highly correlated with the serum concentration of another PFASs... it is impossible to determine which PFAS(s) drive(s) the association”.

Based upon the data provided in the Abraham et al. (2020) publication, the authors of the present article support

the notion that the entire reduction in vaccination response observed by Abraham et al. correlates with PFOA concentration (while the relationship does not seem robust and the individual responses and thus the fitted model may be biased, due to confounding effects, as described above in the discussion of the BMDL₁₀). However, while the EFSA TWI of 4.4 ng/kg bw/week for $\Sigma 4$ PFAS was likely only associated with PFOA effects on vaccination responses, the same does not stand true for liver-related effects assuming the same mixture of the 4 PFAS that the mothers included in the Abraham et al. (2020) study were exposed to. As regards effects on relative liver weight, the relative potencies of the four PFAS are: PFOS – 2, PFOA – 1, PFNA – 10 and PFHxS – 0.6 (Bil et al. 2021; RIVM 2021a, 2025). The maternal exposure to “PFOA&PFNA” and “PFOS&PFHxS” was 0.187:0.444 (page 143 in EFSA CONTAM Panel 2020), i.e. approximately 1:2. **Since PFOS has twice the potency of PFOA and two thirds of the TWI are attributable to PFOS (as per maternal exposure), the majority of the EFSA TWI is attributable to PFOS, and not to PFOA, when considering liver potencies.** Similarly, RIVM (2021a) provided detailed explanations for how the TWI depends on the mixture ratio of “PFOA&PFNA” and “PFOS&PFHxS” and how the potencies of the four PFAS need to be considered in determining whether the TWI is still undercut if the mixture ratio changes.

The relative potency factor approach assumes the concept of dose additivity (Zeilmaker et al. 2018). However, this is only applicable when the substances act in a similar manner with the same MoA and differ only in their potencies. For PFAS, this is not the case as highlighted by Goodrum et al.: “Concentration addition and relative potency factor methods are found to be inappropriate [for the risk assessment of PFAS mixtures] due to differences among sensitive effects and target organ potencies and noncongruent dose–response curves for the same effect endpoints from studies using the same species and protocols” (Goodrum et al. 2022; see also ToxStrategies 2023). The CONTAM Panel also confirmed that relative potencies should be based on effects caused by common MoAs, but that such mechanistic information is currently lacking for PFAS:

“The CONTAM Panel agrees that applying potency factors based on effects on liver or thyroid hormones in animals could be an interesting approach [for the evaluation of PFAS]. However, it is unclear if there are common pathways that lead to the effects on liver, thyroid hormones and the more critical effects on the immune system” (EFSA 2020).

The US EPA *Framework for estimating noncancer health risks associated with mixtures of per- and polyfluoroalkyl substances (PFAS)* includes an example application of the relative potency factor approach to a hypothetical mixture of 5 PFAS (US EPA 2024). This example provided supporting evidence to rebut the view that the PFAS potency rankings for liver vs immune effects may be comparable:

“PFAS can have different potencies across health effect domains. Due to the differences in TK and TD [toxicokinetics and toxicodynamics], PFAS may exhibit complex gradations of potency for different effects, which will be reflected in the corresponding RPFs [relative potency factors]. Some PFAS may be exquisitely potent for some effects and yet virtually inactive in others... calculating RPFs for as many endpoints/effects as possible helps to ensure that subsequent PFAS risk management strategies are health protective” (page 85 in US EPA 2024).

Re. 2b – Relative potency of TFA as compared to PFAS The scientific limitations of comparing PFAS potencies with respect to liver effects caused in rats to vaccination response-related effects in children are equally applicable when aiming to establish the toxic potency of TFA relative to PFOA. Just as TFA should not be grouped together with PFAS, also the potency of TFA should not be established relative to PFOA (further considering their very different potencies (Sect. [“TFA should not be included in the DWD definitions for PFAS”](#))).

Authors’ proposal for a health-based guidance value for TFA in drinking water

Based upon the discussion of the different approaches taken (Sect. [“Guidance values for TFA in drinking water based on toxicological studies”](#) and [“Guidance value for TFA in drinking water based on relative potency to PFAS”](#); Table 6), the following approach is proposed for the science-based derivation of a guidance value for TFA in drinking water.

Firstly, TFA should not be included in the DWD definitions for PFAS (Sect. [“TFA should not be included in the DWD definitions for PFAS”](#)). Also, the approach by RIVM (2023) to base the guidance value for TFA in drinking water on the EFSA TDI/TWI for Σ 4PFAS and the toxic potency of TFA relative to PFOA has substantial scientific shortcomings that invalidate the resulting guidance value (Sect. [“Derivation of a guidance value for TFA in drinking water based upon its relative potency to PFAS”](#)). Instead, the approach proposed by the authors for a science-based derivation of

a guidance value for TFA in drinking water is based upon the WHO standard calculation method (Sect. [“Introduction: derivation of guidance values for substances in drinking water”](#)) and the selection of the below parameters (as presented and discussed in Sect. [“Guidance values for TFA in drinking water based on toxicological studies”](#)). This represents one of several options currently under discussion at the Member State level (Tables 6 and 7).

1. Key study: rat 90-day repeated-dose toxicity study
2. Critical effect: liver adversity
3. NOAEL: 8.4 mkd TFA (in male rats)
4. Assessment factor: 200 (10 each for intraspecies and interspecies variation, 2 for extrapolation from sub-chronic to chronic effects)
5. Average body weight of the relevant human population: 70 kg
6. Daily drinking water consumption: 2 L
7. Fraction of the TDI allocated to drinking water consumption: 0.2

These values yield a TDI of 0.042 mkd TFA (i.e. NOAEL / assessment factor; Sect. [“Introduction: establishment of tolerable or acceptable daily intake levels”](#)) and a health-based guidance value for TFA in drinking water of 0.294 mg/L TFA (i.e. $[0.042 \times 70 \times 0.2]/2$; Sect. [“Introduction: derivation of guidance values for substances in drinking water”](#)) or 294 μ g/L TFA (Table 7). Adherence to this guidance value ensures that the presence of TFA in drinking water does not pose *“a health risk for consumers”* (EP and Council 2020). Generally, the TDI is derived from the most critical adverse endpoint reflecting longer-term exposure encompassing developmental and reproductive toxicity or systemic toxicity, and it is protective for the entire population, representing a science-based threshold for lifetime exposure. The TDI includes a safety factor which addresses any differences in toxicodynamics or toxicokinetics across the population, whatever the age group.

Hence, in agreement with the default WHO approach, a single drinking water guidance value of 294 μ g/L TFA is proposed. Current levels of TFA in drinking water remain well below this value, even in high-concentration samples that were temporarily taken at local hotspots (e.g. > 20 μ g/L in the Bad Wimpfen area near the German river Neckar, Table SI-1). To foster harmonisation and acceptance across agencies, the proposed guidance value should be complemented with appropriate risk management measures (such as introduction of an upper limit for hygienic reasons); however, such further considerations are outside the scope of the present article.

Table 6 Approaches taken in selecting a key study and a critical effect for the derivation of a chronic reference value for trifluoroacetic acid

Study / approach	Critical effect (evaluation)	Selection of critical effect (evaluation)	Reference point	Reference value	Drinking water guidance value	Appraisal by authors of present article
Rat 90-day oral toxicity study	WoE of all liver findings (ALT, HP)	EFSA (2014) Nielson and Beltoft (2021)	NOAEL: 9.9 / 8.4 mkd Na-TFA / TFA	TDI: 0.05 mkd Na-TFA / 0.042 mkd TFA [a]	EFSA (2014): calculated here as 147–294 µg/L TFA [b] Nielson and Beltoft (2021): 100 µg/L TFA	Scientifically meaningful based on WoE evaluation of current toxicological database (Sect. "Liver findings")
	Reduced total bilirubin at d90 [c] Relative liver weight [c]	De Clercq and De Brouwere (2024)	BMDL ₃₀ : 5.2 mkd TFA BMDL ₀₅ : 11 mkd TFA	TDI: 0.0026 mkd TFA [a] TDI: 0.057 mkd TFA [a]	15.6 µg/L TFA 33 µg/L TFA	Selection of BMDL ₃₀ for total bilirubin deviates from established adversity criteria (Sect. "Liver findings", final paragraph)
Rat one-year oral toxicity study	Increased ALT	UBA (2020a) [d]	NOEL: 1.8 mkd Na-TFA	TDI: 0.018 mkd Na-TFA	60 µg/L TFA (target: 10 µg/L)	WoE does not indicate adversity (UBA determined NOEL and not NOAEL; Sect. "Germany")
Rat extended one-generation reproductive toxicity study	Reduced serum thyroxine	EFSA (2025a) [e]	Parental NOAEL: 8.65 mkd Na-TFA	TDI: 0.03 mkd Na-TFA [f]	Calculated here as 87.5–175 µg/L TFA [b]	WoE does not indicate adversity
	(Data on TDAR unavailable)	(Motivation for approaches to derive TDI selected by RIVM (2023) and EFSA (2025a), respectively)	Not applicable			WoE from all available studies does not indicate concern for immunotoxicity
Grouping with PFAS and relative potency factor approach	(Assumption: potency ranking by rat liver effects can be applied to other effects and other species)	RIVM (2023) [g]	BMDL ₁₀ from human study: 17.5 ng/mL for Σ4PFAS	TDI: 0.63 ng/kg bw/day in mothers for Σ4PFAS	2.2 µg/L TFA	TFA is data-rich, grouping with PFAS unnecessary Relative potency factor approach rests on a BMDL ₁₀ that RIVM itself acknowledged as methodologically uncertain (Sect. "Derivation of a guidance value for TFA in drinking water based upon its relative potency to PFAS")

Σ4PFAS: sum of 4 PFAS; ALT: alanine transferase; ARfD: acute reference dose; bw: body weight; BMDL: lower confidence limit on the benchmark dose; d: day; HP: histopathological evaluation; mkd: mg/kg body weight/day; NO(A)EL: no-observed (adverse) effect level; PFAS: per- and polyfluoroalkyl substance(s); TDAR: T-cell dependent and independent antibody responses; TDI: tolerable daily intake; TFA: trifluoroacetic acid; WoE: weight-of-evidence

[a] Calculation by the authors of the present article considering drinking water allocation factors of 10% and 20%, respectively (Table 7)

[b] Absence of 2-year chronic toxicity / carcinogenicity study led to 2× assessment factor (Table 4), see Sect. "Selection of the key study and relevant toxicological endpoint for the derivation of the TDI" for discussion of the scientific limitations of the rat two-year bioassay. Note: The ADI proposed by EFSA (2014) no longer reflects the EFSA (2025a) current draft position, see also Footnote [e]

[c] Data evaluation across 90-day and one-year repeated dose toxicity studies

[d] UBA approach generally supported by Austrian Government (2022), Canton of Zurich (2025), Danish Government (2025), French Government (2025), Hawaii DoH (2025), Luxembourg Health Directorate (2025), Republic of Italy (2025)

[e] In addition, EFSA (2025a) selected eye malformations observed in rabbit prenatal developmental toxicity studies as critical effect for the derivation of an ARfD to cover less than lifetime exposures (foetal NOAEL: 60 mkd Na-TFA, proposed ARfD: 0.6 mg/kg bw Na-TFA). Note: EFSA (2025a) is the *Draft statement on consumer health-based guidance values on trifluoroacetic acid*. Final statement pending at the time of writing this article

[f] "The WG [working group] proposed applying the standard uncertainty factor of 100, along with an additional factor of 3 to account for the remaining uncertainties related to the absence of a long-term toxicity/carcinogenicity study and the absence of TDAR [T-cell dependent and independent antibody responses] assay on TFA" (EFSA 2025a)

[g] RIVM approach generally supported by Walloon SPW Env (2025)

Conclusion and outlook

In this review article, the case study of TFA has been presented as an example to discuss the legal requirements and scientific background to derive guidance values for substances in drinking water under the EU DWD (EP and Council 2020). (By comparison, risk management measures, which also include political and socio-economic considerations, have been excluded from the scope of this scientific publication.)

TFA is a building block in chemical synthesis and a metabolite of a broad spectrum of substances that contain a C-CF₃ moiety (Sect. "TFA: identification, uses and environmental fate"). It is to be considered a non-relevant metabolite of PPPs in drinking water under the DWD since the current exposure levels have not been found to provide a health risk to consumers (Sect. "TFA as a non-relevant metabolite of PPPs in drinking water"). Accordingly, Member States may derive guidance values for TFA in drinking water. However, since there is no EU-wide harmonised concept for the derivation of such guidance values, the different authorities and institutions have proposed a broad spectrum of guidance values for TFA in drinking water (Sect. "Derivation of guidance values for TFA in drinking

water on the Member State level"). Based upon a review of the state-of-the-science (Sect. "Guidance values for TFA in drinking water based on toxicological studies") and considering that TFA should not be included in the DWD definitions for PFAS (Sect. "Guidance value for TFA in drinking water based on relative potency to PFAS"), a harmonised, science-based approach to derive guidance values for TFA has been proposed (Sect. "Authors' proposal for a health-based guidance value for TFA in drinking water"), yielding a TDI of 0.042 mkg TFA and a health-based guidance value of 294 µg/L TFA in drinking water.

The authors of the present article aspire that the proposed science-based approach may lay the foundations for the harmonised derivation of health-based guidance values for substances in drinking water. Scientists from relevant stakeholders like executive and regulatory bodies, academia, water supply companies and industry are invited to engage in discussions to agree on a consistent approach, applicable across Member States, to evaluate whether the presence of substances in drinking water may pose a health risk to consumers and to derive scientifically robust guidance values that truly serve to protect the consumer. Communication to the general public will be facilitated by cohesive messages on the prevention of potential health risks caused by the presence of TFA (or PFAS) in the environment.

Preferably, the science-based guidance values for TFA proposed here should be transposed into substance-specific parametric values laid down in Annex I, Part B, of the DWD. Thereby, it is ensured that they apply to TFA regardless of its source. This is pertinent as TFA can not only reach water bodies via leaching or run-off/drain flow following agricultural use of PPPs containing a C-CF₃ moiety but also, e.g. via direct discharge from industry plants and via precipitation as metabolite from, e.g. refrigerants and propellants that degrade in the atmosphere, further considering that it is difficult to establish the source of TFA (Sect. "Monitoring of TFA in drinking water").

Notably, all harmonisation efforts suggested above, including the proposed health-based guidance value of 294 µg/L TFA are not intended to affect the Member States' right to apply more stringent values under Article 4(4) of the DWD.

A continued monitoring of TFA levels in drinking water (and food) is mandated also since TFA levels in drinking water will likely continue to rise secondary to precipitation of the degradation products of fluorine-containing gases present in the atmosphere. A continued monitoring will ensure that critical concentration levels of TFA in drinking water are reviewed in a timely manner so that the appropriate but resource-intensive countermeasures are implemented when relevant for consumer protection.

Table 7 Dependency of health-based guidance value for TFA in drinking water on selected acceptable daily intake level and allocation of TFA uptake via drinking water

		Drinking water allocation factor [c]	
		10%	20%
Acceptable daily intake level (µg/kg bw/day TFA)	42 [a]	147 µg/L TFA	294 µg/L TFA [d]
	25 [b]	87.5 µg/L TFA	175 µg/L TFA

Health-based guidance value for TFA in drinking water derived using the WHO standard calculation method (Sect. "Introduction: derivation of guidance values for substances in drinking water") and different acceptable daily intake levels and allocation factors (all four approaches: assessment factor: 200; body weight: 70 kg; daily drinking water consumption: 2 L)

[a] Published by EFSA (2014); see Sect. "Hazard assessment: establishment of an ADI for TFA by EFSA" for discussion

[b] An acceptable daily intake level of 0.03 mg/kg body weight (bw)/day Na-TFA (corresponding to 0.025 mg/kg bw/day TFA and 25 µg/kg bw/day TFA) was proposed by EFSA (2025a); see Sect. "Hazard assessment: establishment of an ADI for TFA by EFSA" for discussion. Note: EFSA (2025a) is the *Draft statement on consumer health-based guidance values on trifluoroacetic acid*. Final statement pending at the time of writing this article

[c] See Sect. "Allocation of TFA uptake via water" for discussion

[d] Approach proposed by authors; see Sect. "Guidance values for TFA in drinking water based on toxicological studies" for discussion of the underlying parameters

Appendix

Glossary

Acceptable daily intake (ADI): By analogy to the tolerable daily intake (TDI), “an estimate of the amount of a substance in food and drinking-water, expressed on a body weight basis (milligram or microgram per kilogram of body weight), that can be ingested over a lifetime without appreciable health risk, and with a margin of safety.” ADIs “are established for food additives and pesticide residues that occur in food for necessary technological purposes or plant protection reasons” (adapted from Sect. 8.2.2 in WHO 2022).

Acute reference dose (ARfD): “an estimate of the amount of a substance in food or drinking water that can be ingested over a short period of time, usually during one meal or one day, without appreciable health risk to the consumer” (JMPR 2002 [Food and Agriculture Organization of the United Nations (FAO)/World Health Organization (WHO) Joint Meeting on Pesticide Residues]; ECHA 2016).

Benchmark dose: “The benchmark dose (BMD) is a dose level, estimated from the fitted dose–response curve, associated with a specified change in response relative to the control group (background response), the benchmark response... The BMDL is the lower bound of the BMD’s credible interval, and this value is normally used as the RP [reference point]” (EFSA SC 2022).

Benchmark dose level (BMDL): “the lower confidence limit of the dose that produces a small increase (e.g. 5% or 10%) in the level of adverse effects. The BMDL is derived on a quantitative basis using data from the entire dose–response curve for the critical effect rather than from a single dose at the NOAEL or LOAEL and accounts for the statistical power and quality of the data (IPCS 2009)” (Sect. 8.2.2 in WHO 2022).

Benchmark response: “A specified change in response relative to the control group (background response)” (EFSA SC 2022).

Drinking water: The water at the tap, i.e. the water that humans consume.

Exposure assessment: “The process of estimating or measuring the magnitude, frequency, and duration of exposure to an agent, along with the number and characteristics of the population exposed. Ideally, it describes the sources, pathways, routes, and the uncertainties in the assessment” (WHO IPCS 2004).

Guidance value / guideline value: see health-based guidance value / guideline value and provisional guideline value and Table 8.

Hazard: “means a biological, chemical, physical or radiological agent in water, or another aspect of the condition of water, with the potential to cause harm to human health” (Article 2(7) in EP and Council 2020).

Hazard identification: “The identification of the type and nature of adverse effects that an agent has an inherent capacity to cause in an organism, system, or (sub)population. Hazard identification is the first stage in hazard assessment” (WHO IPCS 2004).

Hazard characterisation (dose–response assessment): “The qualitative and, wherever possible, quantitative description of the inherent property of an agent or situation having the potential to cause adverse effects. This should, where possible, include a dose–response assessment and its attendant uncertainties. Hazard characterization is the second stage in the process of hazard assessment” (WHO IPCS 2004).

Health-based guidance value / health-based guideline value: “The concentration of a constituent that does not result in any significant risk to health. This is for a lifetime of consumption unless indicated.”—“For adaptation into national drinking-water standards if warranted by local circumstances including data on occurrence” (WHO 2022; see also Table 8).

Lowest-observed adverse effect level (LOAEL): “the lowest observed dose or concentration of a substance at which there is a detectable adverse health effect. When a LOAEL is used instead of a NOAEL, an additional uncertainty factor is normally applied...” (Sect. 8.2.2 in WHO 2022).

- **Metabolite / relevant metabolite** EU DWD: “...metabolites as defined in point (32) of Article 3 of Regulation (EC) No 1107/2009 of the European Parliament and of the Council... that are considered relevant for water intended for human consumption. A pesticide metabolite shall be deemed relevant for water intended for human consumption if there is reason to consider that it has intrinsic properties comparable to those of the parent substance in terms of its pesticide target activity or that either itself or its transformation products generate a health risk for consumers” (Note in Annex I [Part B] in EP and Council 2020).
- **EU PPP Regulation:** “Metabolite means any metabolite or a degradation product of an active substance, safener or synergist, formed either in organisms or in the environment. A metabolite is deemed relevant if there is a reason to assume that it has intrinsic properties comparable to the parent substance in terms of its biological target activity, or that it poses a higher or comparable risk to organisms than the parent substance or that it has certain toxicological properties that are considered unacceptable. Such a metabolite is relevant for the overall

Table 8 Terminology related to water quality targets

Organisation / authority / reference	Term (original language, if applicable)	Note
World Health Organization (WHO 2022)	Health-based guideline value	Page 172: “ <i>The concentration of a constituent that does not result in any significant risk to health. This is for a lifetime of consumption unless indicated.</i> ”—“ <i>For adaptation into national drinking-water standards if warranted by local circumstances including data on occurrence.</i> ”
	Provisional guideline value	Page 172: Similar to health-based guideline value, but “ <i>takes into account practical issues associated with monitoring, management or competing risks</i> ”
	Provisional reference value	Page 171: “ <i>Where data are inadequate to establish a formal guideline value, a provisional reference value may be proposed. These values are derived from the limited data and are too uncertain to be used for developing regulations or standards. Nevertheless these “bounding values” may be useful to guide actions... in the event of need.</i> ”
European Union (DWD; EP and Council 2020)	Parametric value	Used in relation to the values provided for the specific parameters listed in Annex I of the DWD
	(Health-based) guidance value	Used in Article 13 in relation to non-relevant metabolites and substances included in “watch list”
	Threshold value	Used in Article 8(2)c(iii) in relation to the values provided in Annex II of Directive 2006/118/EC for groundwater pollutants
Danish DTU Food (Nielsen and Beltoft 2021)	Health-based quality criterion (<i>sundhedsmæssigt baseret kvalitetskriterium</i>)	Corresponds to health-based guidance value
Netherlands (RIVM 2023)	Indicative drinking water guideline value (<i>Drinkwaterrichtwaarde</i>)	For the derivation of <i>indicative</i> guideline values, existing evaluations are in principle used, whereas the underlying studies are generally not assessed
Flemish VITO (De Clercq and De Brouwere 2024)	Health-based advisory value (<i>gezondheidskundige advieswaarde</i>)	Corresponds to health-based guidance value
France (French Government 2025)	Indicative health value (<i>valeur sanitaire indicative</i>)	Aligned with UBA (2020a)
Germany (UBA 2020a, b)	Health-related orientation value (<i>gesundheitlicher Orientierungswert</i>)	Corresponds to provisional reference value as per WHO definition
	Health-based guidance value (<i>gesundheitlicher Leitwert</i>)	Health-based guidance value
Luxembourg Health Directorate (2024)	Guidance value (<i>valeur guide</i>)	Health-based guidance value
Walloon SPW Env (2025)	Guidance value (<i>valeur guide</i>)	Aligned with RIVM (2023)

approval decision or for the definition of risk mitigation measures.” (Article 3(32) in EP and Council 2009a).

- **REACH Regulation:** No distinction between relevant and non-relevant metabolites of chemical substances; instead, only general reference to metabolites (see e.g. Point 8.4.5, Column B, in Annex IX of the REACH Regulation; EP and Council 2006b).

No-observed adverse effect level (NOAEL): “*the highest dose or concentration of a chemical in a single study, found by experiment or observation, that causes no detectable adverse health effect. Wherever possible, the NOAEL is based on long-term studies, preferably of ingestion in drinking-water. However, NOAELs obtained from short-term studies and studies using other sources of exposure (e.g. food, air) may also be used. If a NOAEL is not available, a LOAEL may be used*” (Sect. 8.2.2 in WHO 2022).

- **Per- and polyfluoroalkyl substance(s) (PFAS):** Buck et al. (2011): “*aliphatic substances containing one or more C atoms on which all the H substituents present in the nonfluorinated analogues from which they are notionally derived have been replaced by F atoms, in such a manner that PFASs contain the perfluoroalkyl moiety C_nF_{2n+1} .*”
- EP and Council (2020); DWD, ‘**PFAS Total**’: “*substances that contain a perfluoroalkyl moiety with three or more carbons (i.e. $-C_nF_{2n-}$, $n \geq 3$) or a perfluoroalkylether moiety with two or more carbons (i.e. $-C_nF_{2n}OCmF_{2m-}$, n and $m \geq 1$)*”.
- EP and Council (2020); DWD, ‘**Sum of PFAS**’: A subset of ‘PFAS Total’ that currently includes twenty PFAS listed in point 3 of Part B of Annex III to the DWD.
- OECD (2021): “*are defined as fluorinated substances that contain at least one fully fluorinated methyl or methylene carbon atom (without any H/Cl/Br/I atom*

attached to it), i.e. with a few noted exceptions, any chemical with at least a perfluorinated methyl group ($-CF_3$) or a perfluorinated methylene group ($-CF_2-$) is a PFAS". (Hence, while the definition by Buck et al. generally includes substances with a perfluorinated methyl group [$-CF_3-$], the OECD definition additionally covers substances with a perfluorinated methylene group [$-CF_2-$].)

- **US EPA (2023):** "PFAS is defined as including at least one of these three structures: (1) $R-(CF_2)_n-CF(R')R''$, where both the CF_2 and CF moieties are saturated carbons; (2) $R-CF_2OCF_2-R'$, where R and R' can either be F , O , or saturated carbons; and (3) $CF_3C(CF_3)R'R''$, where R' and R'' can either be F or saturated carbons... Under this rule's definition of PFAS, TFA is not within scope" (US EPA 2023; similarly in US EPA 2022).
- **ECHA (2026c):** a PFAS is as "any substance that contains at least one fully fluorinated methyl (CF_3-) or methylene ($-CF_2-$) carbon atom (without any $H/Cl/Br/I$ attached to it)."

Pesticide: "A pesticide means a plant protection product..." (Article 3(10) in Directive 2009/128/EC on the sustainable use of pesticides; EP and Council 2009b).

Plant protection products (PPPs): "Products used to protect, preserve or influence the growth of desirable plants or to destroy or control the growth of unwanted plants or parts of plants" (EFSA 2026).

Pollutant: "means any substance liable to cause pollution, in particular those listed in Annex VIII" (EP and Council 2000; in this Directive establishing a framework for Community action in the field of water policy, Annex VIII provides an indicative list of the main pollutants).

Primary crops: "Crops listed on pesticide product labels intended for protection against specific pests (insects, fungi, weeds, etc.) by direct treatment (soil, foliar, seed treatment, etc.) with the pesticide" (OECD 2018d).

Provisional guideline value: Similar to health-based guideline value, but "takes into account practical issues associated with monitoring, management or competing risks" (WHO 2022; see also Table 8).

Provisional reference value: "Where data are inadequate to establish a formal guideline value, a provisional reference value may be proposed. These values are derived from the limited data and are too uncertain to be used for developing regulations or standards. Nevertheless these "bounding values" may be useful to guide actions... in the event of need" (WHO 2022; see also Table 8).

Raw water: the water extracted by the water suppliers before further processing to become drinking water (see definition).

Residues: "one or more substances present in or on plants or plant products, edible animal products, drinking

water or elsewhere in the environment and resulting from the use of a plant protection product, including their metabolites, breakdown or reaction products" (Article 3(1) in EP and Council 2009a).

Risk: "means a combination of the likelihood of a hazardous event and the severity of the consequences if the hazard and hazardous event occur in the supply system of water intended for human consumption" (Article 2(9) in EP and Council 2020).

Risk assessment: "A process intended to calculate or estimate the risk to a given target organism, system, or (sub)population, including the identification of attendant uncertainties, following exposure to a particular agent, taking into account the inherent characteristics of the agent of concern as well as the characteristics of the specific target system" (WHO IPCS 2004).

Risk characterisation: "The qualitative and, wherever possible, quantitative determination, including attendant uncertainties, of the probability of occurrence of known and potential adverse effects of an agent in a given organism, system, or (sub)population, under defined exposure conditions" (WHO IPCS 2004).

Risk management: "Decision-making process involving considerations of political, social, economic, and technical factors with relevant risk assessment information relating to a hazard so as to develop, analyse, and compare regulatory and non-regulatory options and to select and implement appropriate regulatory response to that hazard. Risk management comprises three elements: risk evaluation; emission and exposure control; and risk monitoring" (WHO IPCS 2004).

Rotational crops: "Crops planted following the harvest or destruction of primary crops treated with a pesticide" (OECD 2018d).

Substance of concern (as defined the EU PPP Regulation): "means any substance which has an inherent capacity to cause an adverse effect on humans, animals or the environment and is present or is produced in a plant protection product in sufficient concentration to present risks of such an effect. Such substances include, but are not limited to, substances meeting the criteria to be classified as hazardous in accordance with Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures [EP and Council 2008b], and present in the plant protection product at a concentration leading the product to be regarded as dangerous within the meaning of Article 3 of Directive 1999/45/EC" [EP and Council 1999; i.e. depending on toxicological endpoint: ≥ 0.1 or $\geq 1\%$ weight by weight] (Article 3(4) in EP and Council 2009a).

Tolerable daily intake (TDI): "an estimate of the amount of a substance in food and drinking-water, expressed on a

body weight basis (milligram or microgram per kilogram of body weight), that can be ingested over a lifetime without appreciable health risk, and with a margin of safety” (Sect. 8.2.2 in WHO 2022).

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00204-026-04530-9>.

Acknowledgements The authors wish to thank the secretariat of CropLife Europe (Brussels, Belgium) for organisational support and CropLife Europe Expert Groups for technical support.

Data availability This article does not include hitherto unpublished data. All data included herein are from publicly available sources and appropriately referenced.

Declarations

Conflict of interest With the exception of the freelance scientific writer (author UGS), all authors are employed by manufacturers of crop protection products that contain active substances potentially metabolising to trifluoroacetate (TFA) in the environment, i.e. Bayer AG (authors RS, BB, AH), BASF SE (authors FMK, VL, SM-K], Syngenta (authors RB, ES), ADAMA Agricultural Solutions (author JM) and ISK Biosciences Europe (author DFW). The derivation of a health-based guidance value for TFA in drinking water directly concerns the regulatory and commercial conditions under which these products are authorised and marketed in the European Union. Administrative support, article publishing charges and fees for writing assistance were provided by CropLife Europe, the industry association representing the manufacturers named above. The authors declare that, notwithstanding the above, the scientific analysis and conclusions presented here reflect their independent expert judgment. Readers are encouraged to weigh this declaration in light of the financial relationships described.

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